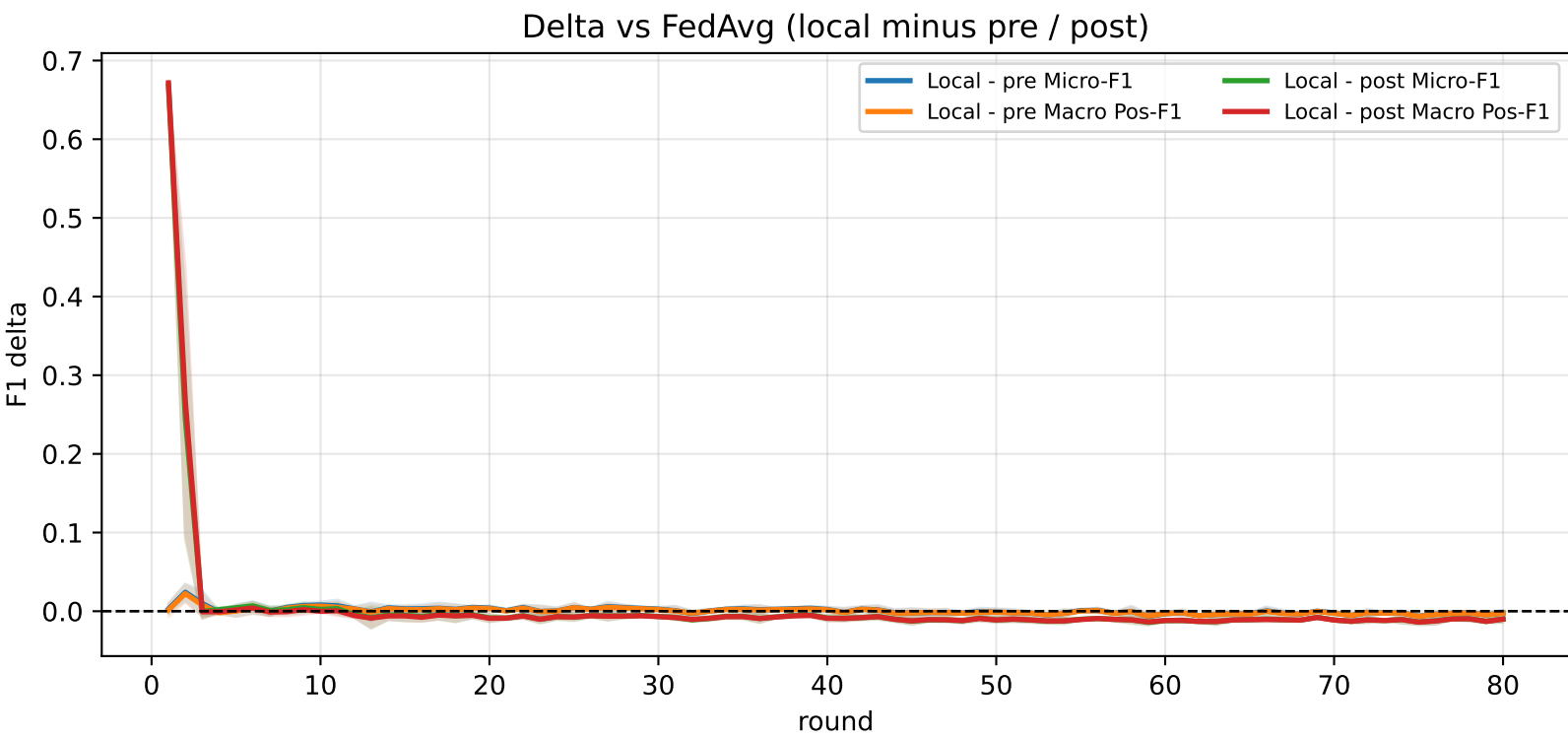
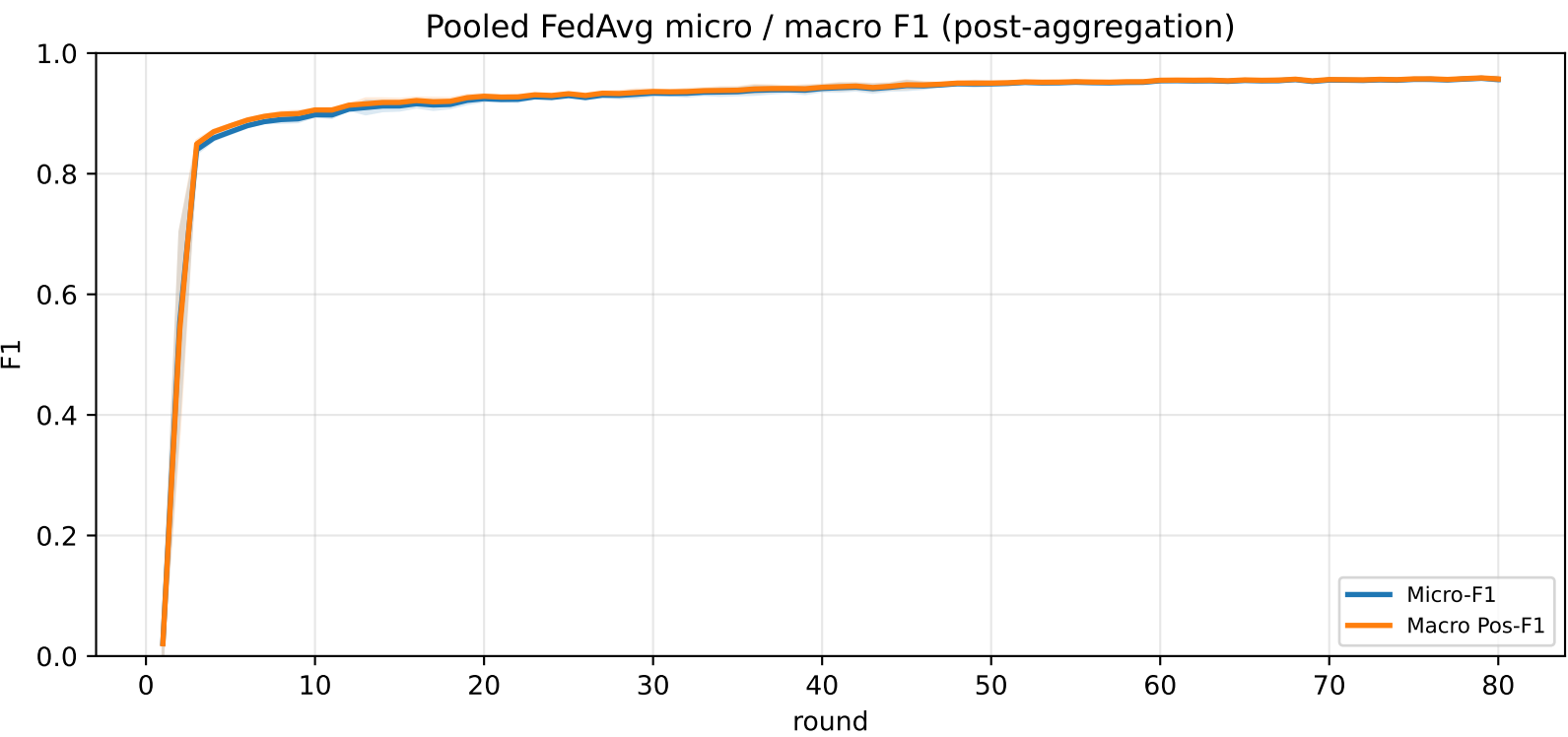
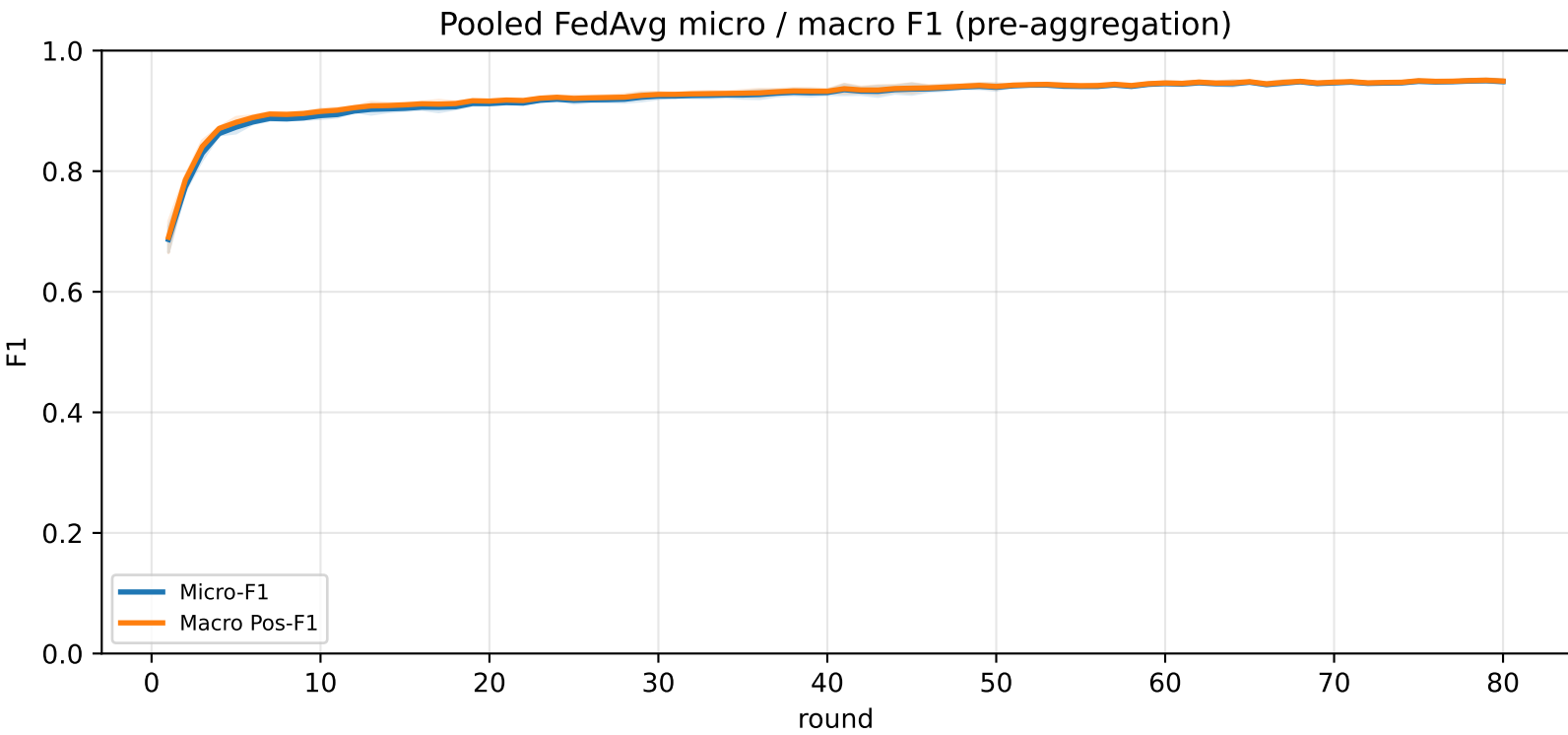
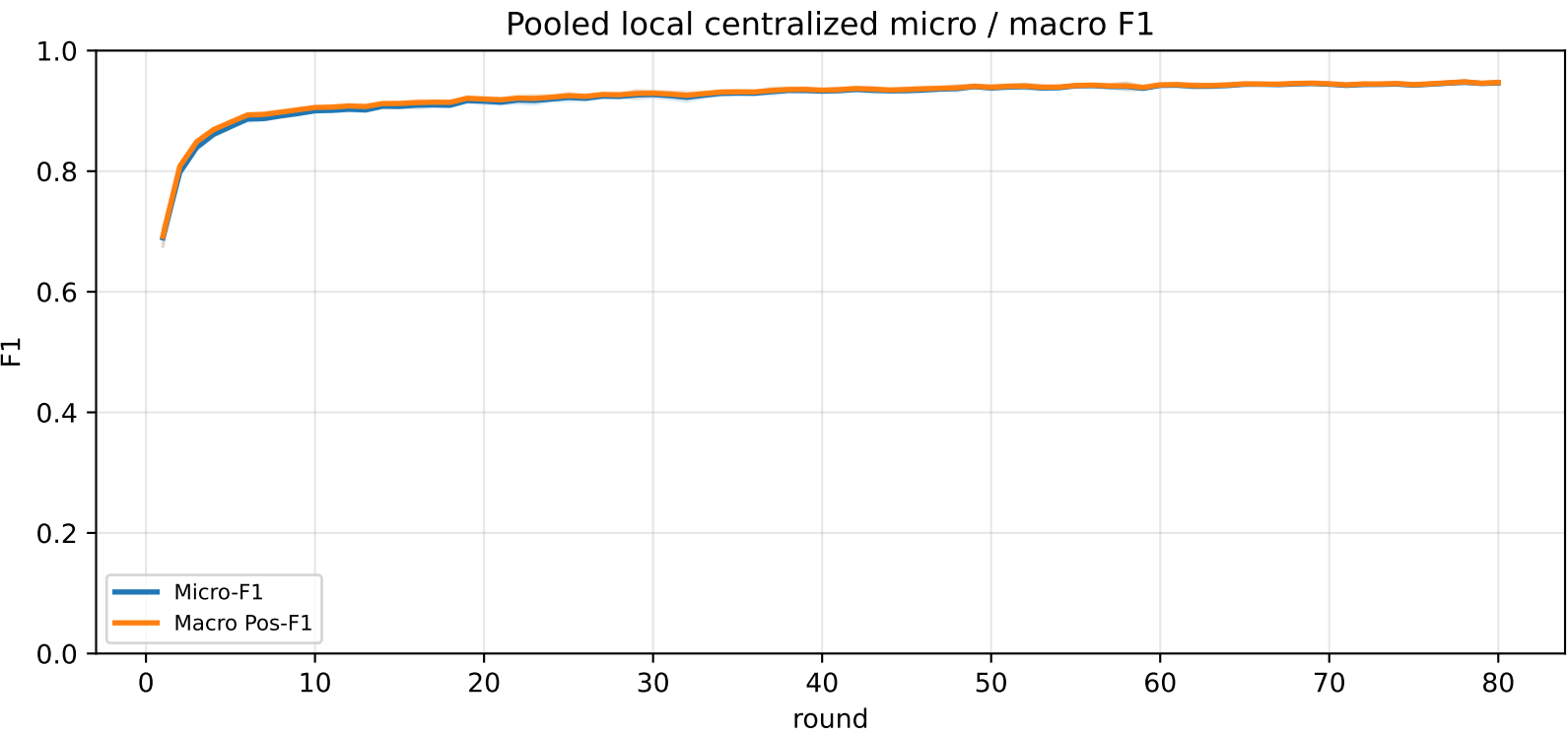
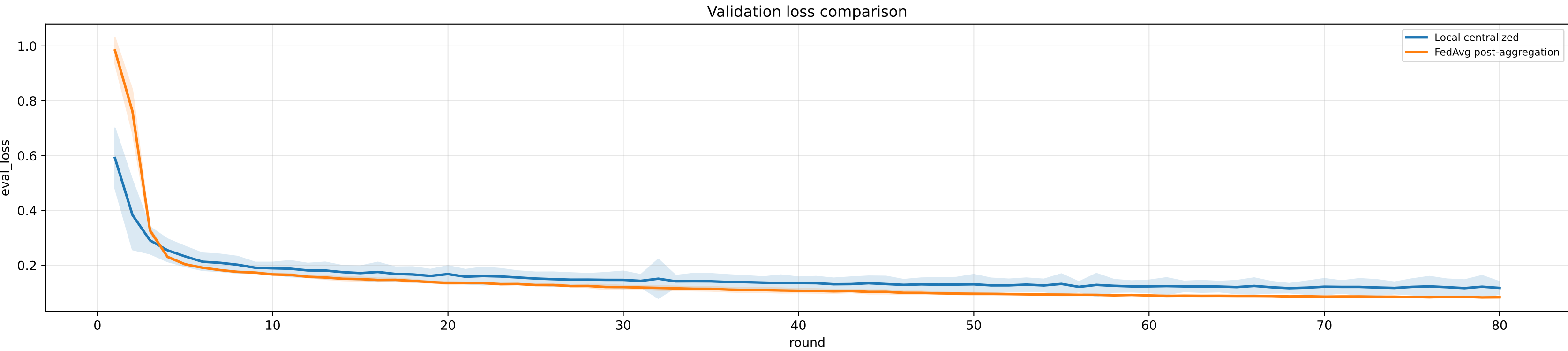


Heterogeneity level: Low | FedAvg diagnostics | aggregated / pooled over 20 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

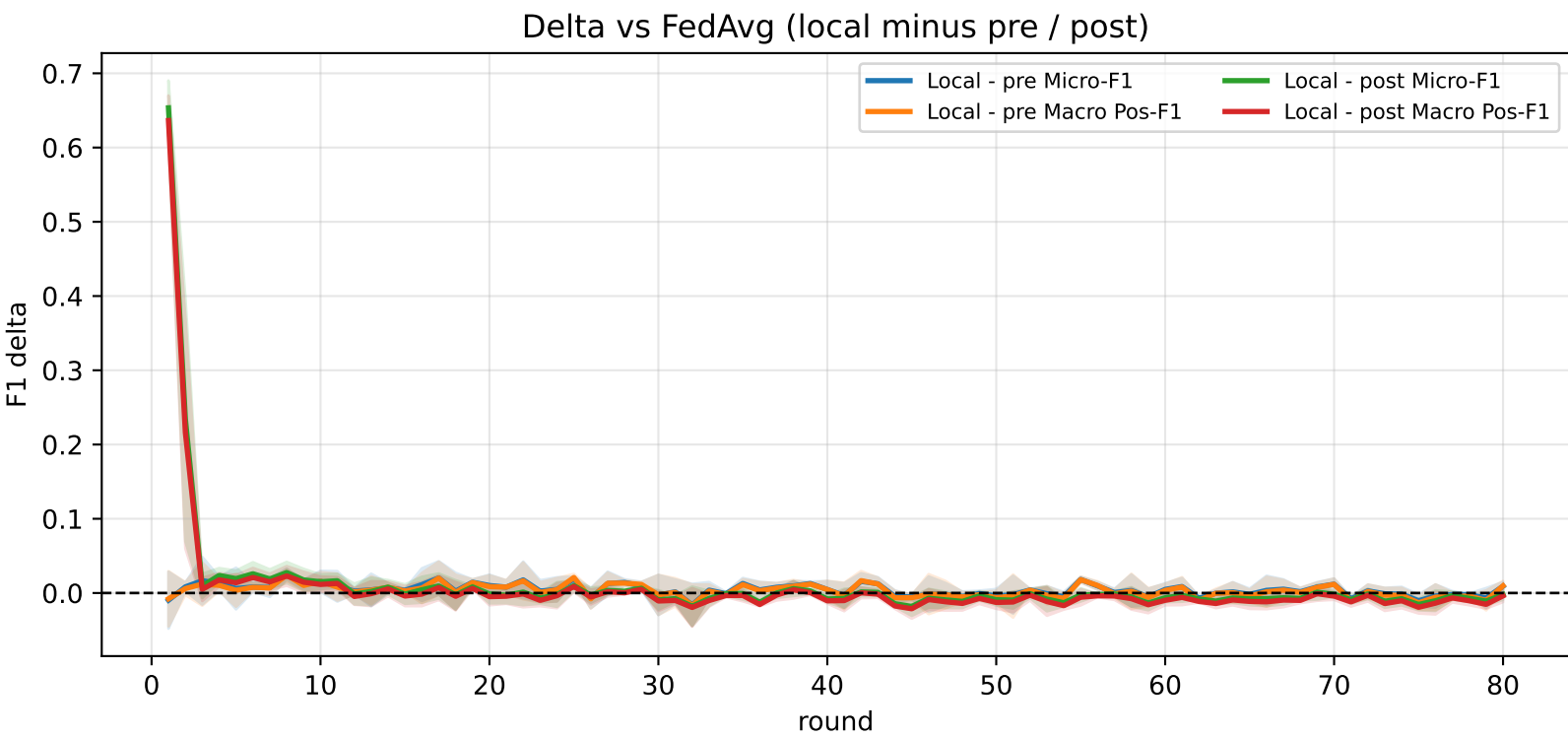
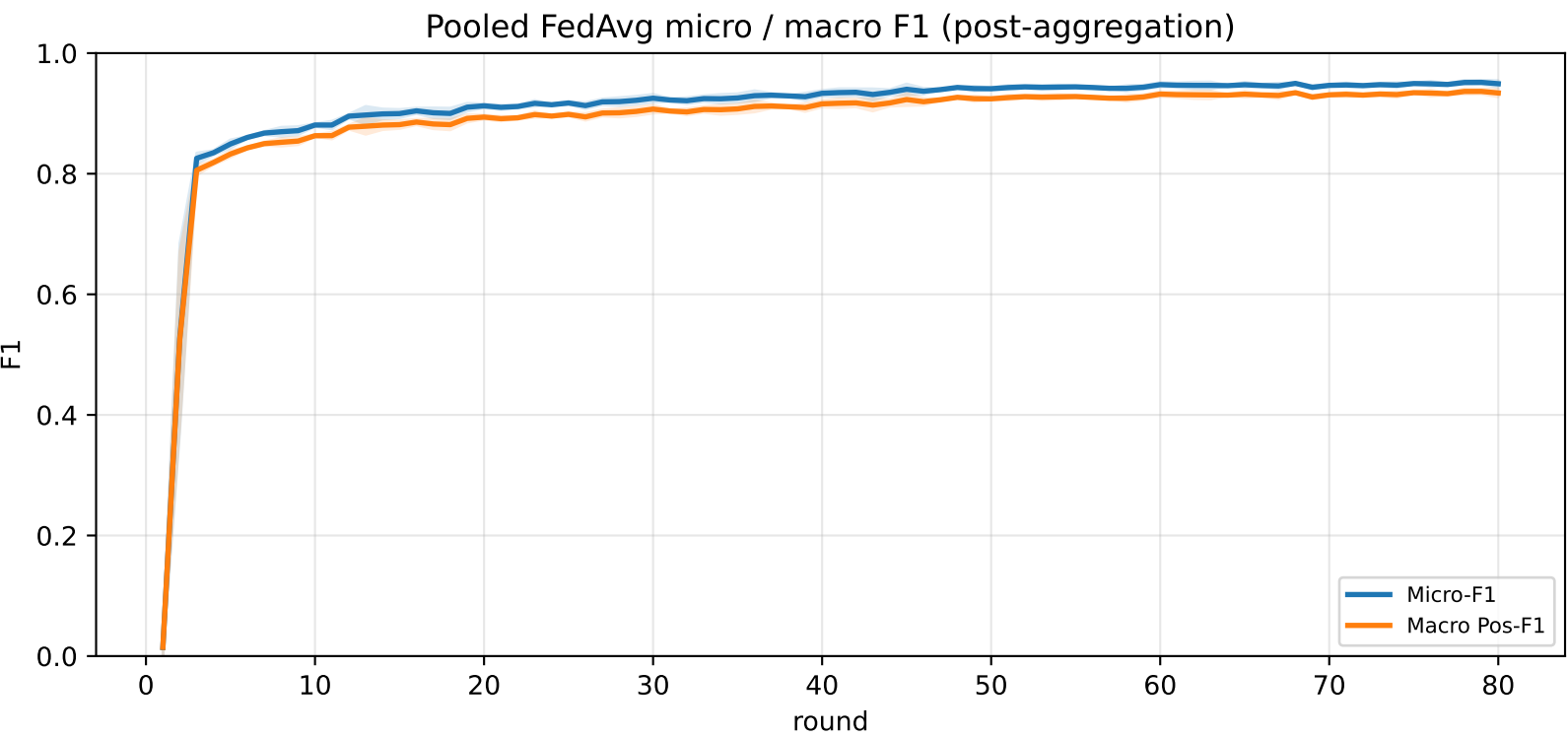
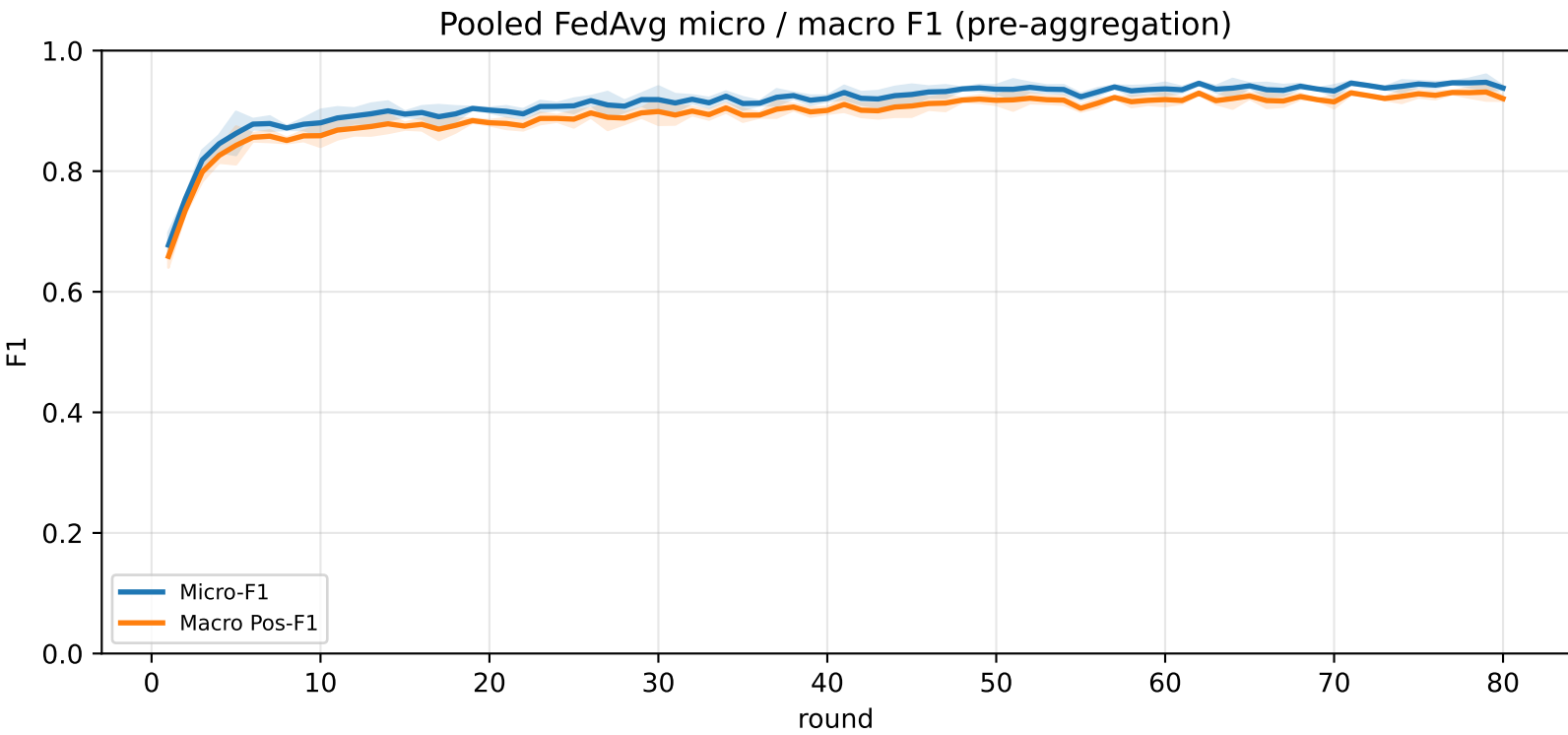
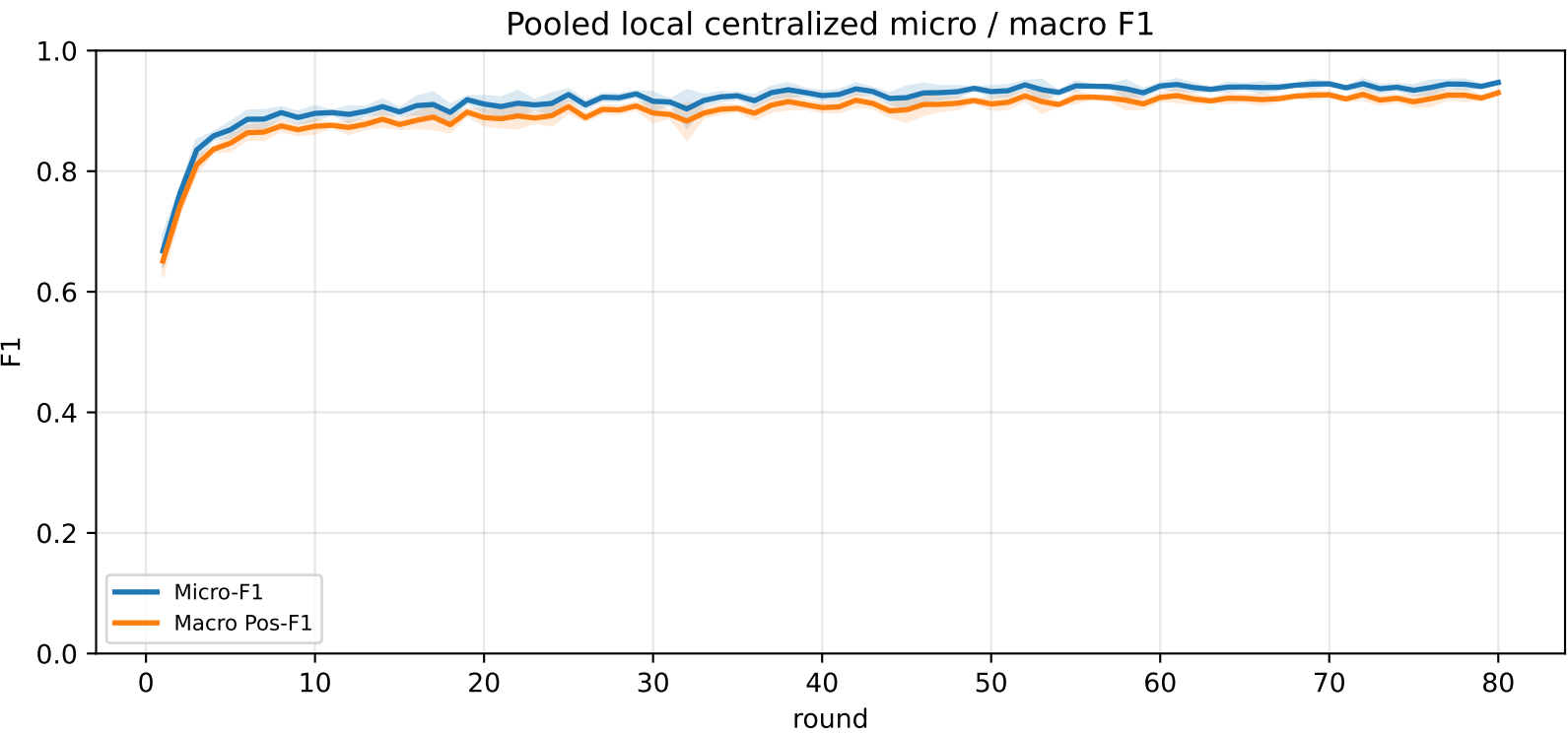
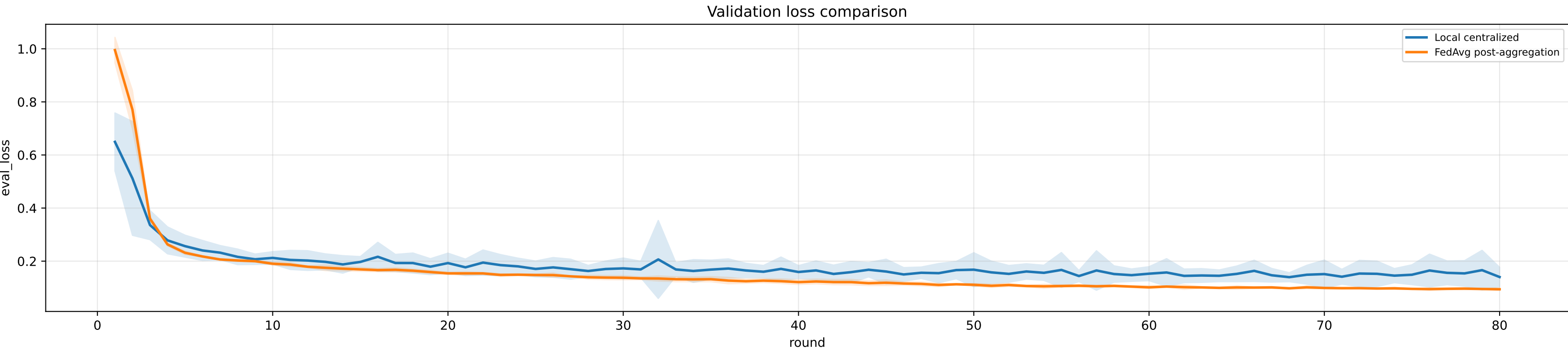
Best logged test performance (FedAvg vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.7% ± 0.2%	94.4% ± 0.2%	99.2% ± 0.0%	98.6% ± 0.1%	96.8% ± 0.2%	93.1% ± 0.2%	86.2% ± 0.5%
FedAvg	95.9% ± 0.1%	95.7% ± 0.1%	99.2% ± 0.1%	98.7% ± 0.1%	97.4% ± 0.1%	95.2% ± 0.2%	89.5% ± 0.2%



Heterogeneity level: Low | FedAvg diagnostics | family=strong_decreasing | n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=strong_decreasing | cycle2=35.9%±5.1% | cycle3=31.5%±1.3% | cycle4=25.3%±2.1% | cycle5=19.6%±3.4% | cycle6=15.3%±3.9%
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

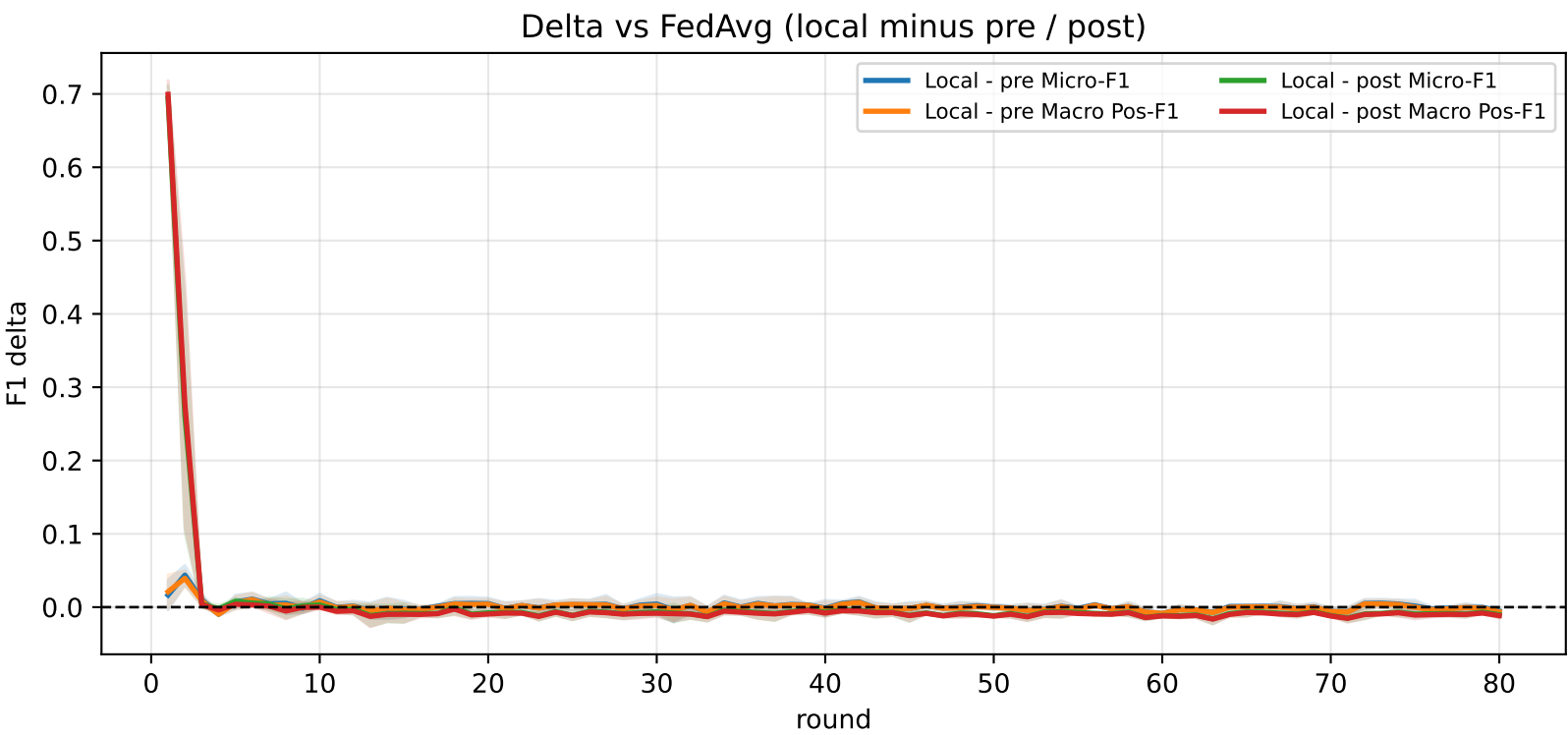
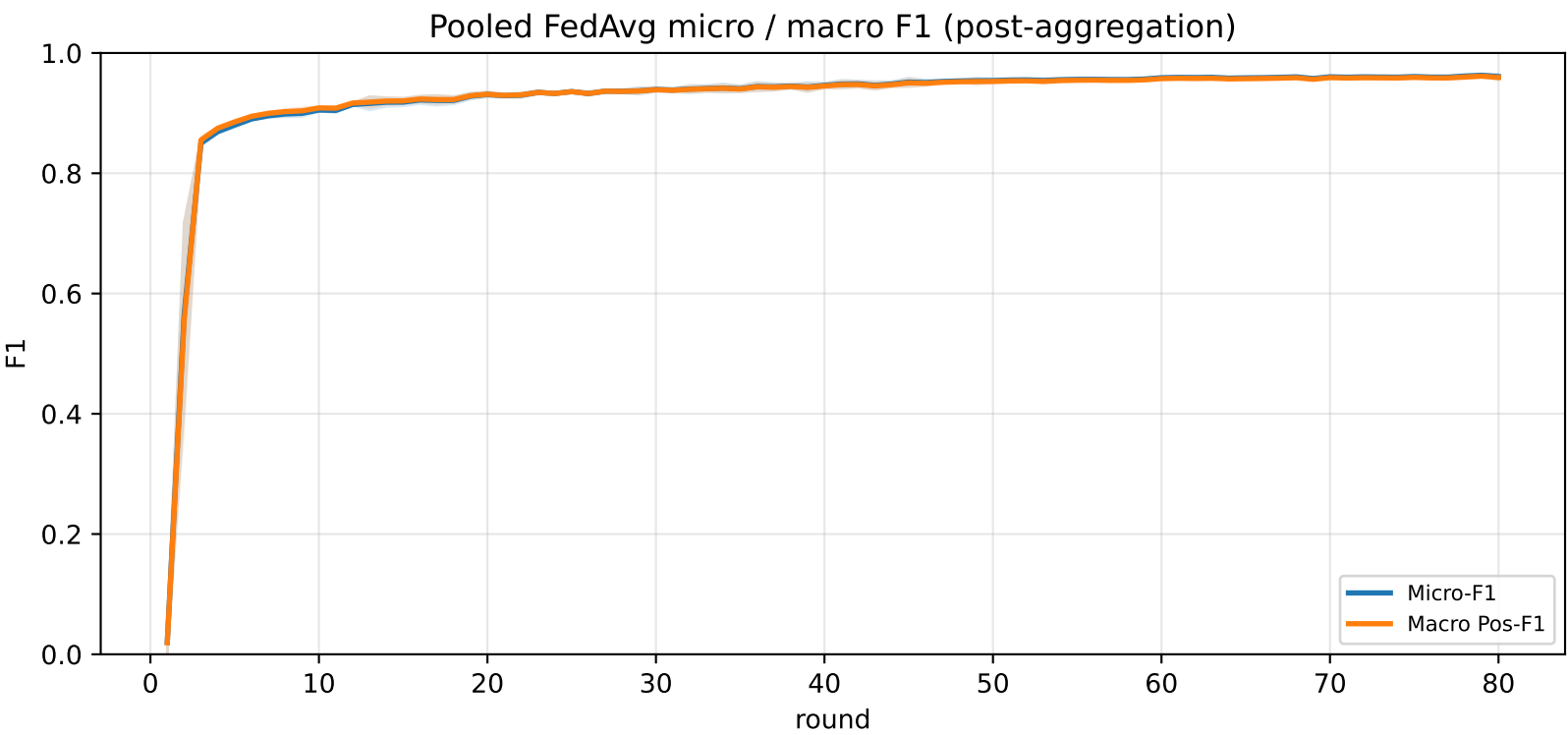
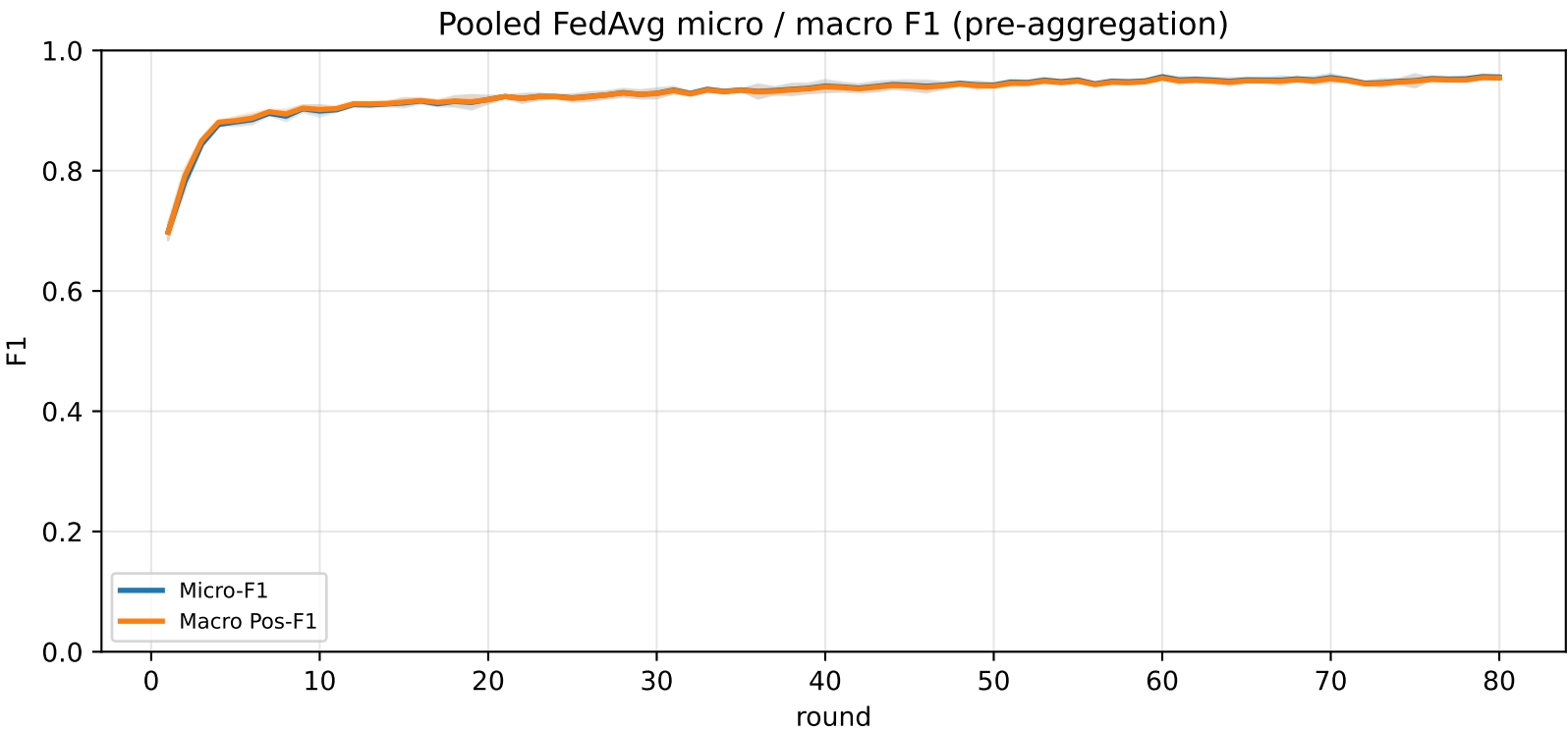
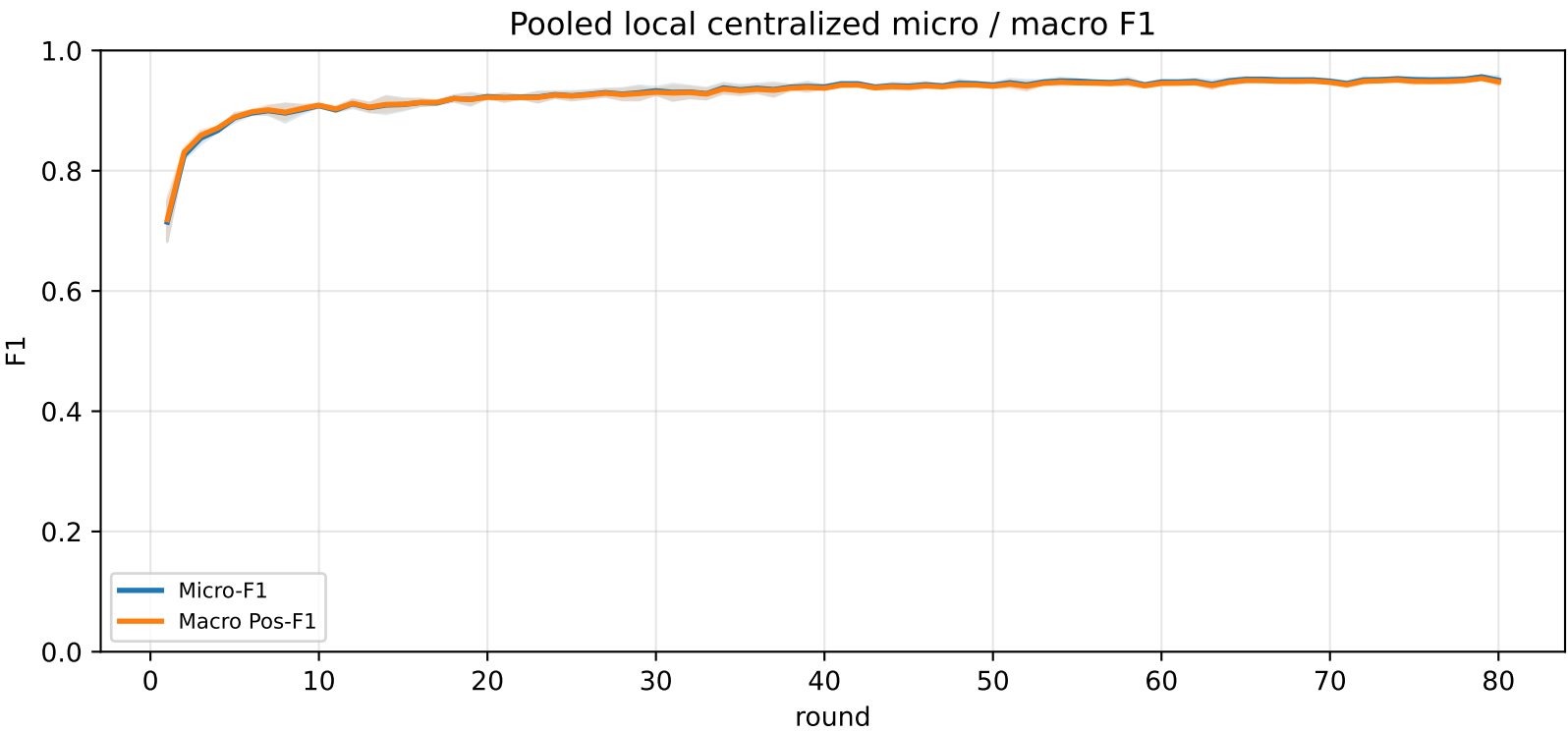
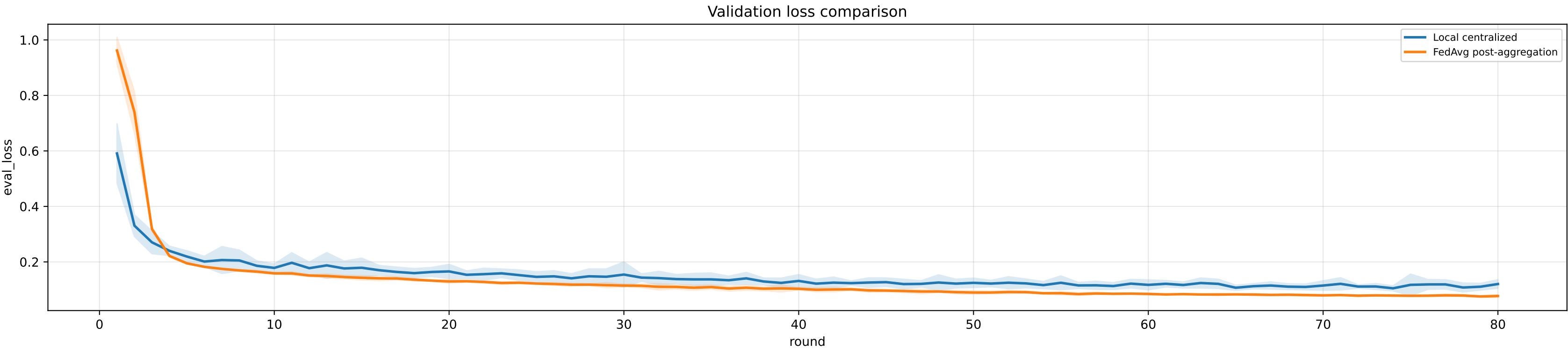
Best logged test performance (FedAvg vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.2% ± 0.3%	91.8% ± 0.3%	99.6% ± 0.1%	99.0% ± 0.1%	96.5% ± 0.2%	90.1% ± 1.0%	76.7% ± 0.6%
FedAvg	95.4% ± 0.0%	93.2% ± 0.0%	99.6% ± 0.1%	99.2% ± 0.1%	97.1% ± 0.2%	92.7% ± 0.2%	80.3% ± 0.3%



Heterogeneity level: Low | FedAvg diagnostics | family=mild_decreasing | n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=mild_decreasing | cycle2=28.4%±0.4% | cycle3=28.5%±0.9% | cycle4=28.1%±0.7% | cycle5=25.6%±0.3% | cycle6=23.4%±0.3%
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

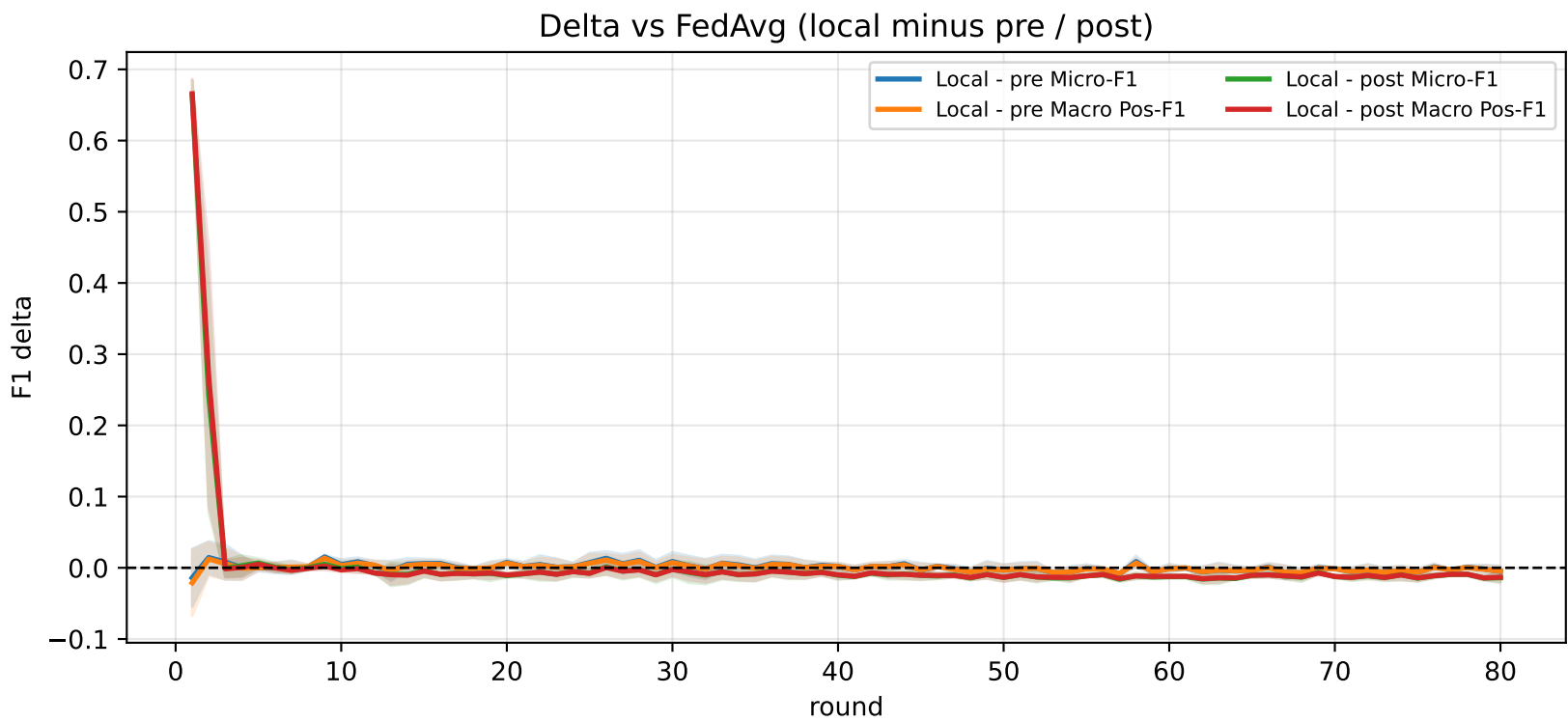
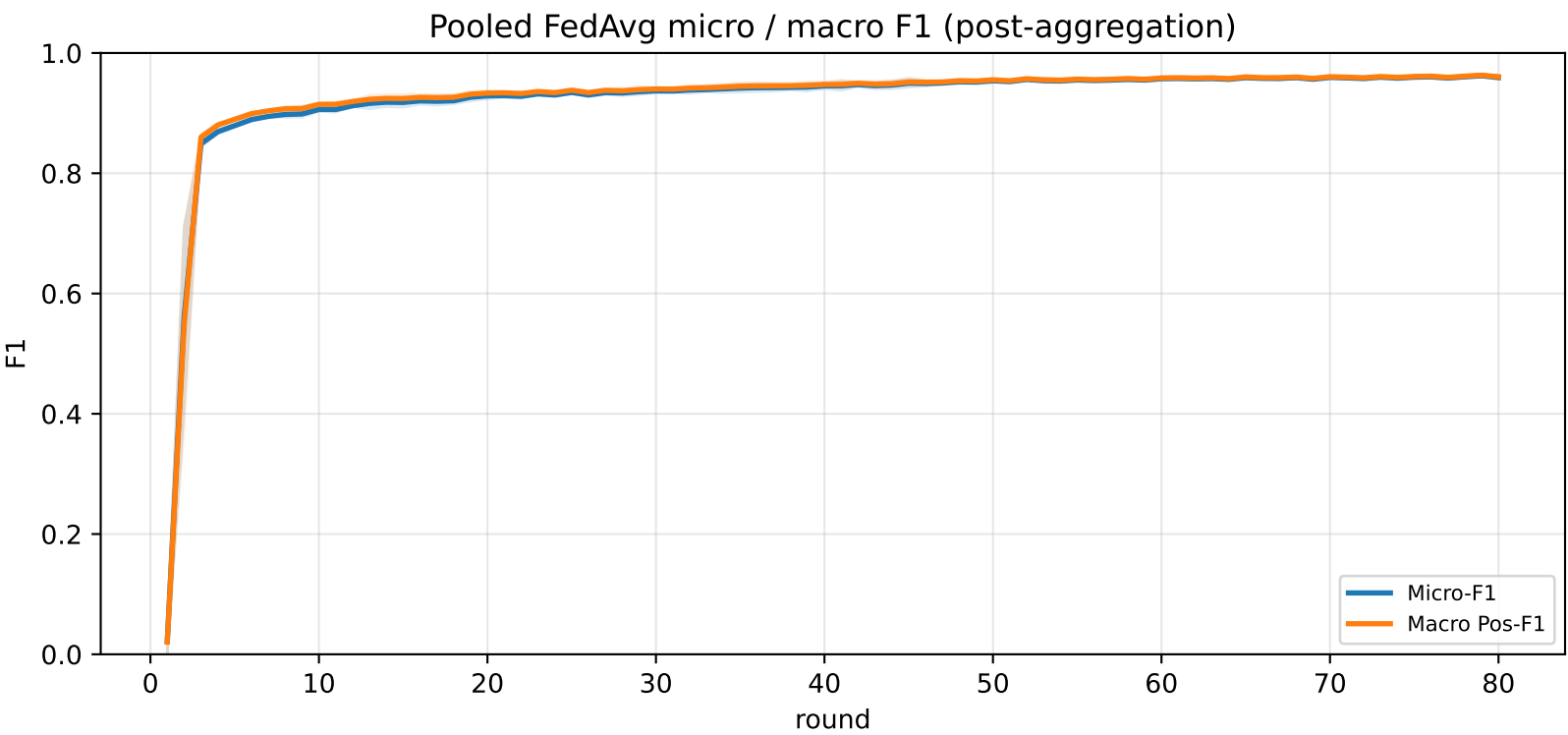
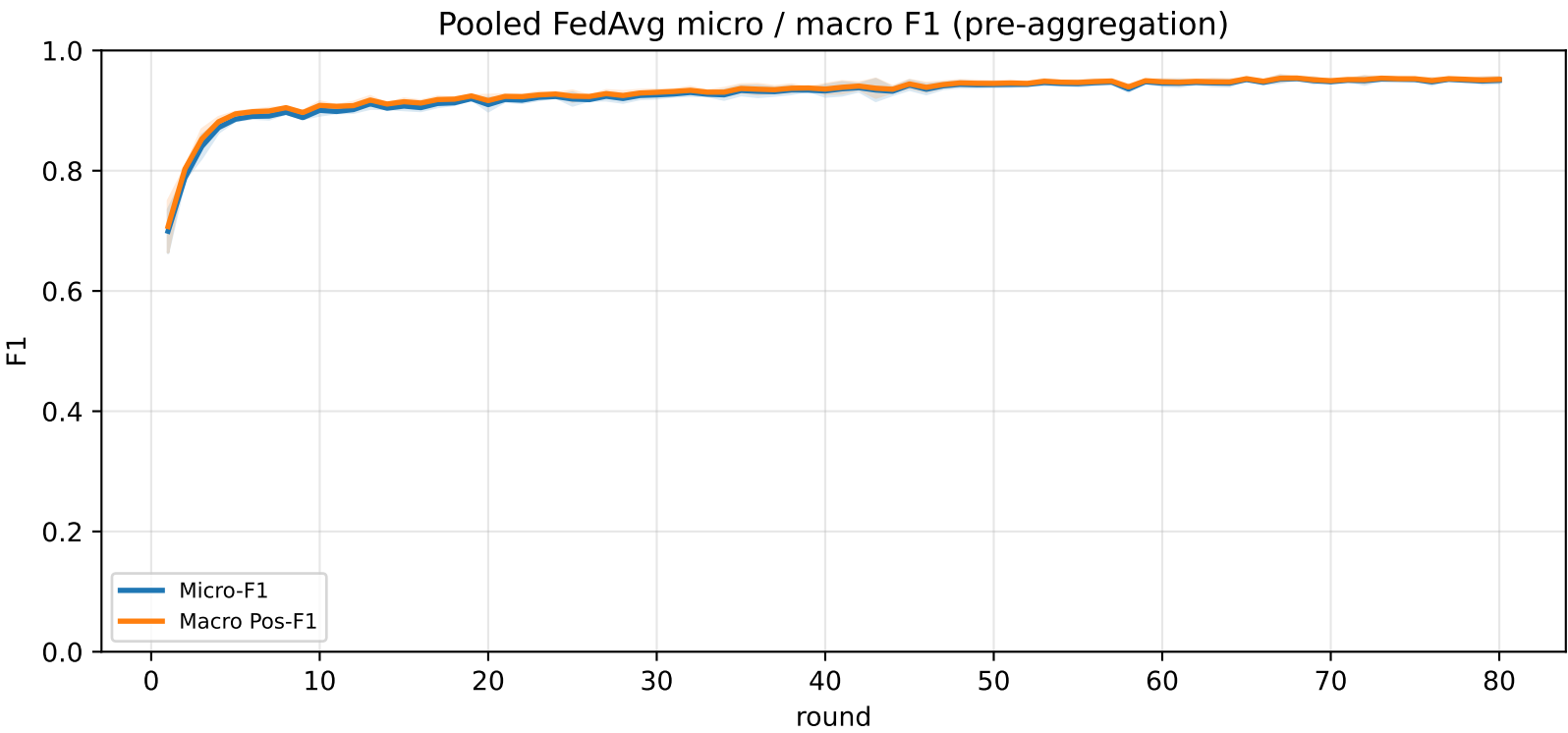
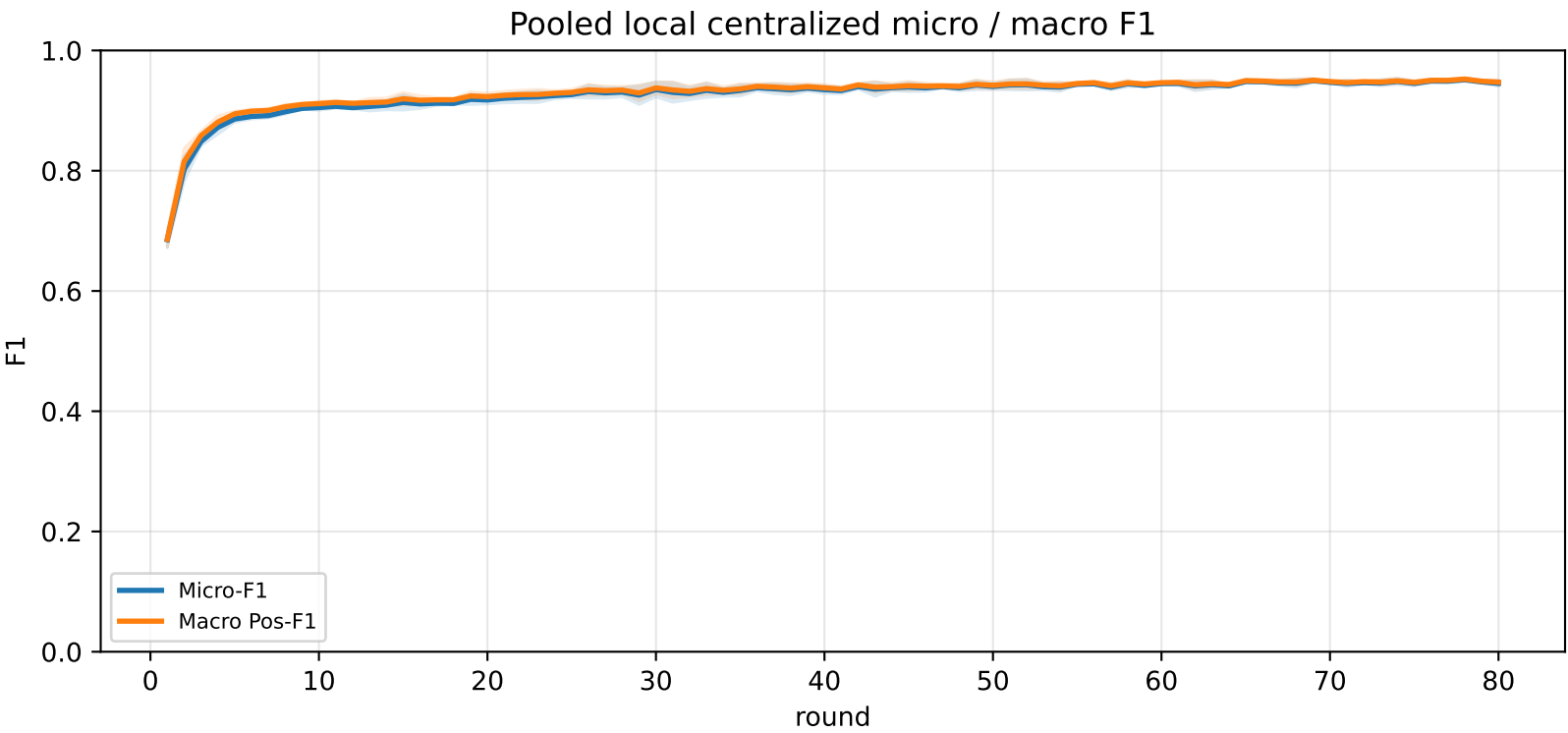
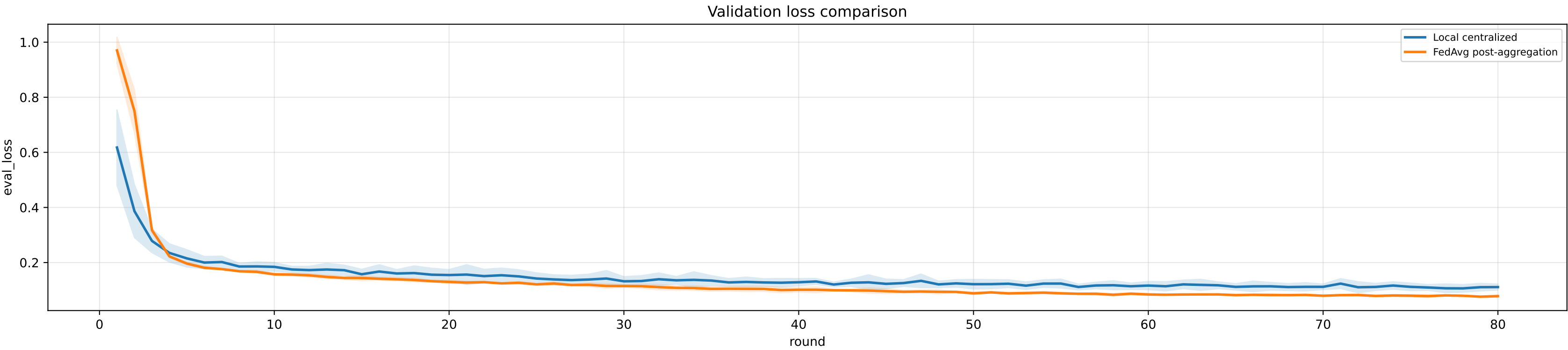
Best logged test performance (FedAvg vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	95.3% ± 0.2%	95.2% ± 0.2%	99.3% ± 0.1%	98.9% ± 0.1%	97.4% ± 0.3%	93.3% ± 0.4%	86.9% ± 0.7%
FedAvg	96.4% ± 0.2%	96.2% ± 0.2%	99.3% ± 0.0%	99.0% ± 0.1%	97.9% ± 0.3%	95.6% ± 0.3%	89.4% ± 0.2%



Heterogeneity level: Low | FedAvg diagnostics | family=flat_balanced | n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=flat_balanced | cycle2=25.3%±0.6% | cycle3=26.4%±0.5% | cycle4=26.9%±0.4% | cycle5=26.6%±0.7% | cycle6=26.2%±0.9%
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

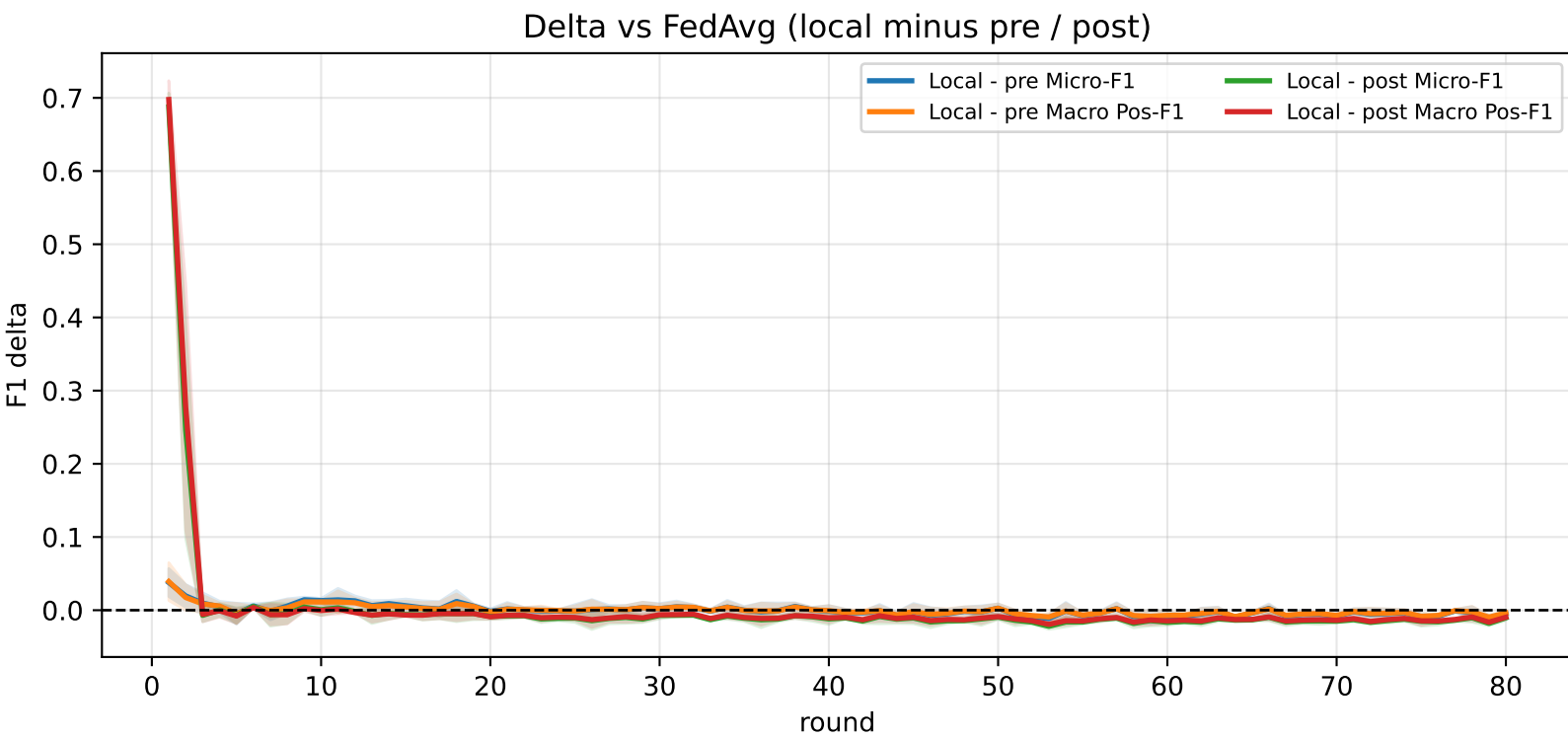
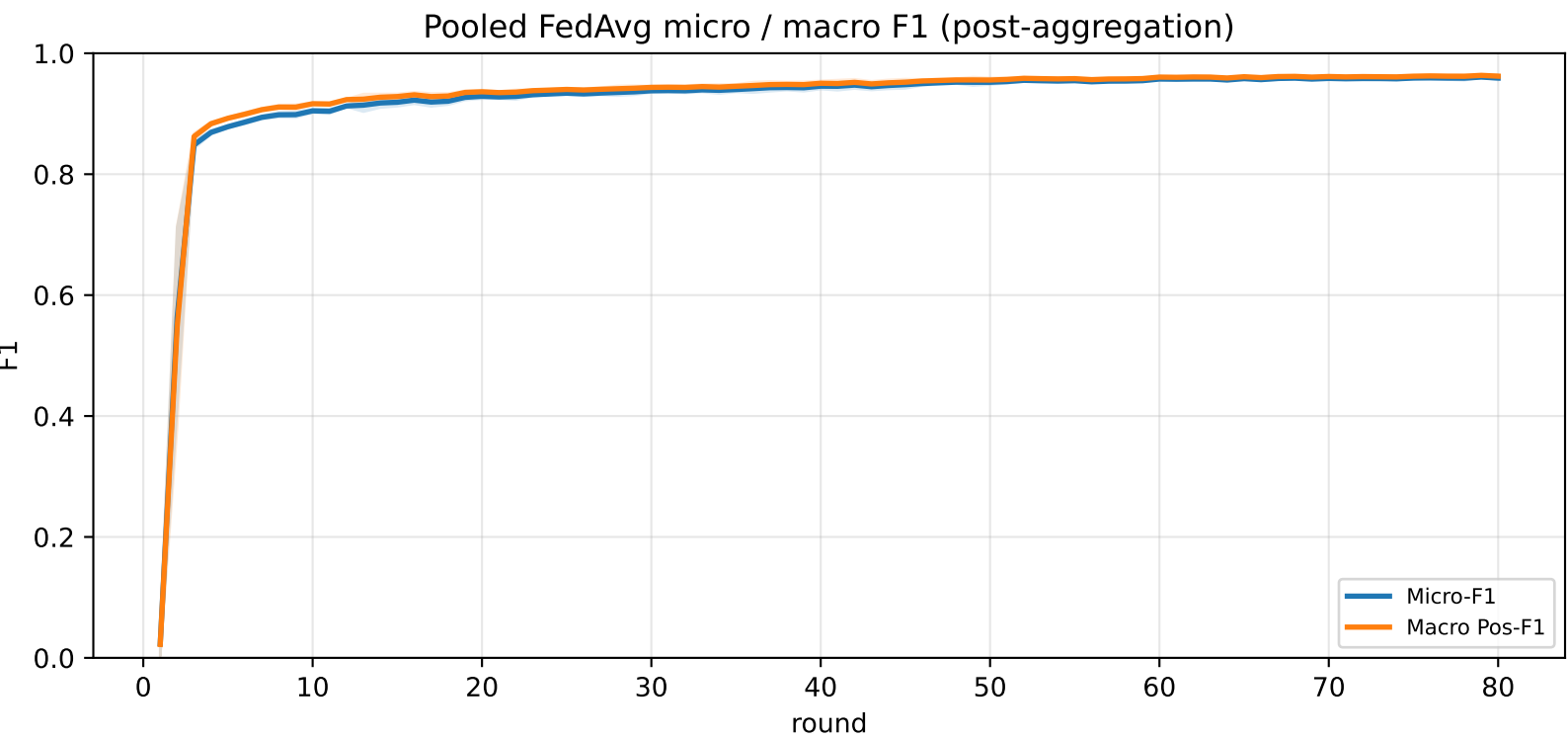
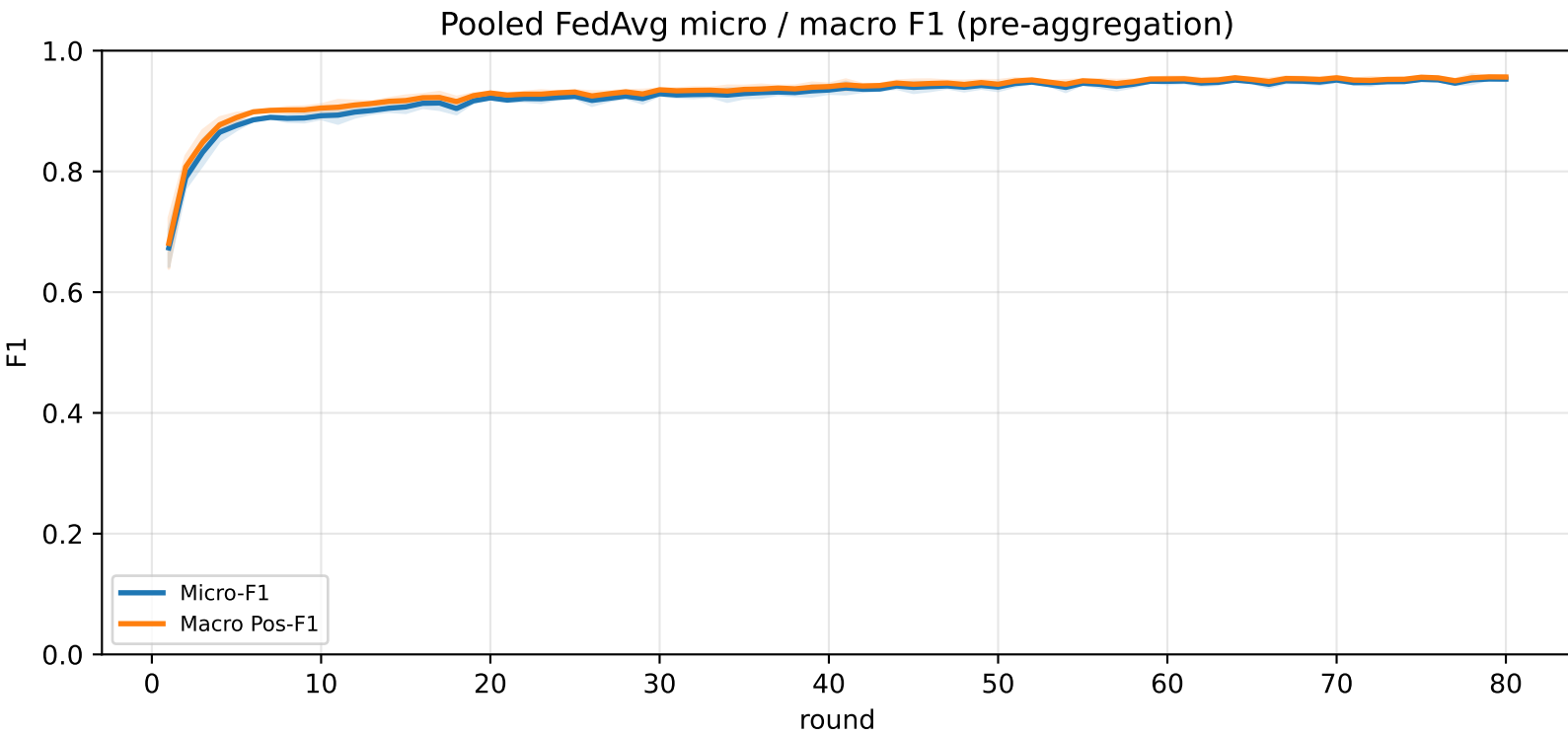
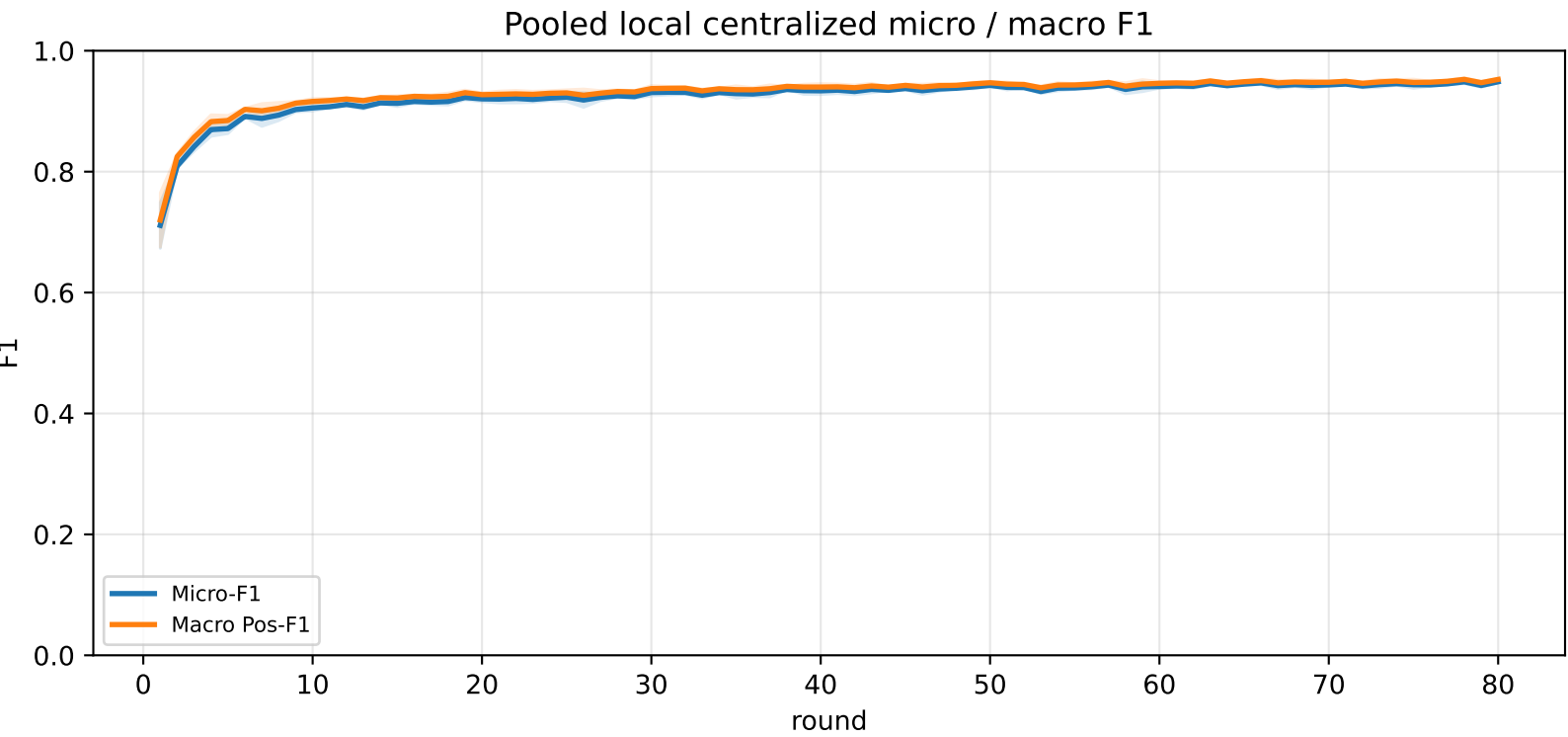
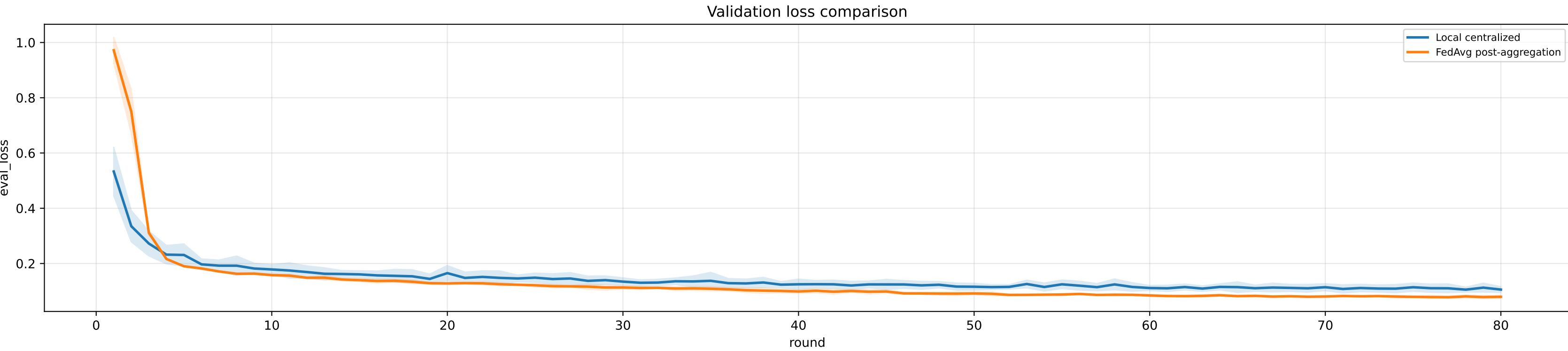
Best logged test performance (FedAvg vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	95.0% ± 0.1%	95.2% ± 0.1%	99.2% ± 0.1%	98.8% ± 0.2%	97.4% ± 0.2%	94.1% ± 0.2%	86.6% ± 0.4%
FedAvg	96.3% ± 0.1%	96.4% ± 0.1%	99.2% ± 0.1%	98.7% ± 0.1%	97.7% ± 0.0%	95.9% ± 0.0%	90.5% ± 0.4%



Heterogeneity level: Low | FedAvg diagnostics | family=mild_increasing | n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=mild_increasing | cycle2=22.3%±1.0% | cycle3=24.2%±0.4% | cycle4=25.3%±1.1% | cycle5=27.0%±0.5% | cycle6=27.3%±1.3%
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

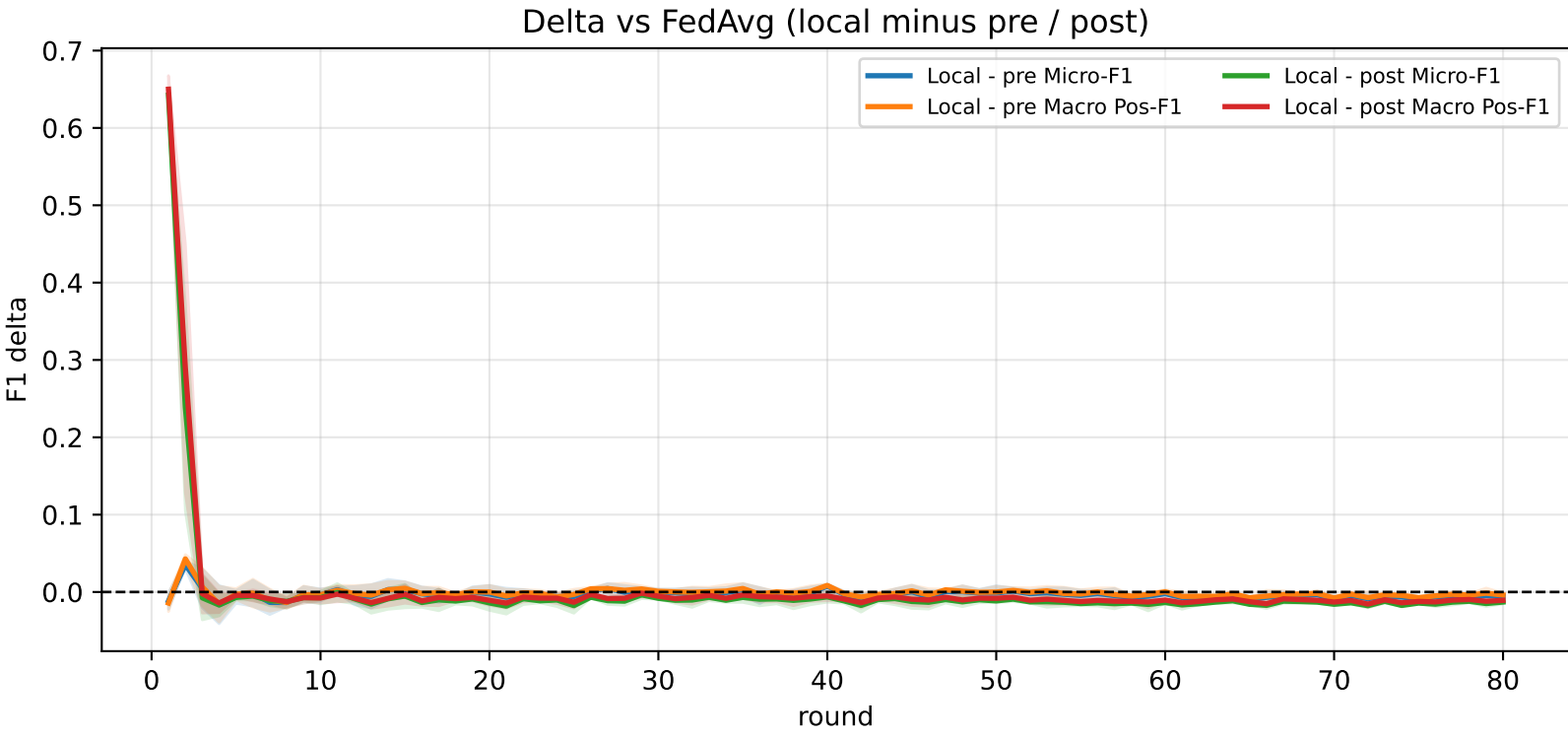
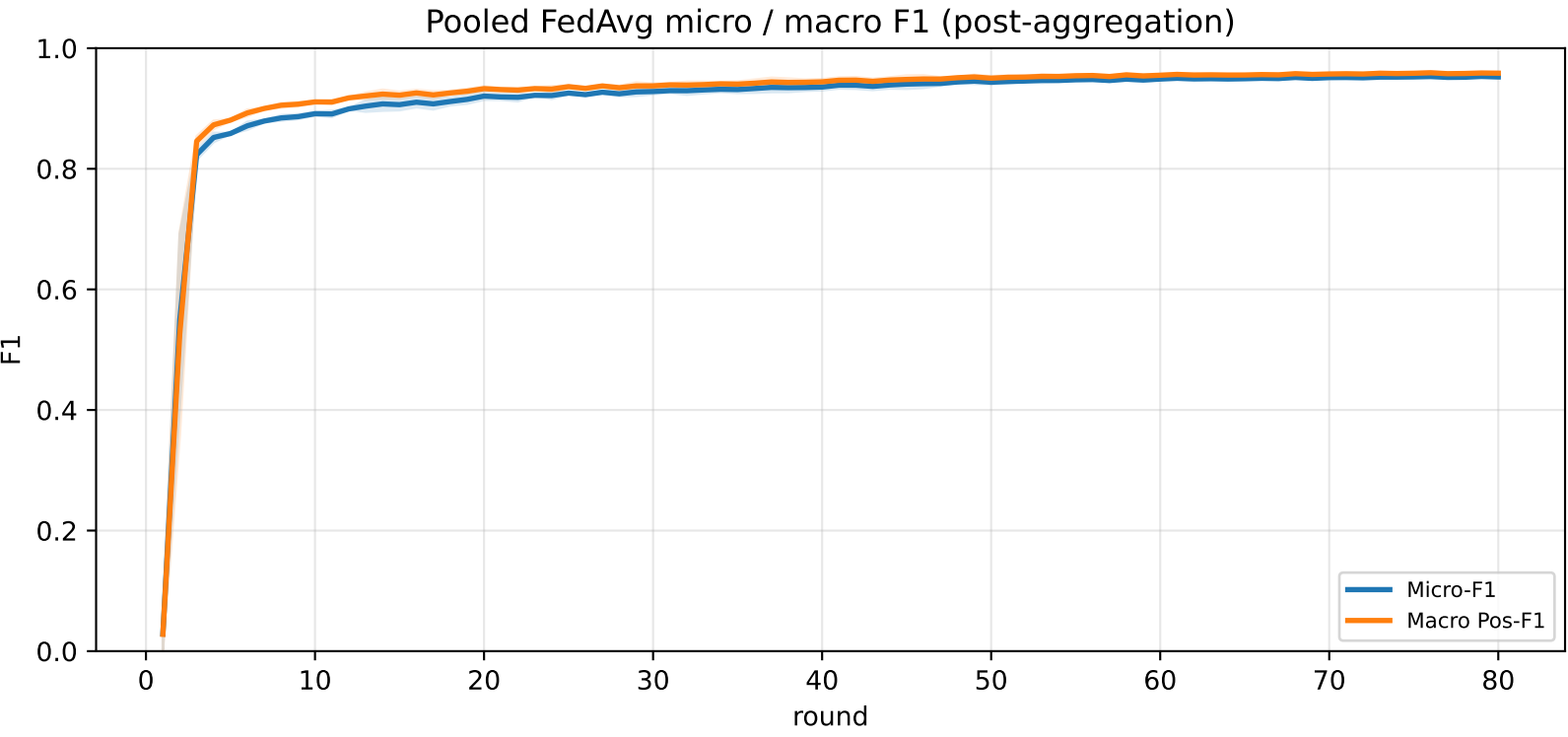
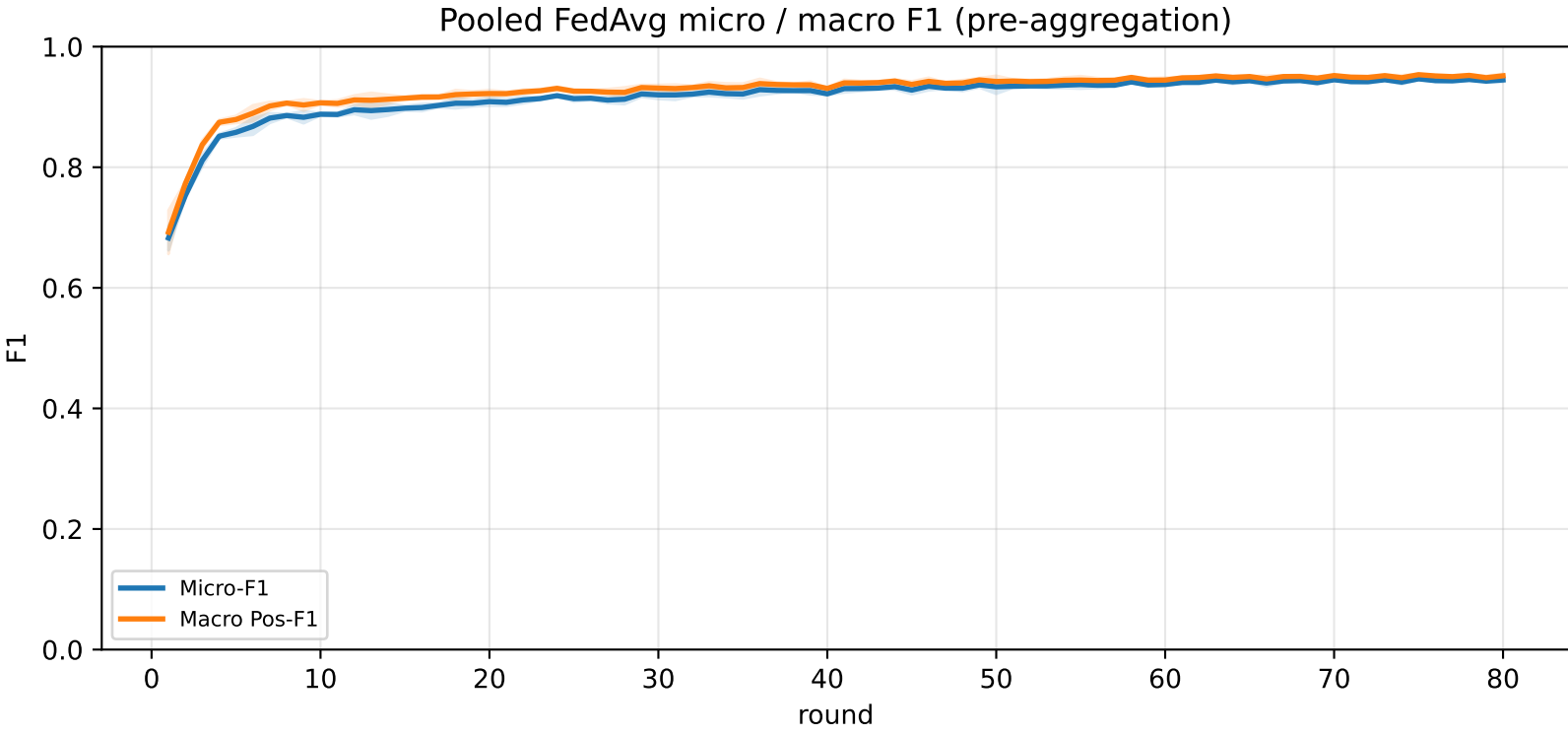
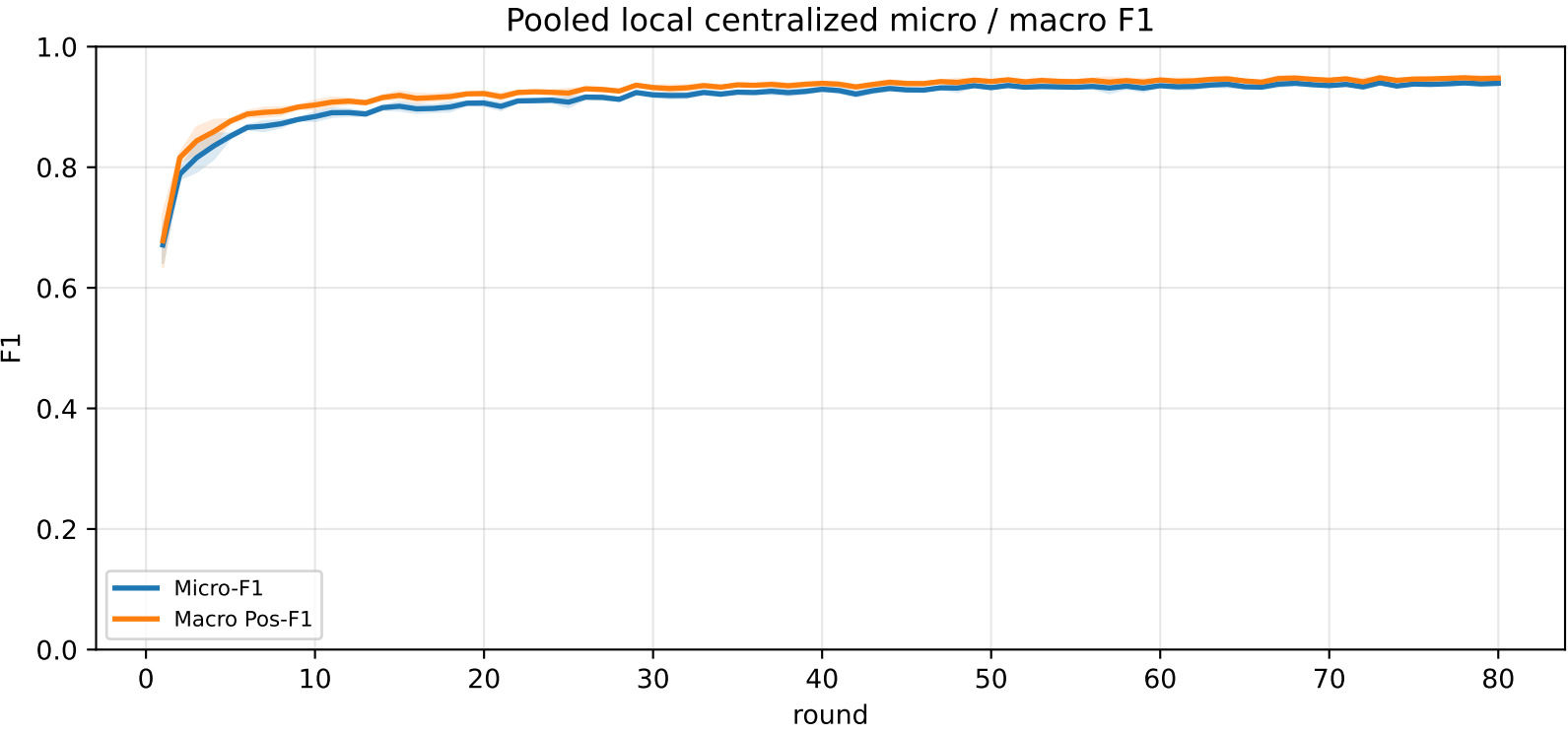
Best logged test performance (FedAvg vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.8% ± 0.3%	95.2% ± 0.3%	98.9% ± 0.1%	98.4% ± 0.2%	96.7% ± 0.3%	94.2% ± 0.2%	87.9% ± 0.9%
FedAvg	96.2% ± 0.1%	96.4% ± 0.1%	98.9% ± 0.2%	98.4% ± 0.1%	97.5% ± 0.1%	96.0% ± 0.1%	91.3% ± 0.1%



Heterogeneity level: Low | FedAvg diagnostics | family=strong_increasing | n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=strong_increasing | cycle2=15.0%±1.0% | cycle3=19.4%±0.7% | cycle4=22.9%±1.2% | cycle5=25.6%±1.2% | cycle6=27.6%±0.4%
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

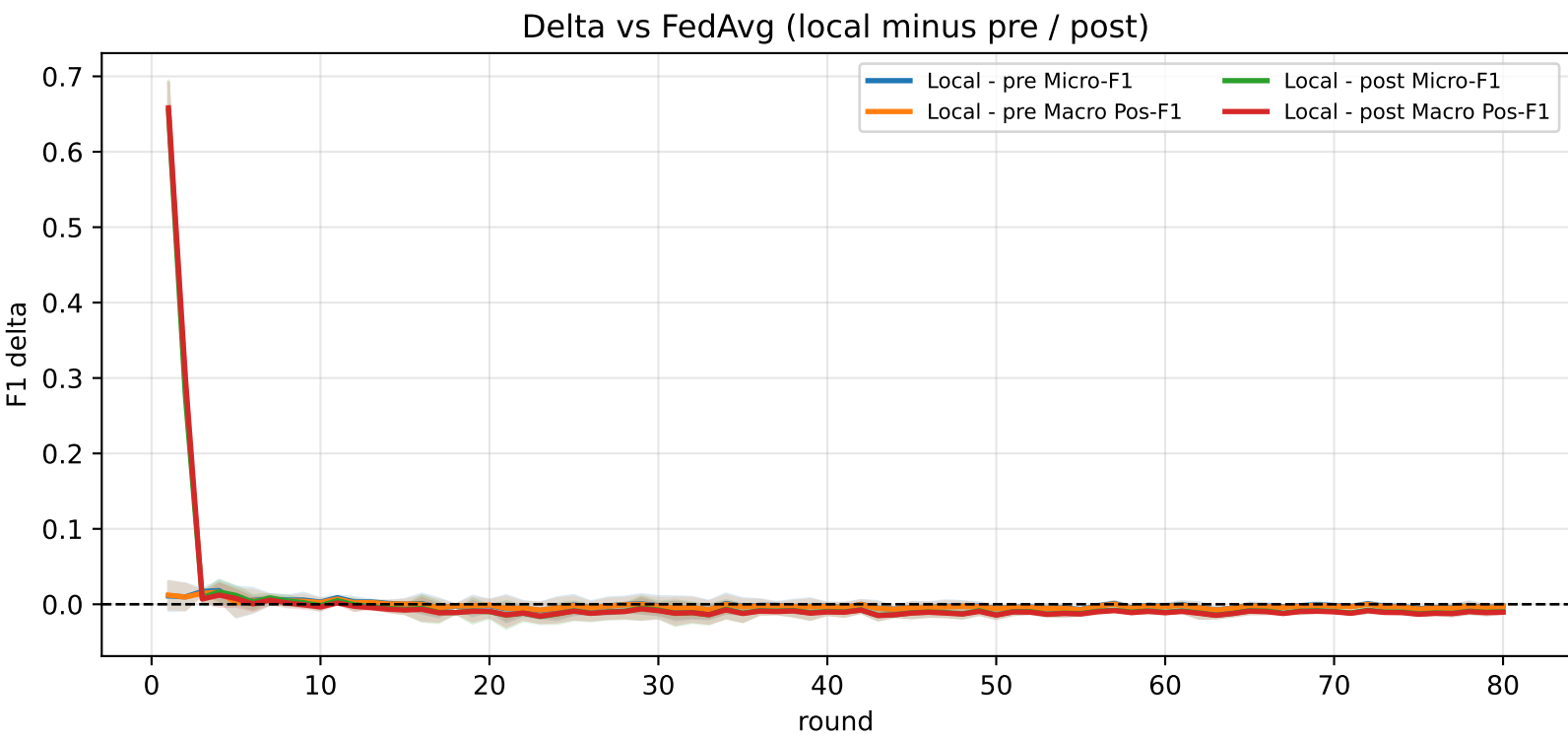
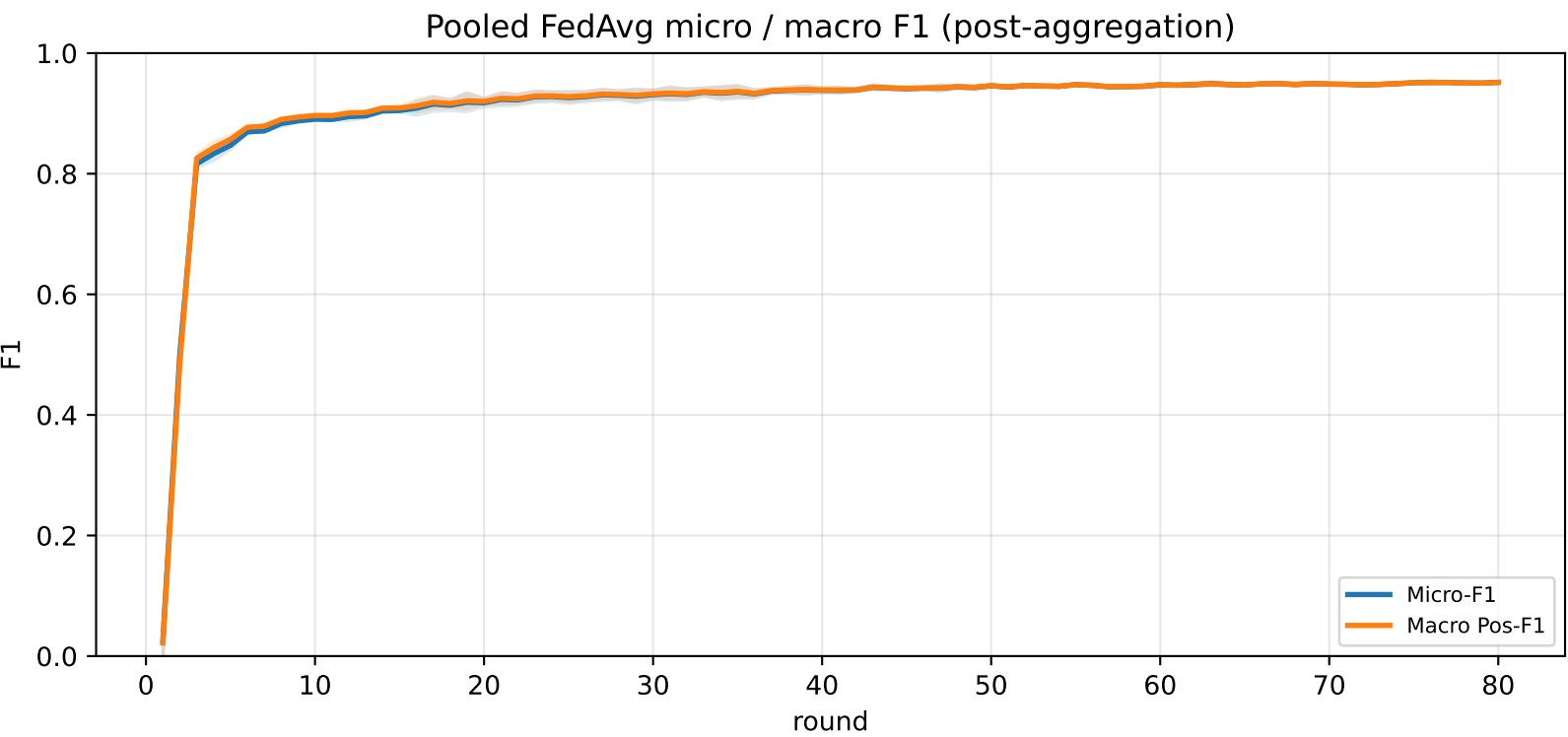
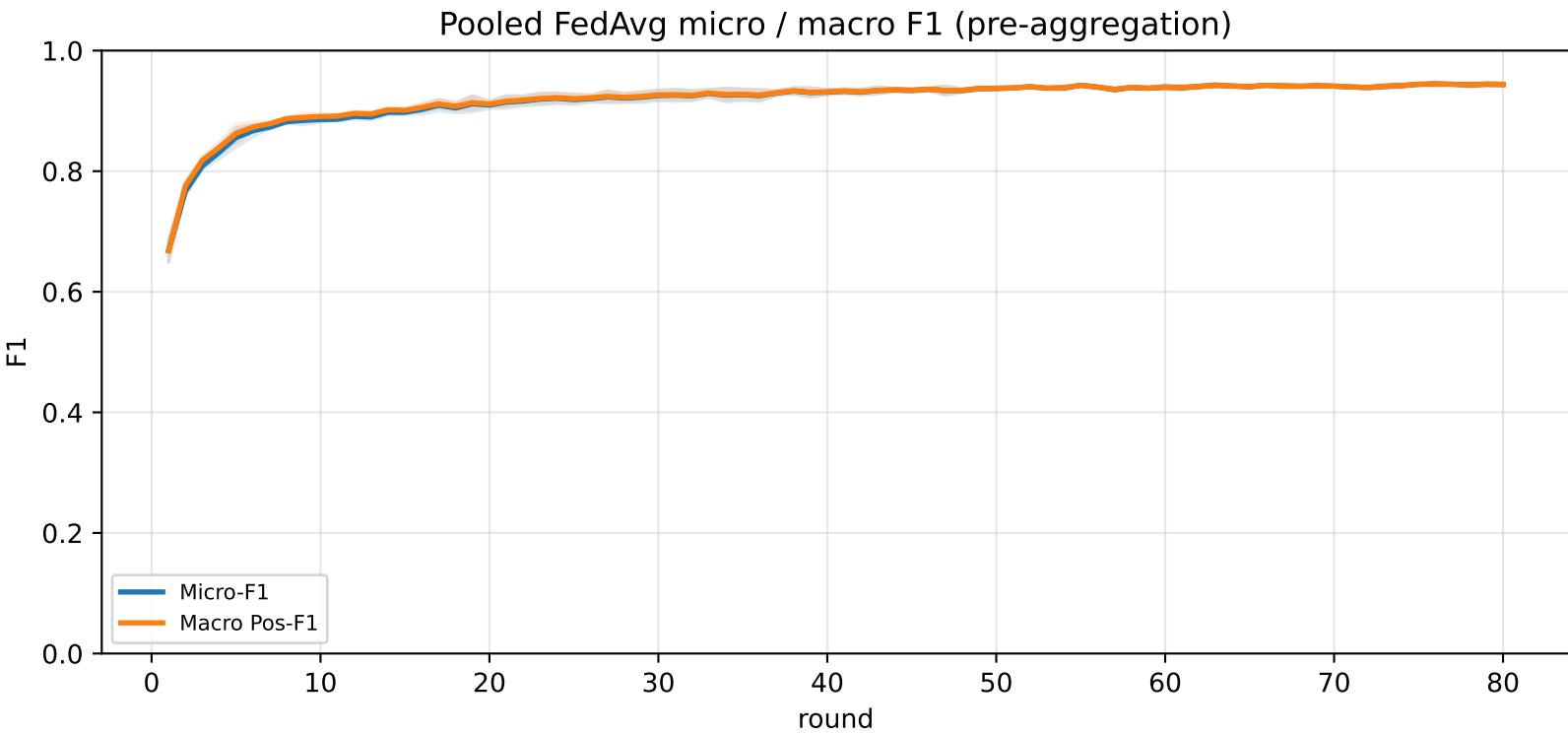
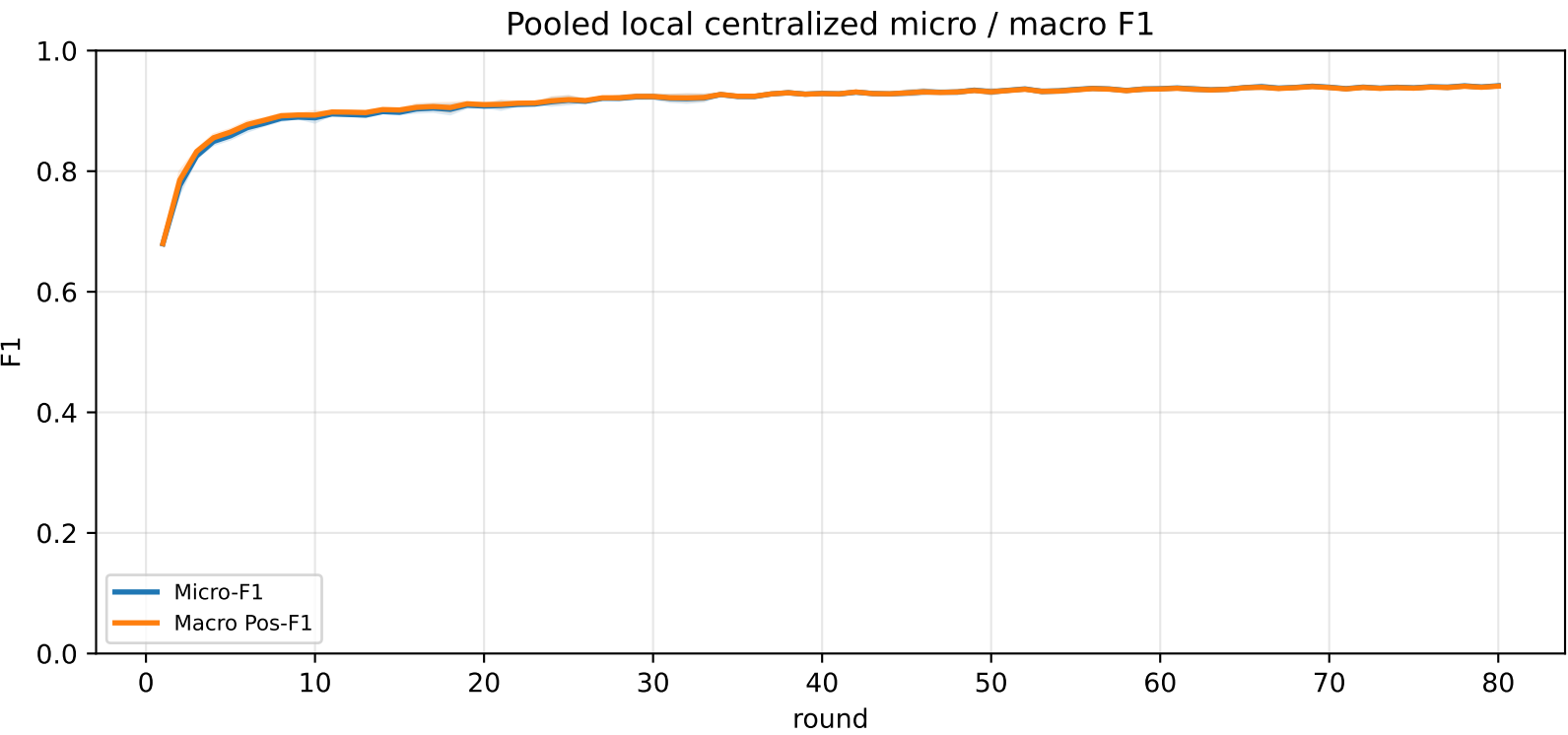
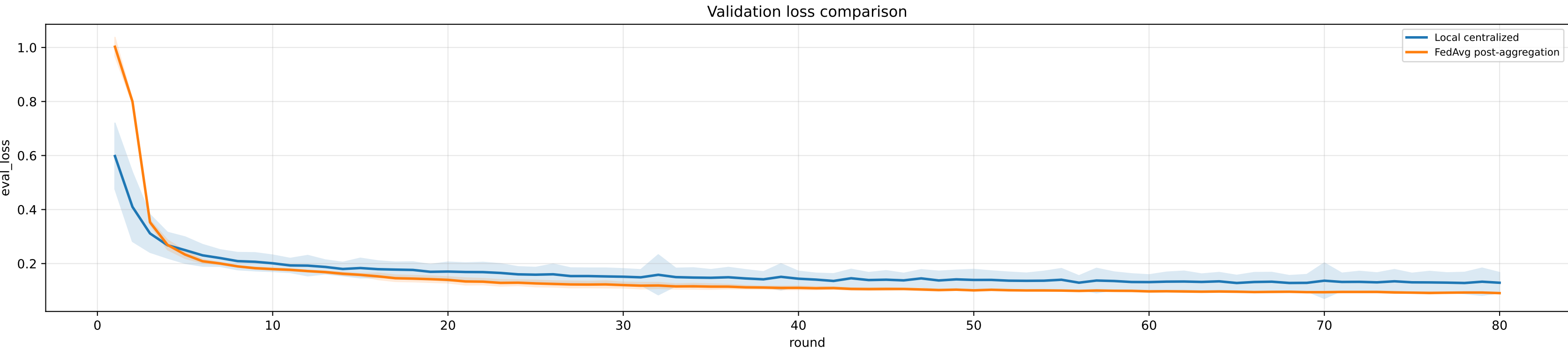
Best logged test performance (FedAvg vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	93.8% ± 0.2%	94.8% ± 0.2%	98.3% ± 0.3%	97.7% ± 0.2%	96.0% ± 0.1%	92.9% ± 0.3%	88.9% ± 0.5%
FedAvg	95.5% ± 0.1%	96.0% ± 0.1%	98.5% ± 0.2%	97.7% ± 0.3%	96.7% ± 0.1%	95.1% ± 0.4%	92.2% ± 0.1%



Heterogeneity level: Mid | FedAvg diagnostics | aggregated / pooled over 20 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

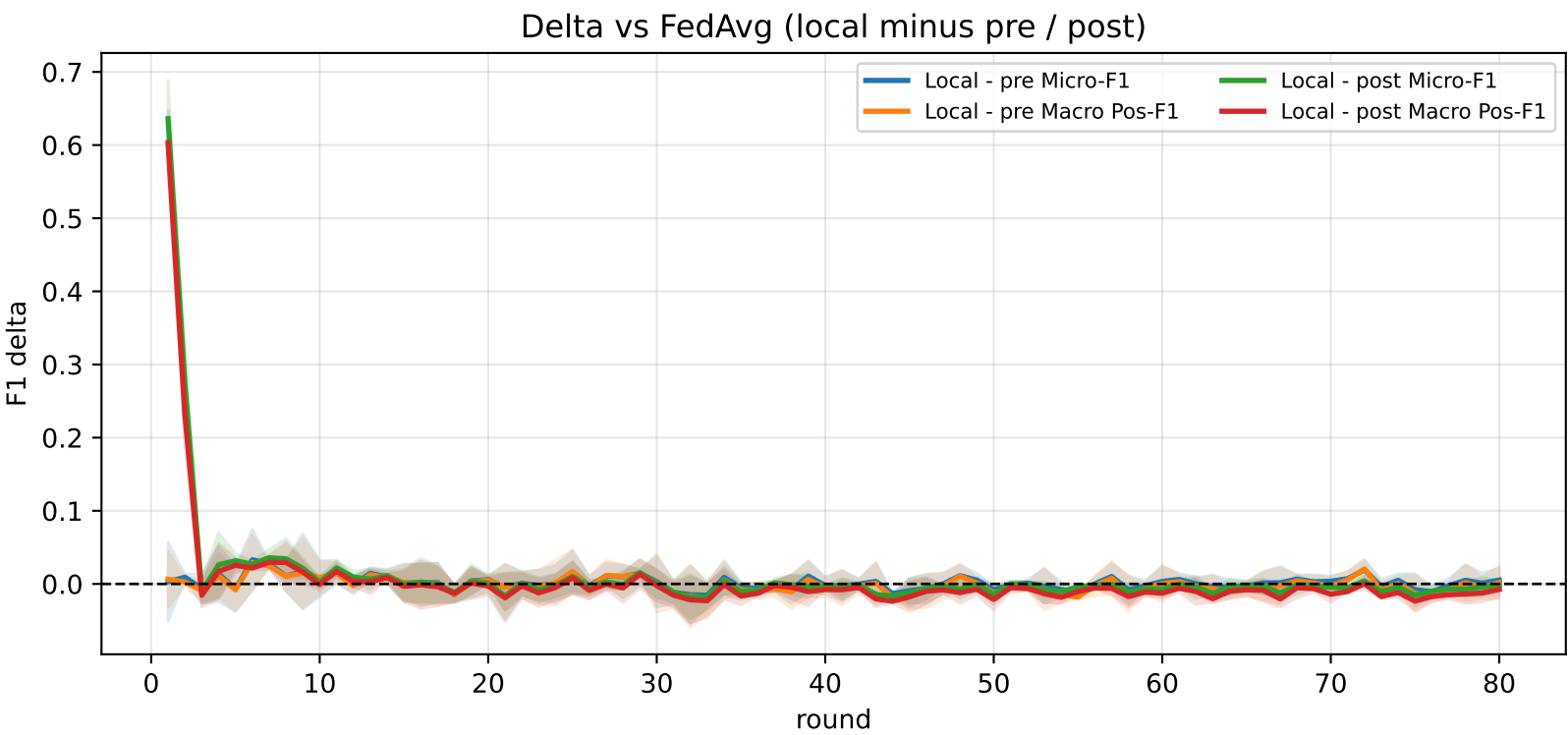
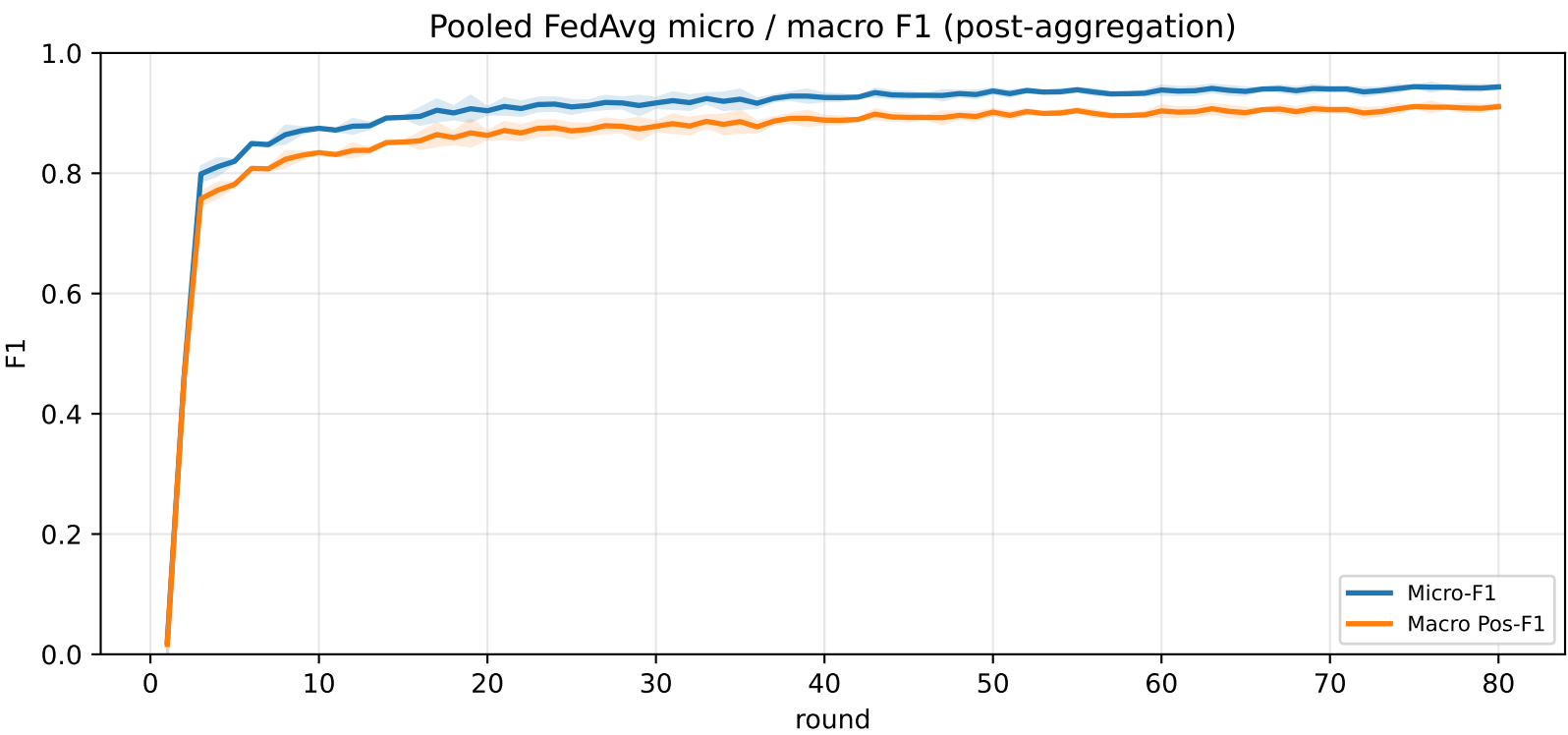
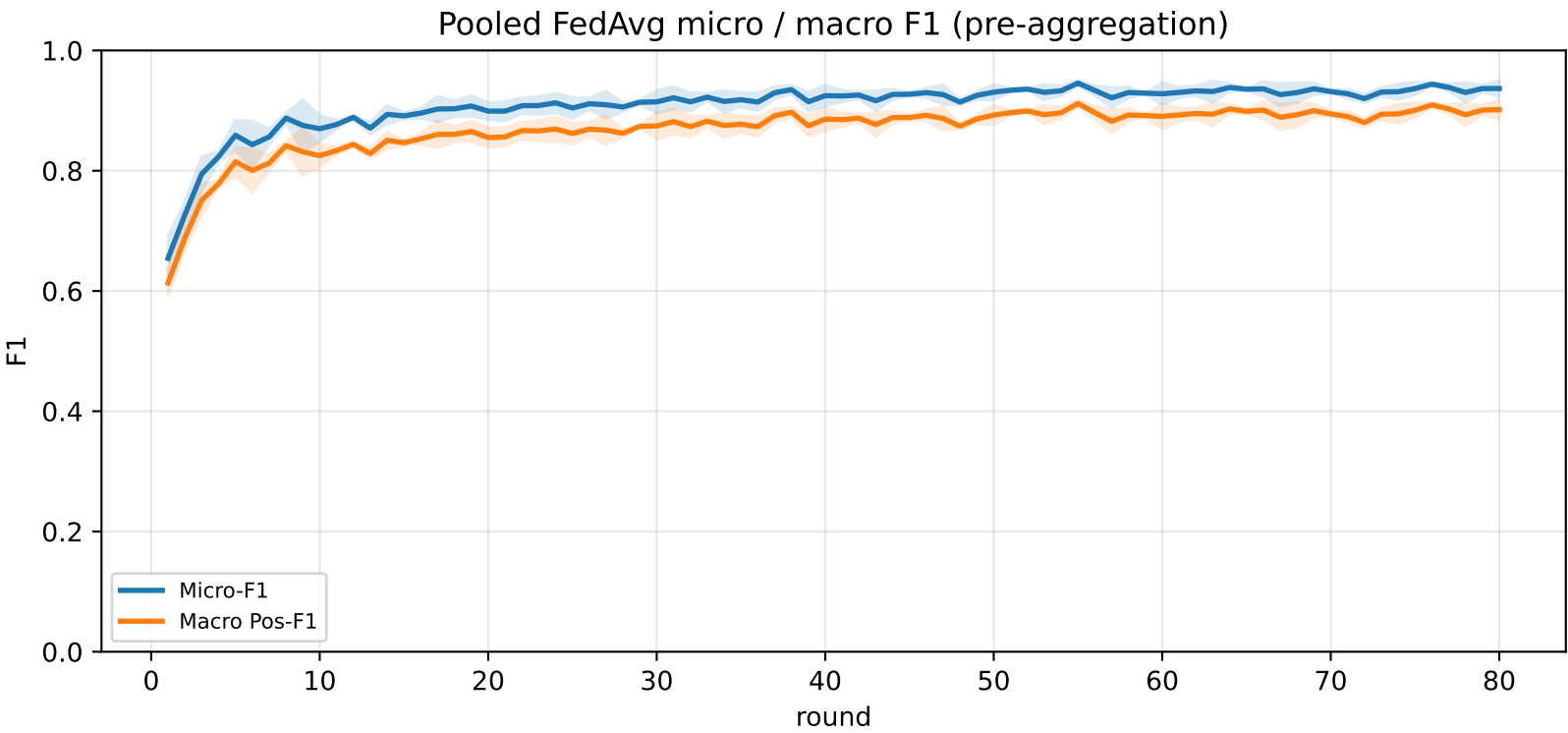
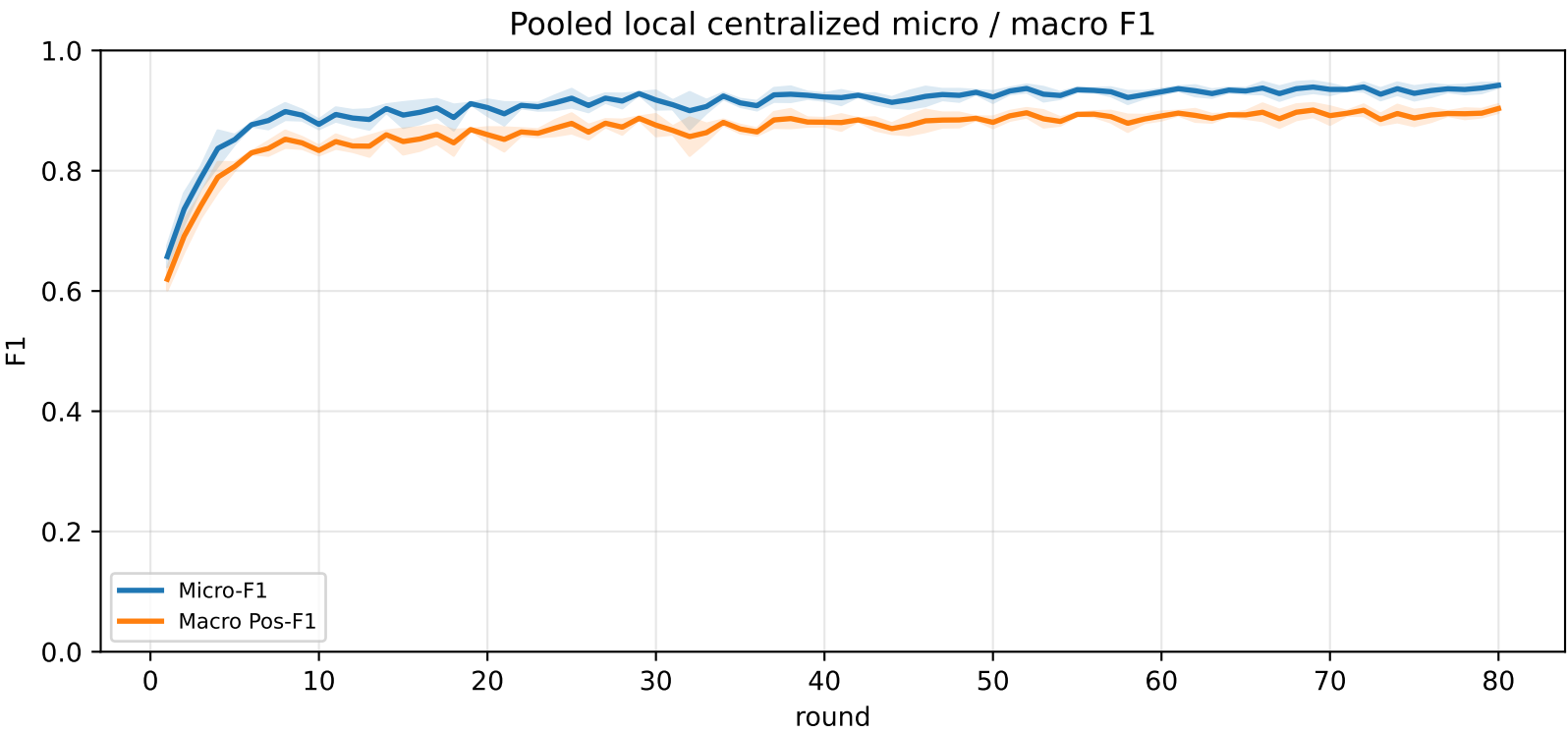
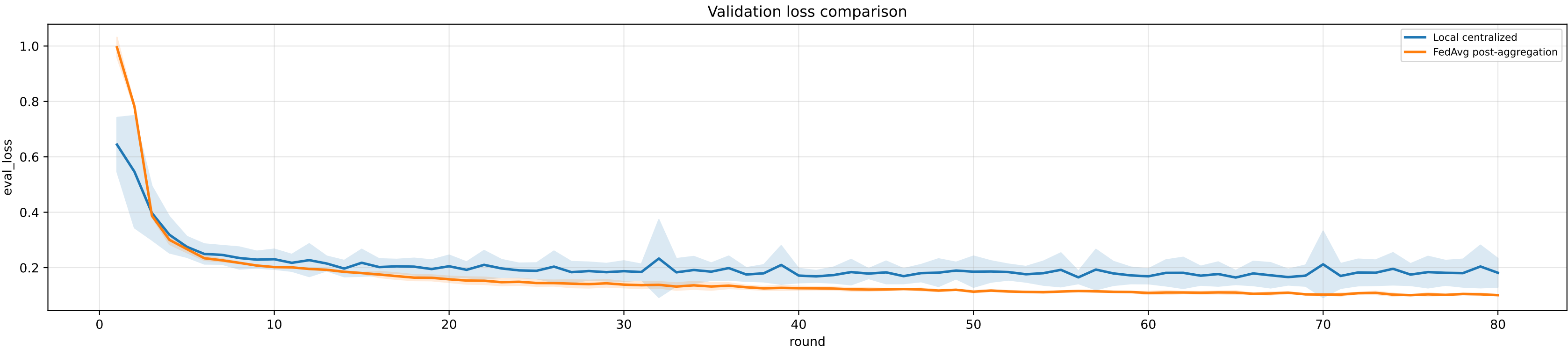
Best logged test performance (FedAvg vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.0% ± 0.1%	93.5% ± 0.1%	99.2% ± 0.1%	98.5% ± 0.0%	96.4% ± 0.0%	92.1% ± 0.4%	84.6% ± 0.3%
FedAvg	95.3% ± 0.0%	94.9% ± 0.0%	99.4% ± 0.0%	98.7% ± 0.0%	97.2% ± 0.1%	94.2% ± 0.2%	87.5% ± 0.2%



Heterogeneity level: Mid | FedAvg diagnostics | family=strong_decreasing | n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=strong_decreasing | cycle2=40.2%±4.0% | cycle3=32.9%±0.8% | cycle4=24.4%±1.8% | cycle5=16.8%±2.6% | cycle6=11.2%±3.3%
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

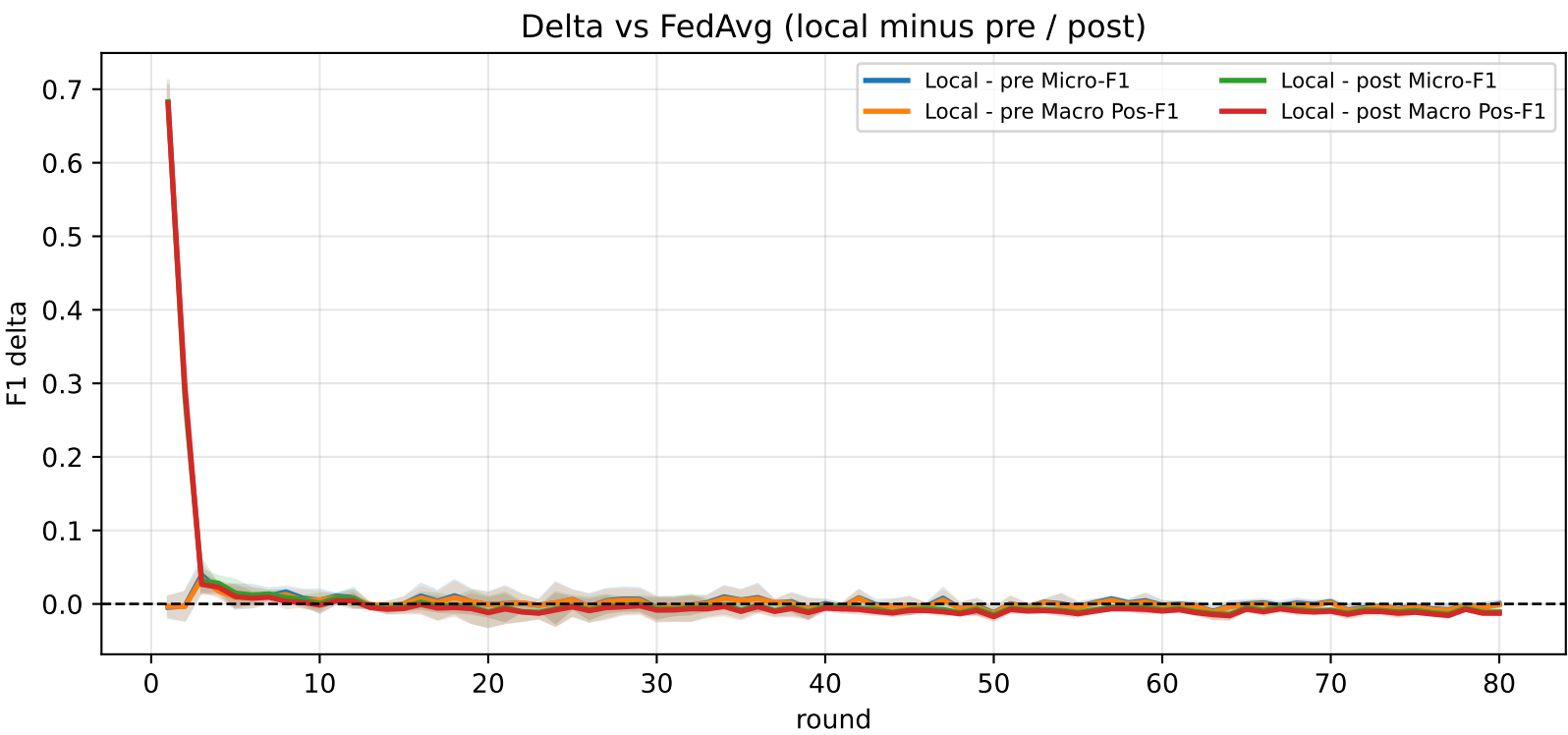
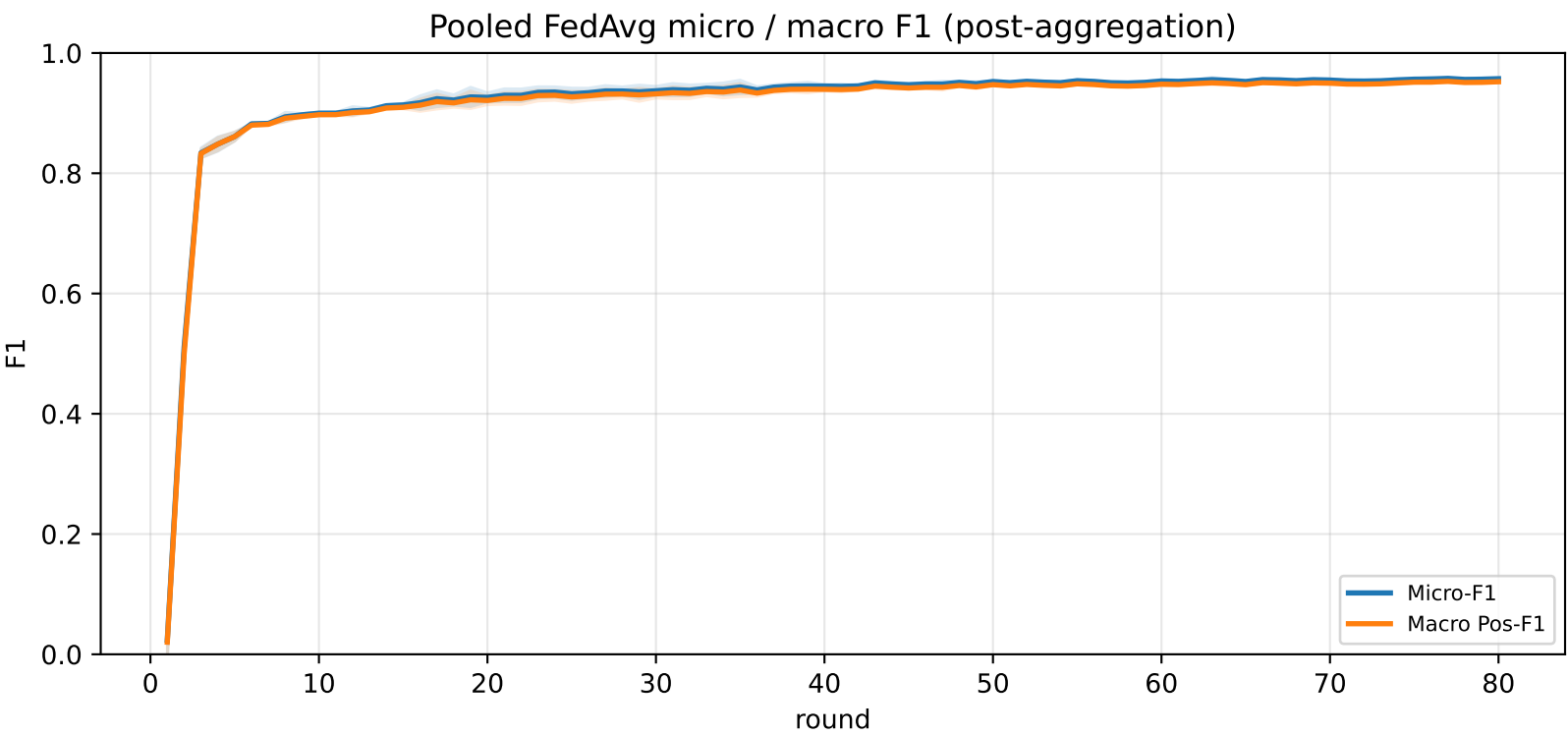
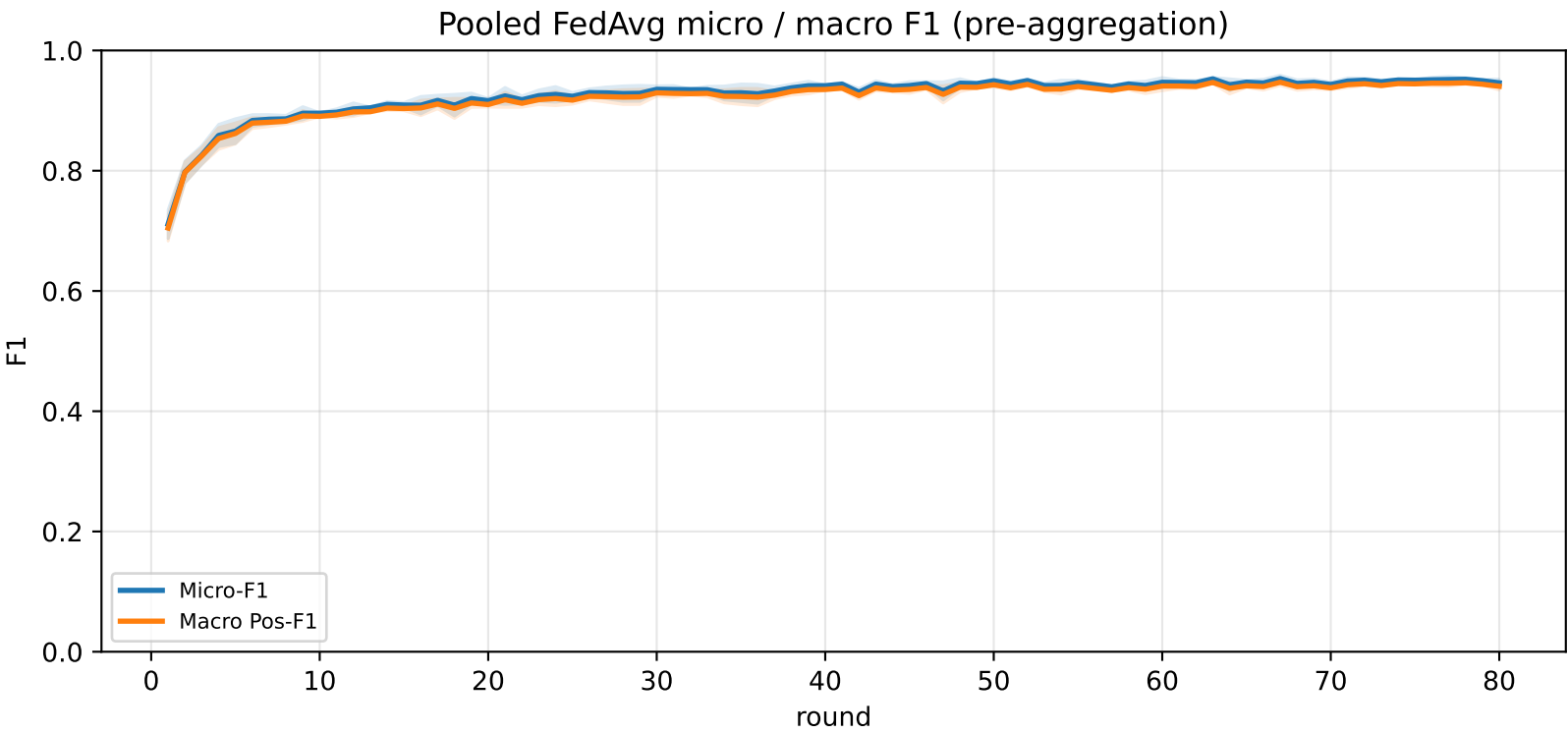
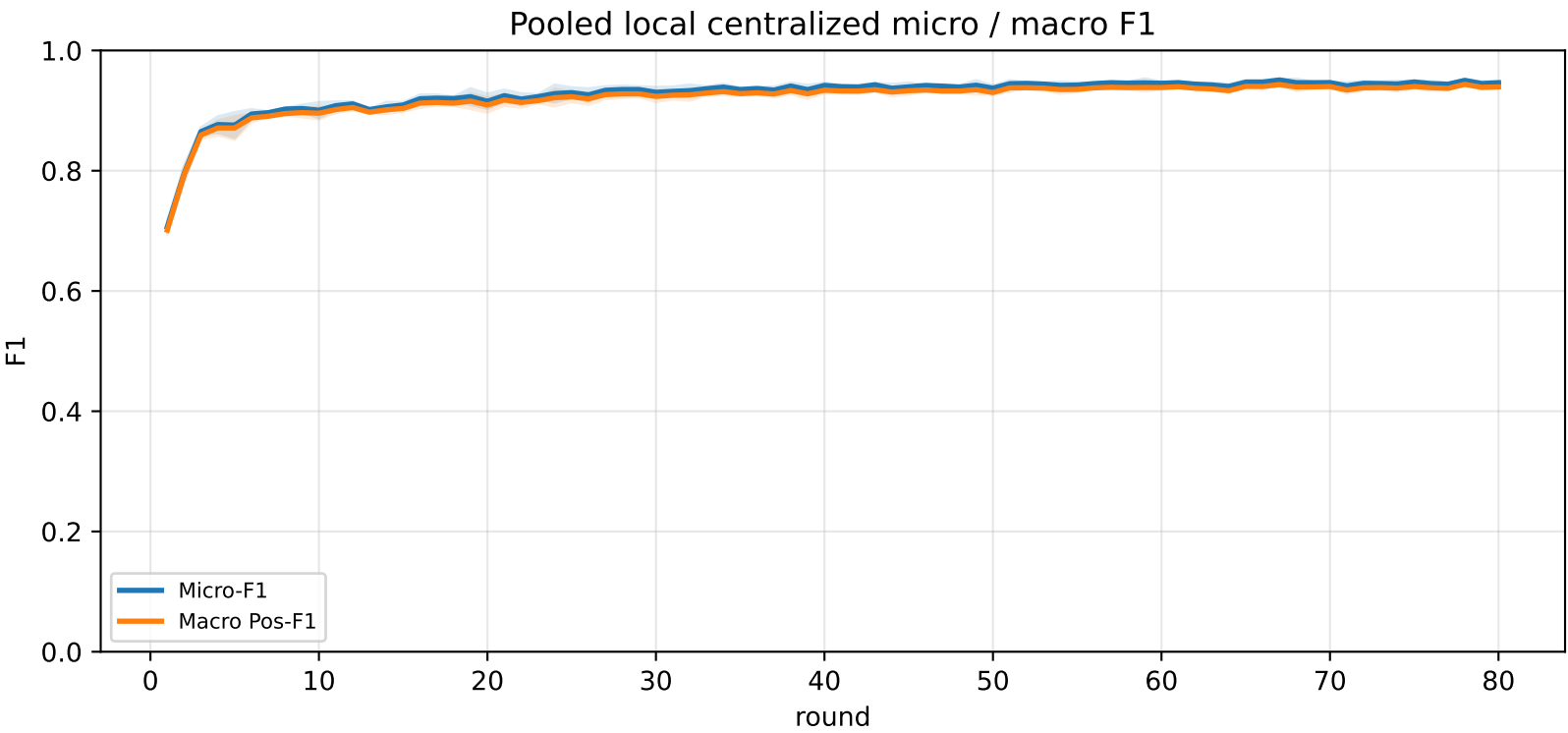
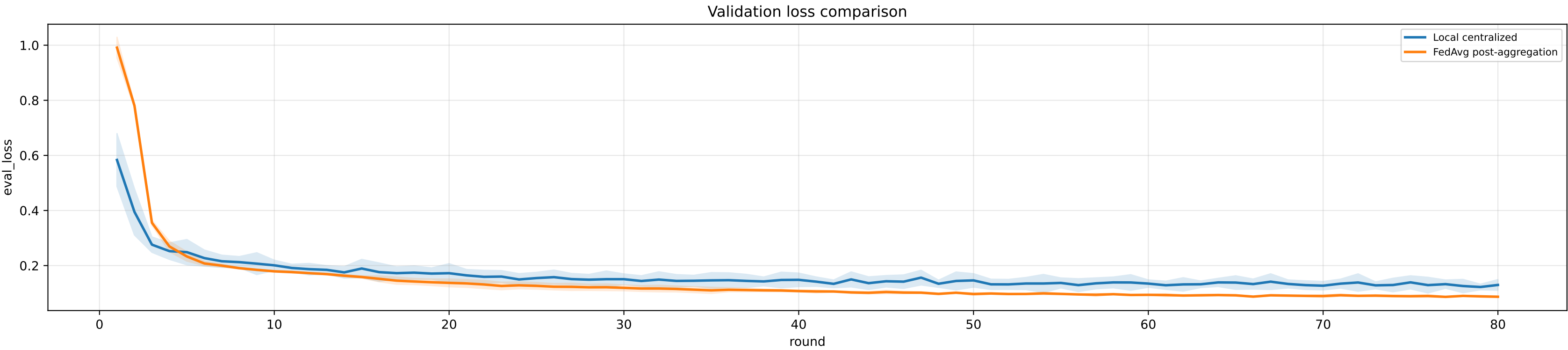
Best logged test performance (FedAvg vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	93.1% ± 0.5%	89.3% ± 0.6%	99.6% ± 0.1%	99.0% ± 0.0%	95.8% ± 0.5%	86.3% ± 0.9%	68.1% ± 2.2%
FedAvg	94.9% ± 0.1%	91.7% ± 0.2%	99.8% ± 0.0%	99.4% ± 0.0%	96.8% ± 0.3%	91.0% ± 0.5%	73.8% ± 0.7%



Heterogeneity level: Mid | FedAvg diagnostics | family=mild_decreasing | n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=mild_decreasing | cycle2=31.7%±0.3% | cycle3=30.3%±0.5% | cycle4=28.0%±0.8% | cycle5=25.5%±0.5% | cycle6=20.2%±1.0%
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

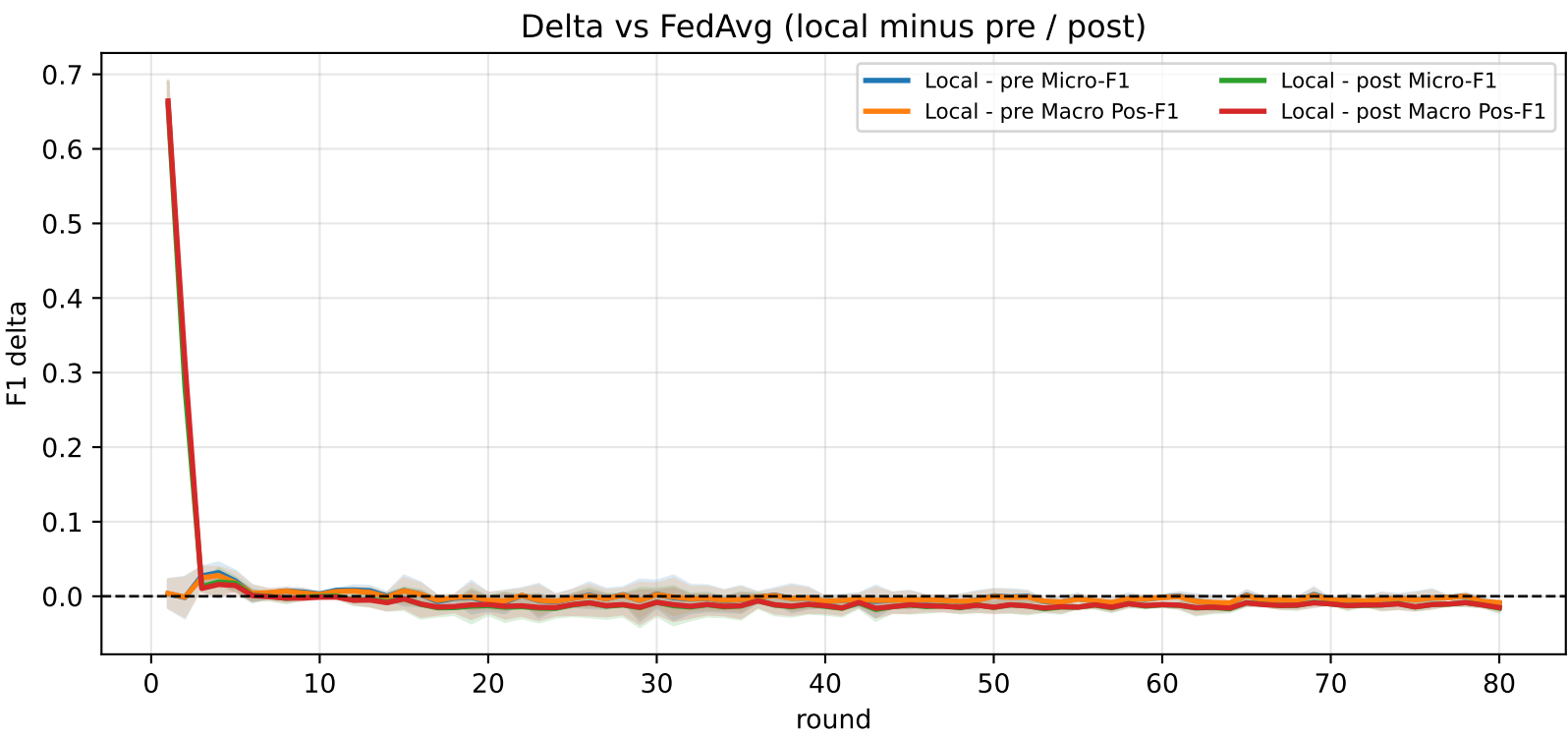
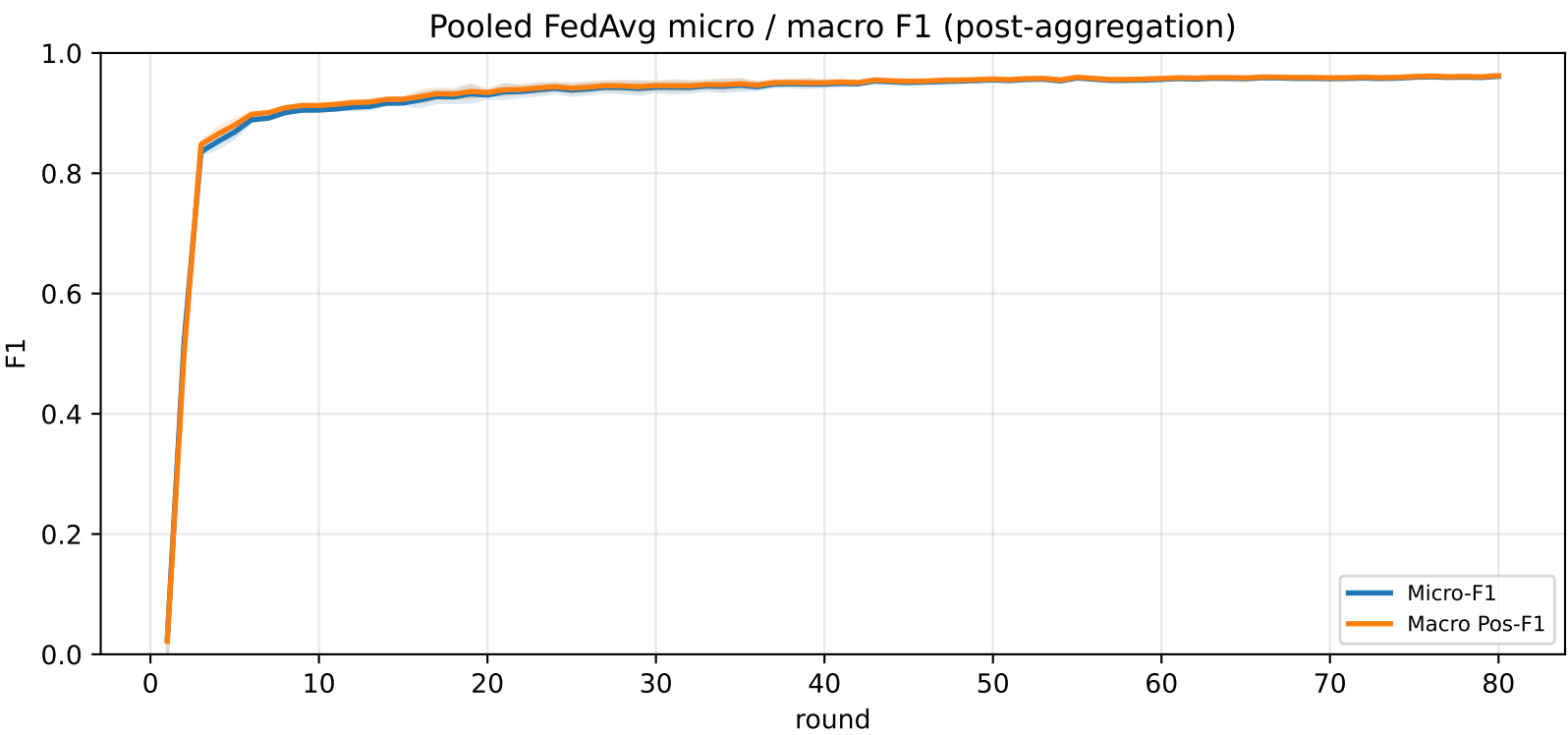
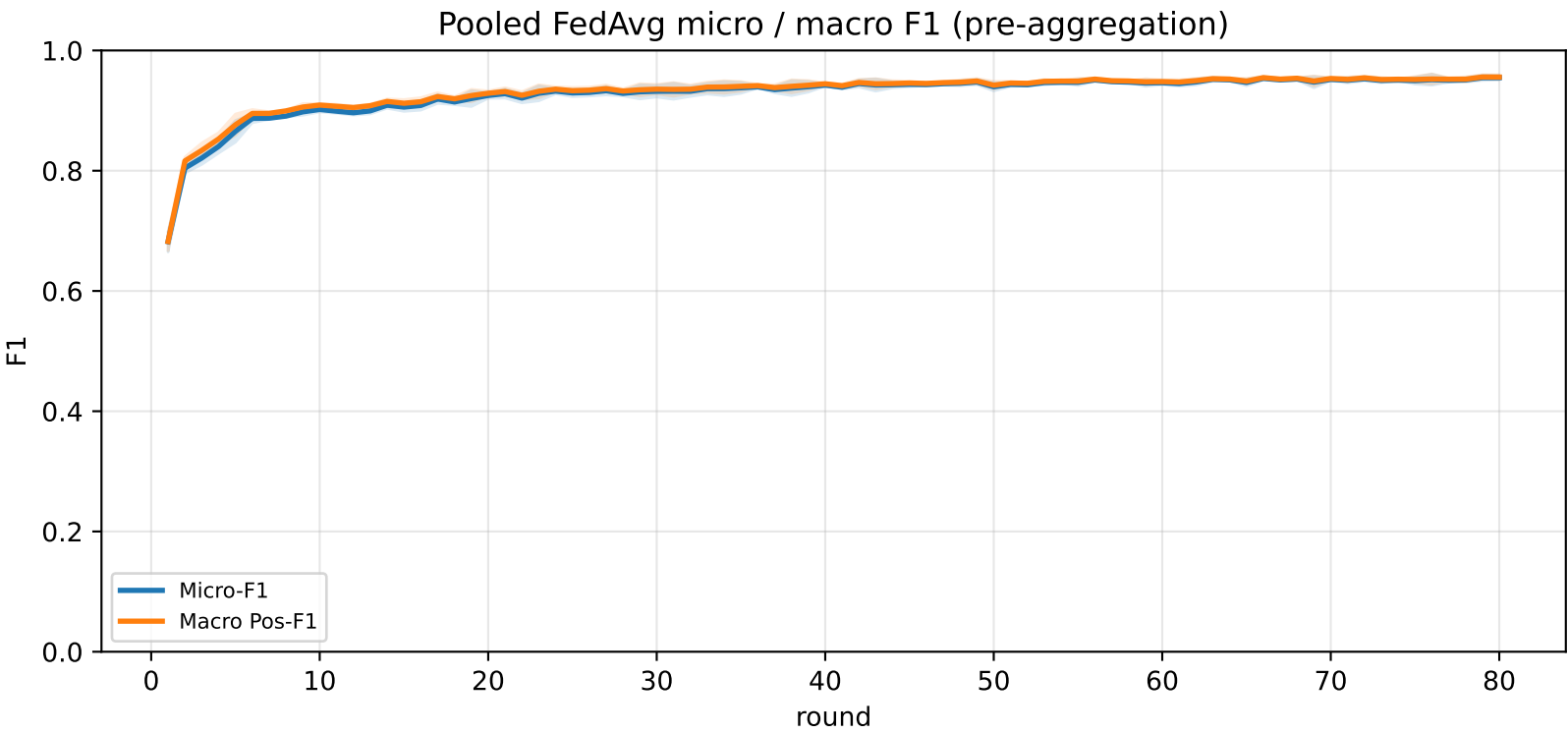
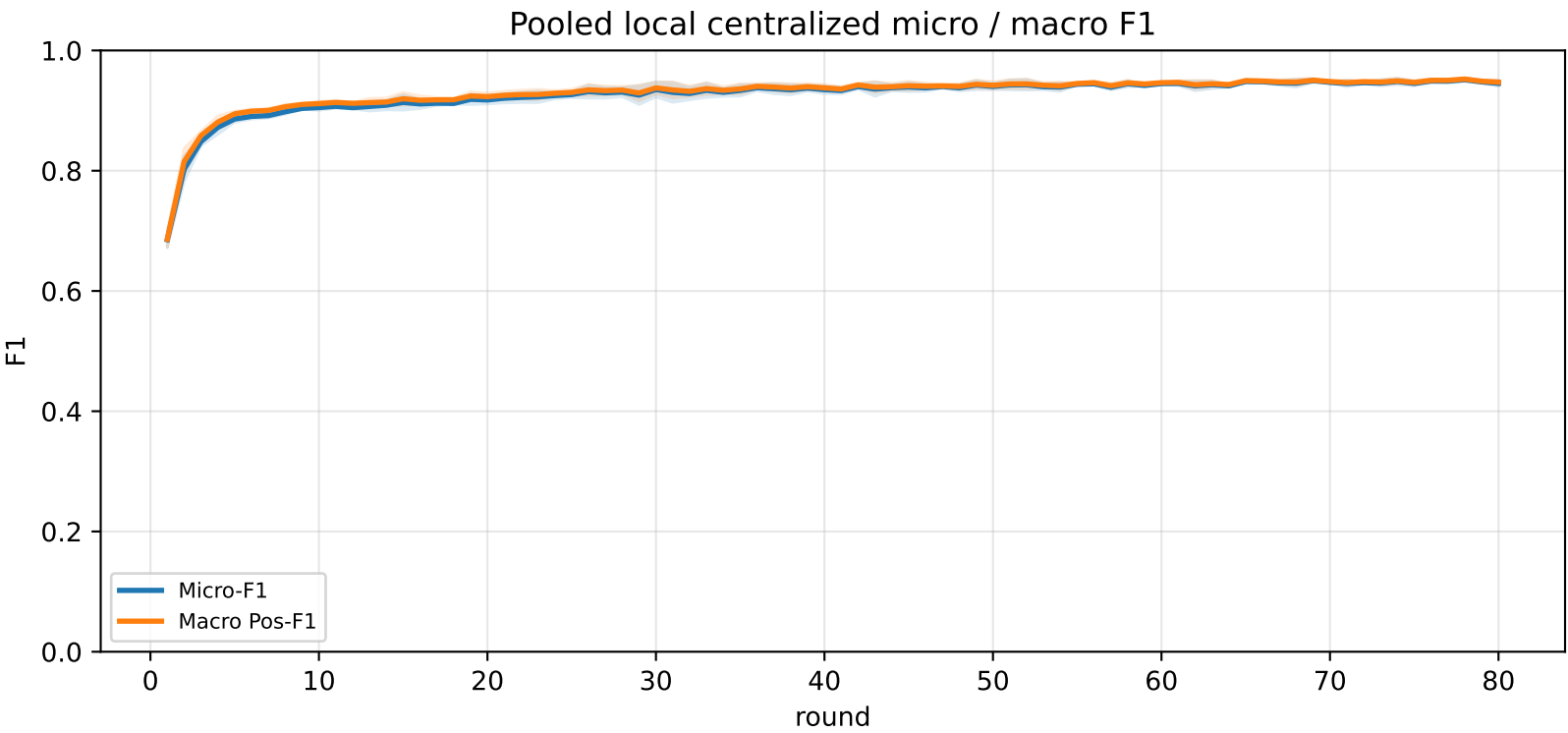
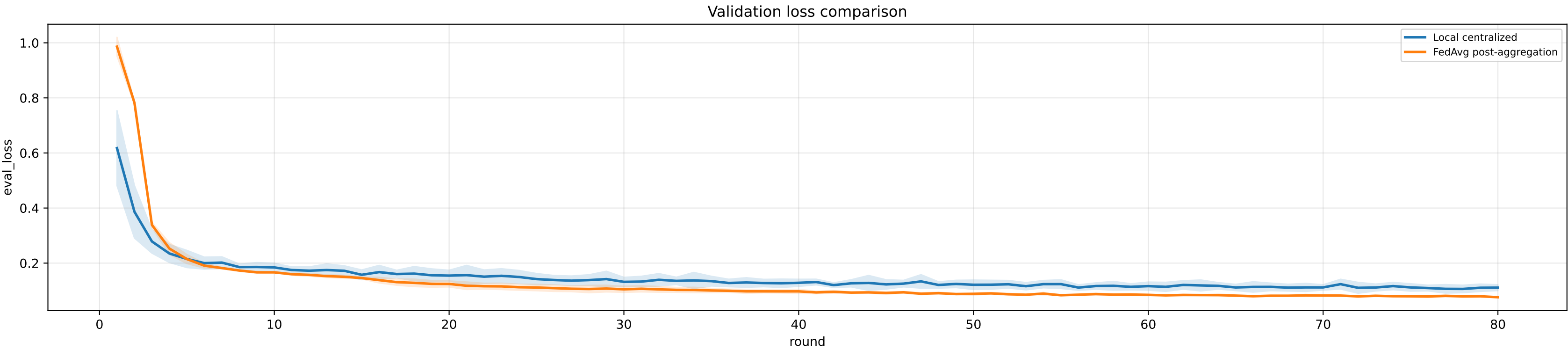
Best logged test performance (FedAvg vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.9% ± 0.2%	94.3% ± 0.2%	99.4% ± 0.0%	99.1% ± 0.0%	97.2% ± 0.1%	92.7% ± 0.7%	83.0% ± 0.3%
FedAvg	95.9% ± 0.1%	95.3% ± 0.1%	99.5% ± 0.0%	99.2% ± 0.0%	97.8% ± 0.1%	94.9% ± 0.4%	85.2% ± 0.6%



Heterogeneity level: Mid | FedAvg diagnostics | family=flat_balanced | n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=flat_balanced | cycle2=25.3%±0.6% | cycle3=26.4%±0.5% | cycle4=26.9%±0.4% | cycle5=26.6%±0.7% | cycle6=26.2%±0.9%
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

Best logged test performance (FedAvg vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	95.0% ± 0.1%	95.2% ± 0.1%	99.2% ± 0.1%	98.8% ± 0.2%	97.4% ± 0.2%	94.1% ± 0.2%	86.6% ± 0.4%
FedAvg	96.2% ± 0.1%	96.4% ± 0.1%	99.3% ± 0.0%	98.8% ± 0.2%	97.9% ± 0.1%	95.8% ± 0.3%	90.1% ± 0.3%



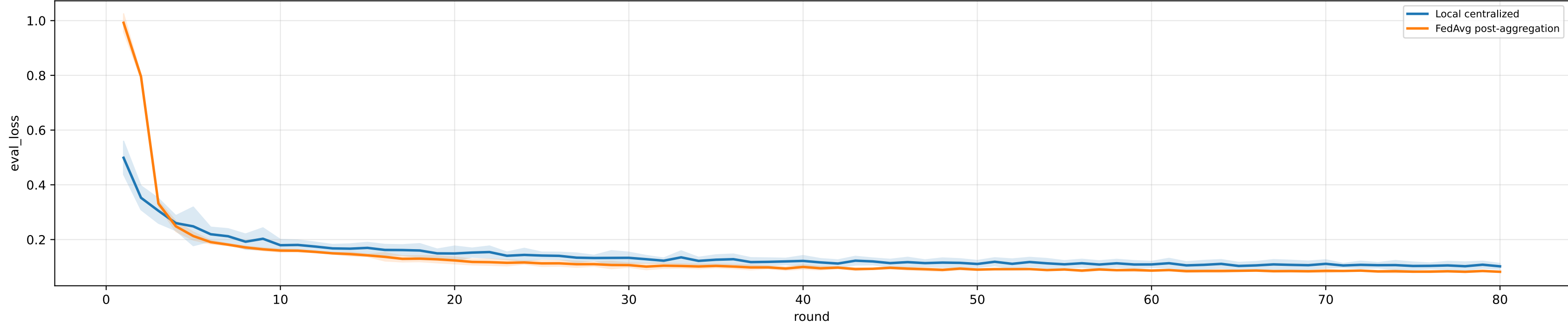
Heterogeneity level: Mid | FedAvg diagnostics | family=mild_increasing |
n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=mild_increasing | cycle2=18.6%±0.8% | cycle3=22.7%±1.2% | cycle4=25.7%±0.5% | cycle5=27.9%±0.9% | cycle6=28.0%±0.4%
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

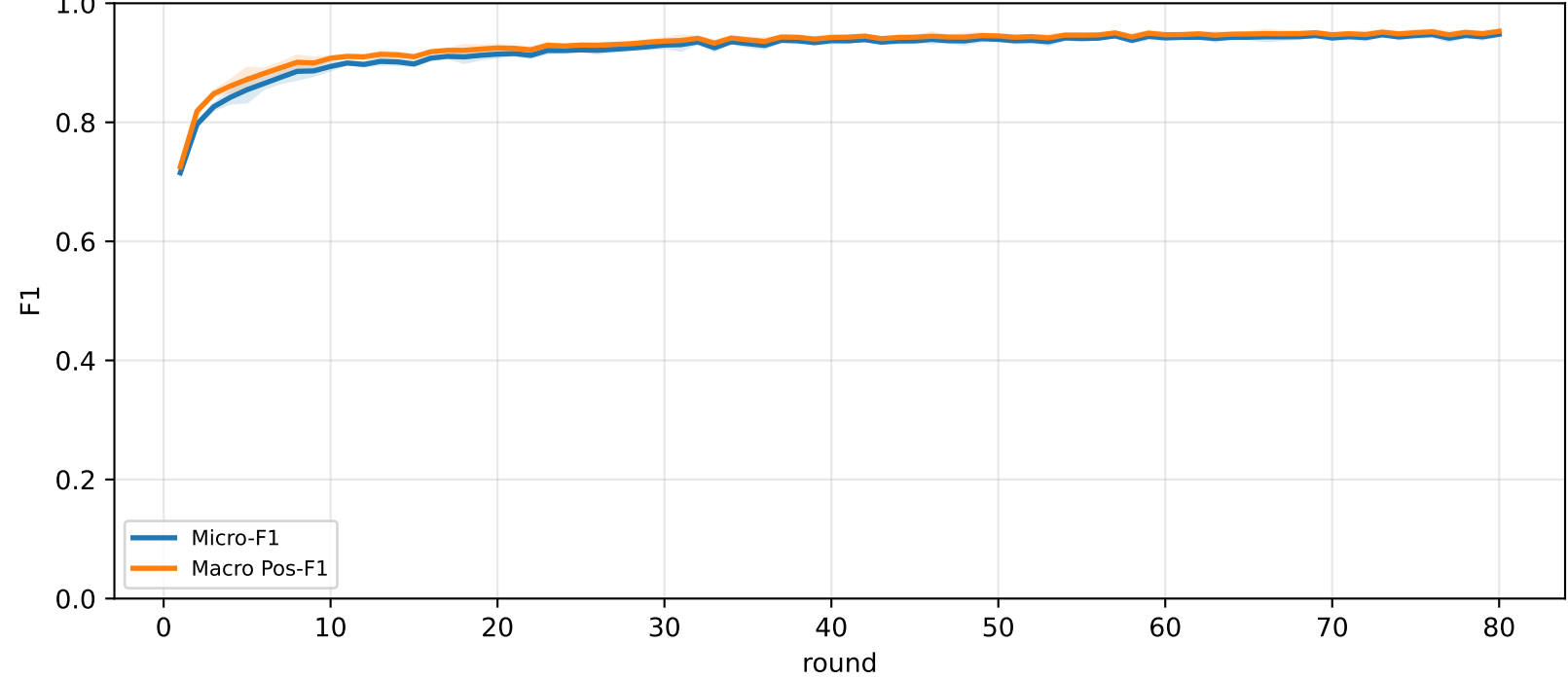
Best logged test performance (FedAvg vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.9% ± 0.2%	95.4% ± 0.2%	98.9% ± 0.1%	98.2% ± 0.2%	96.8% ± 0.3%	94.1% ± 0.5%	89.2% ± 0.3%
FedAvg	95.9% ± 0.0%	96.3% ± 0.0%	99.0% ± 0.1%	98.3% ± 0.1%	97.5% ± 0.2%	95.1% ± 0.2%	91.6% ± 0.1%

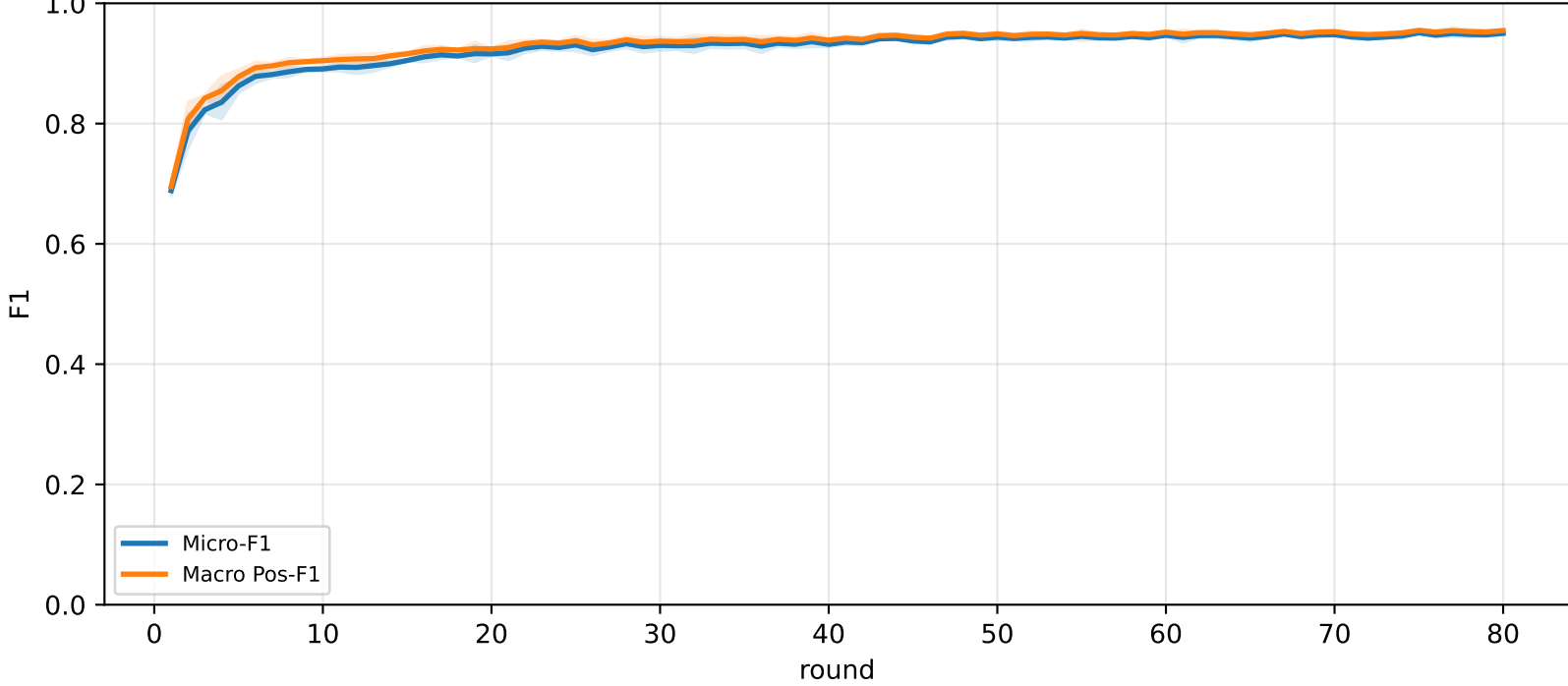
Validation loss comparison



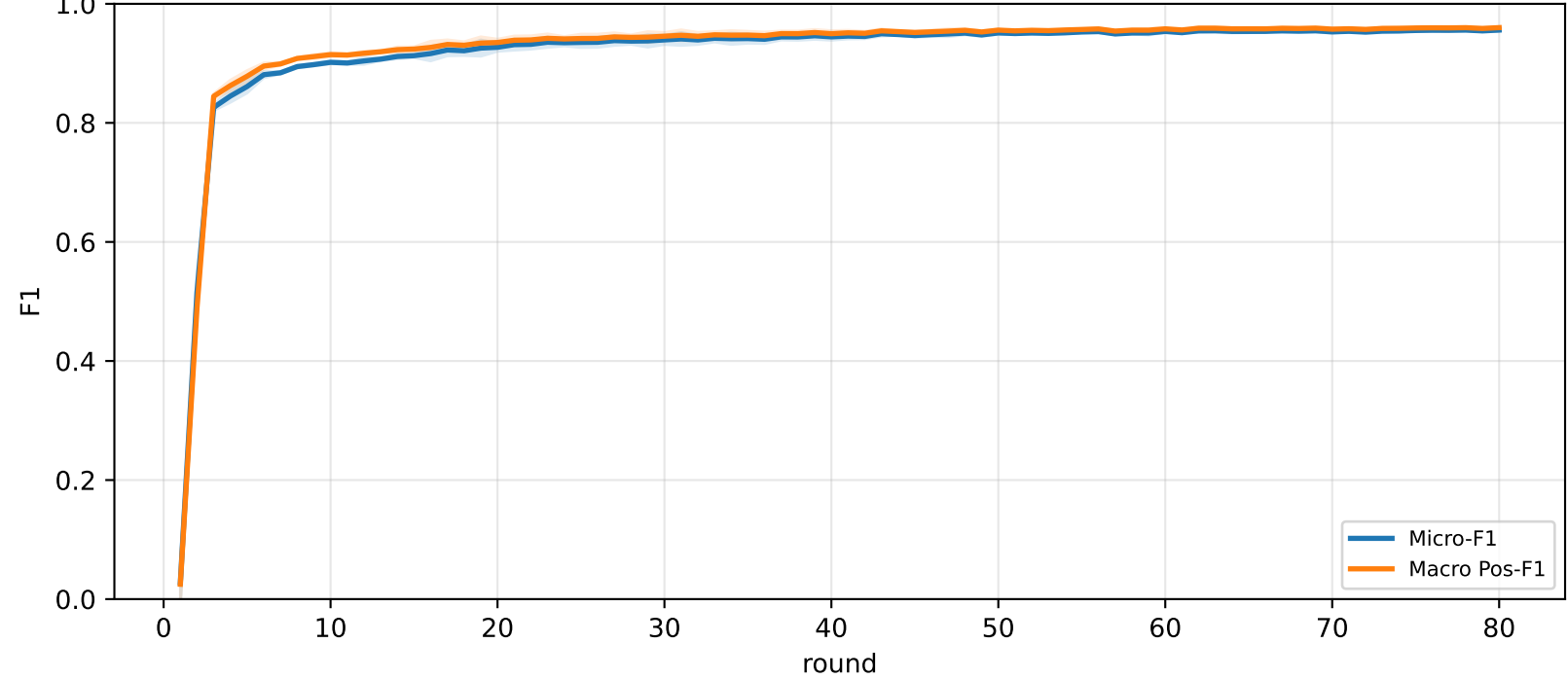
Pooled local centralized micro / macro F1



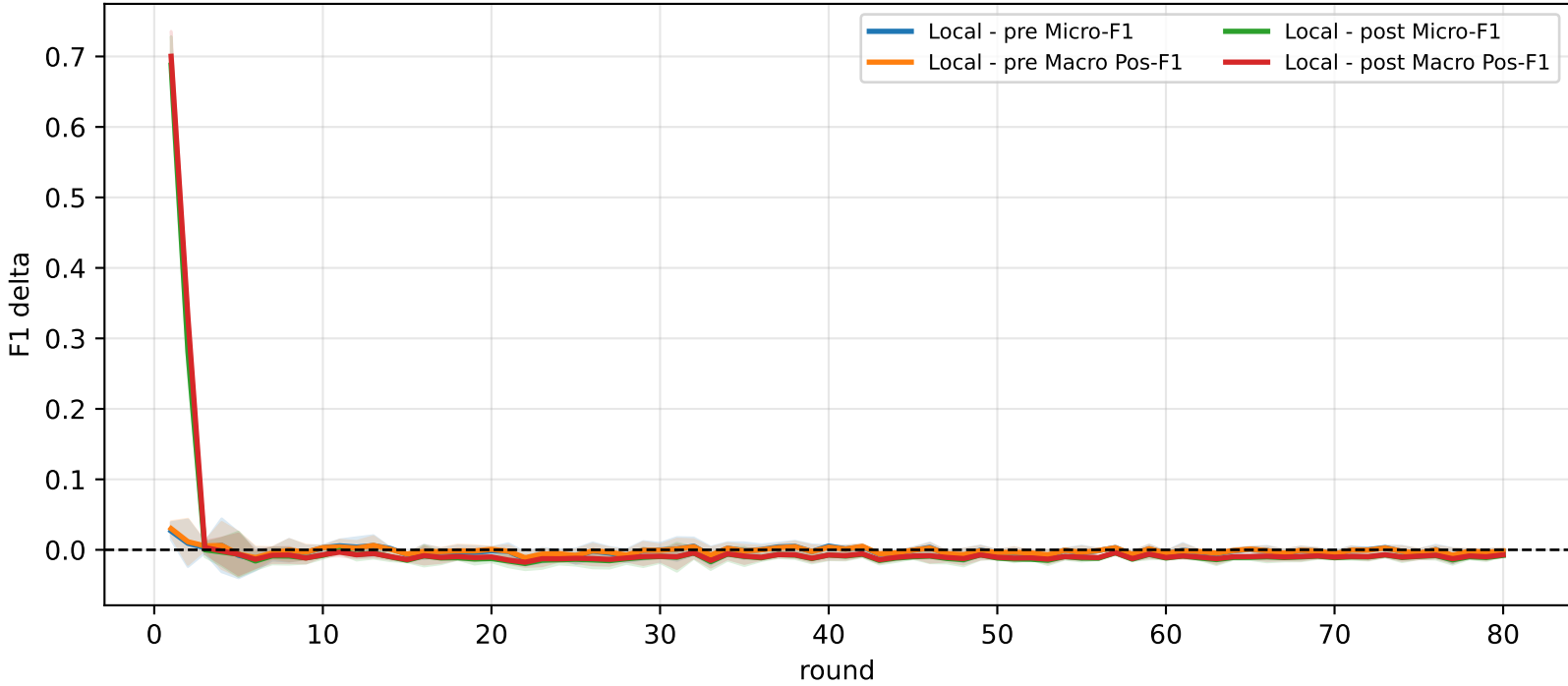
Pooled FedAvg micro / macro F1 (pre-aggregation)



Pooled FedAvg micro / macro F1 (post-aggregation)



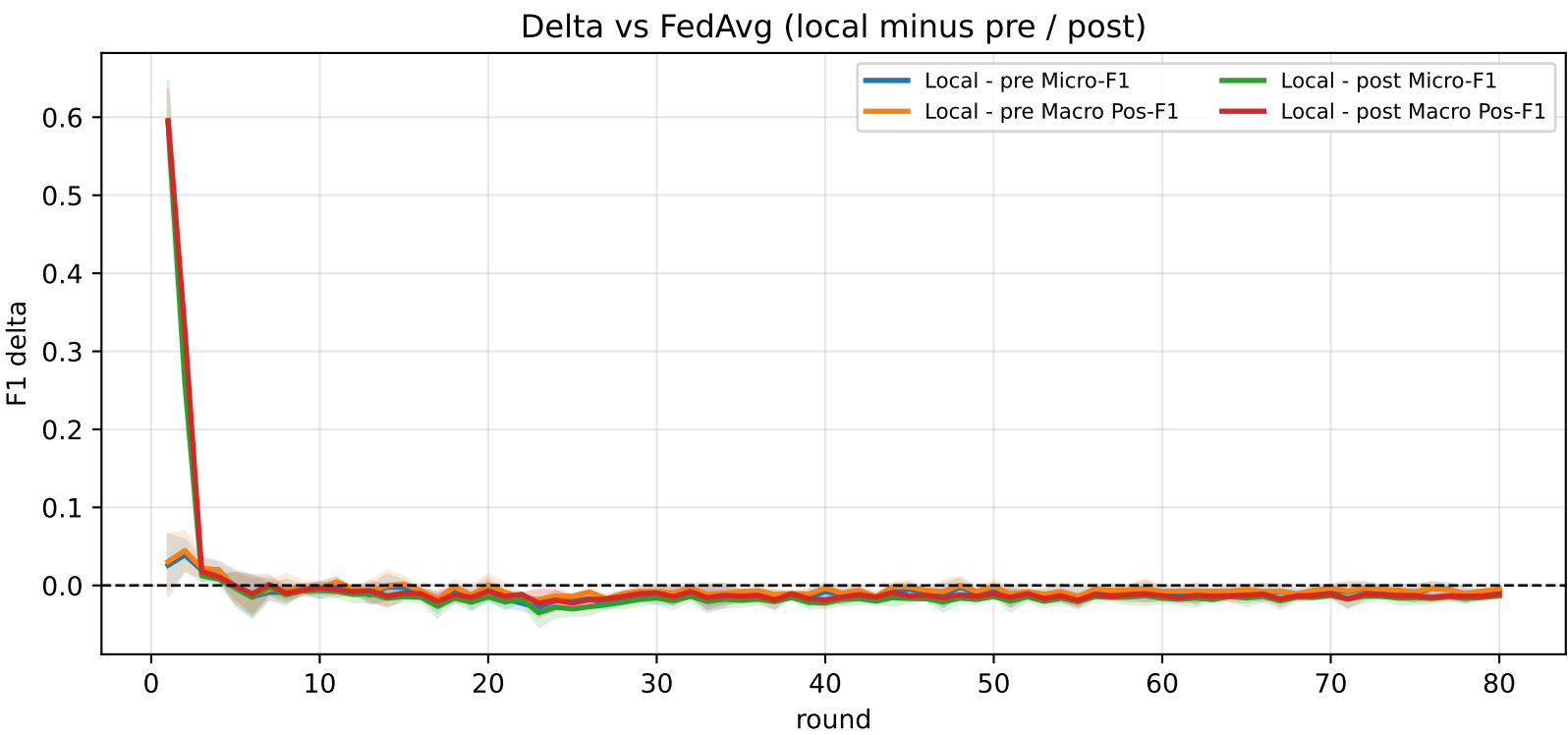
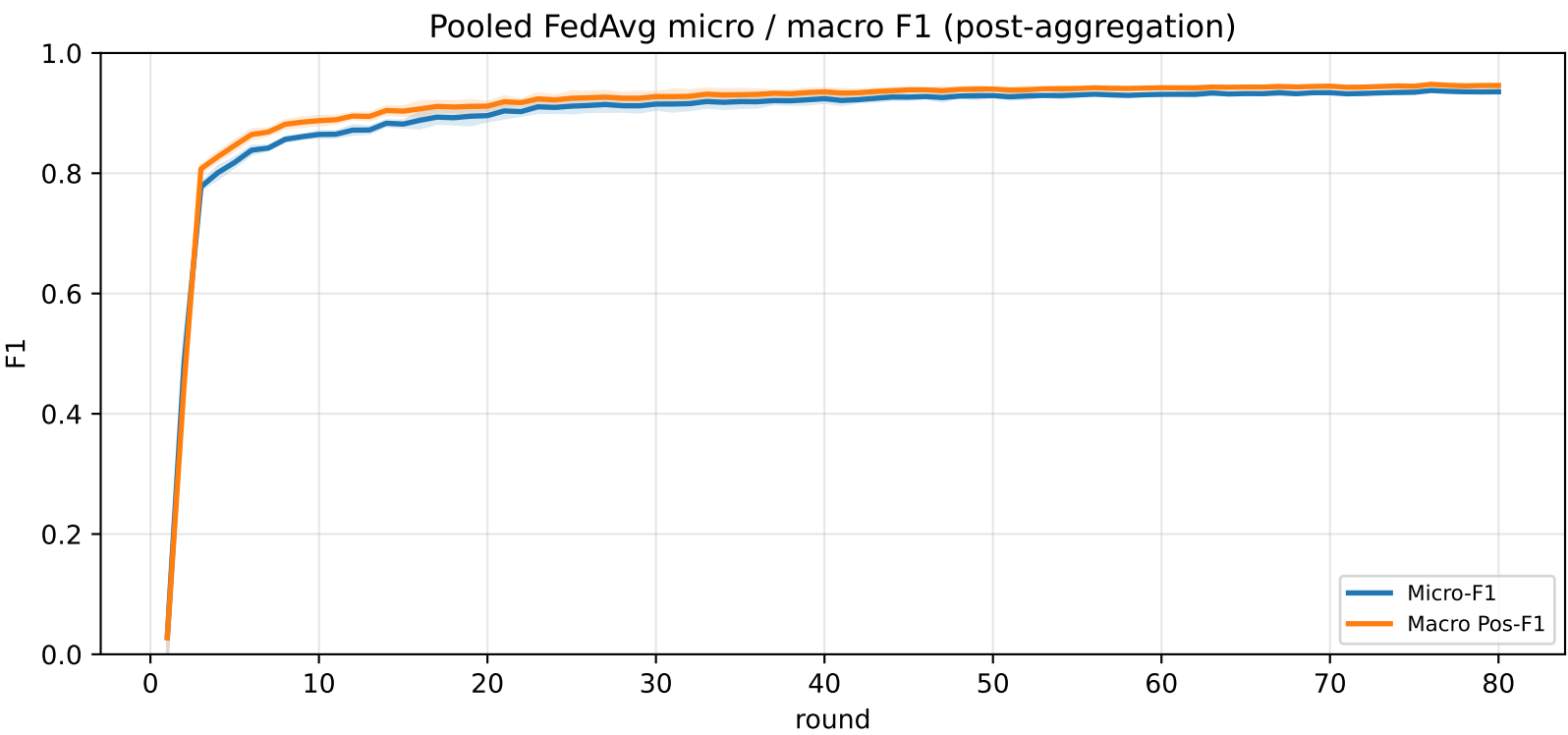
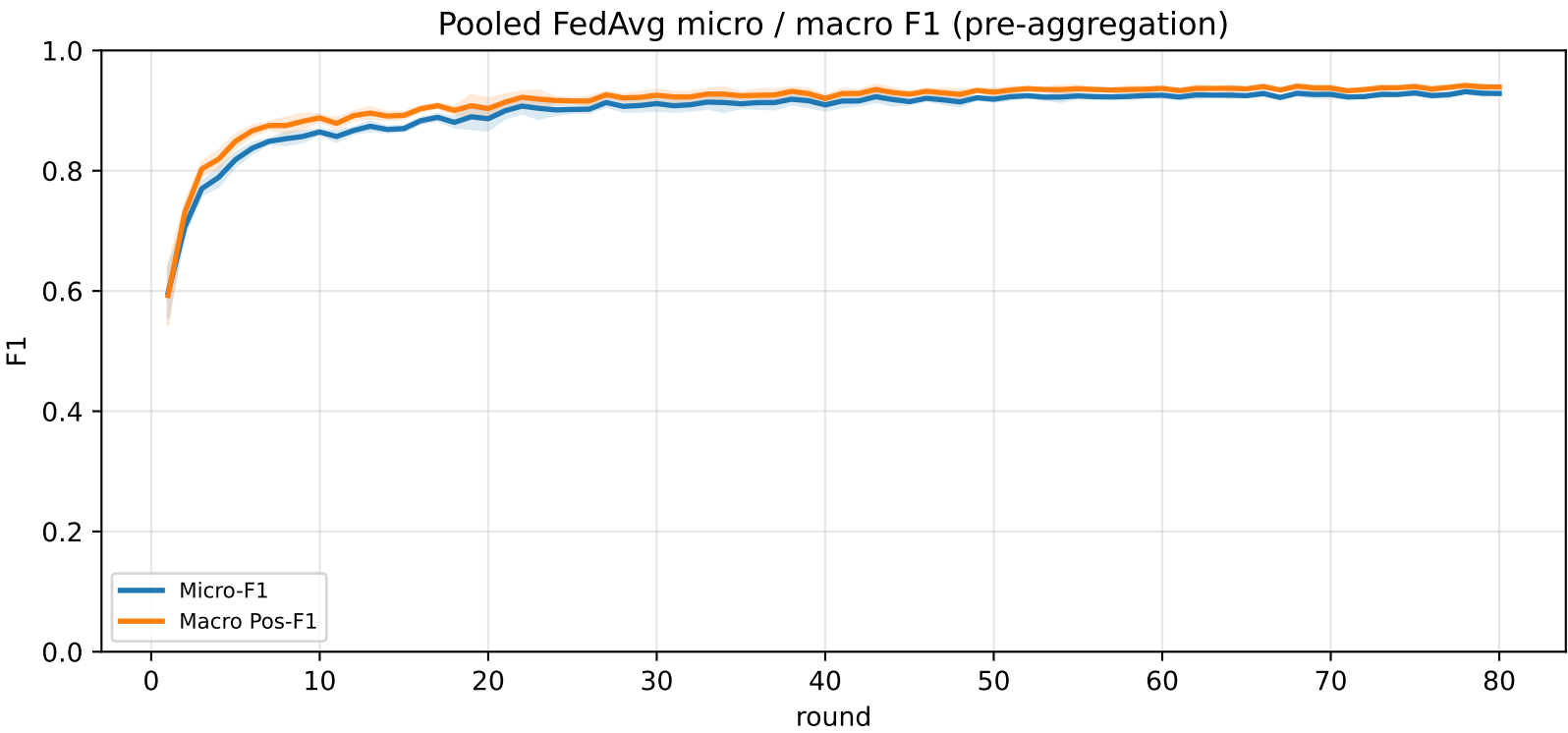
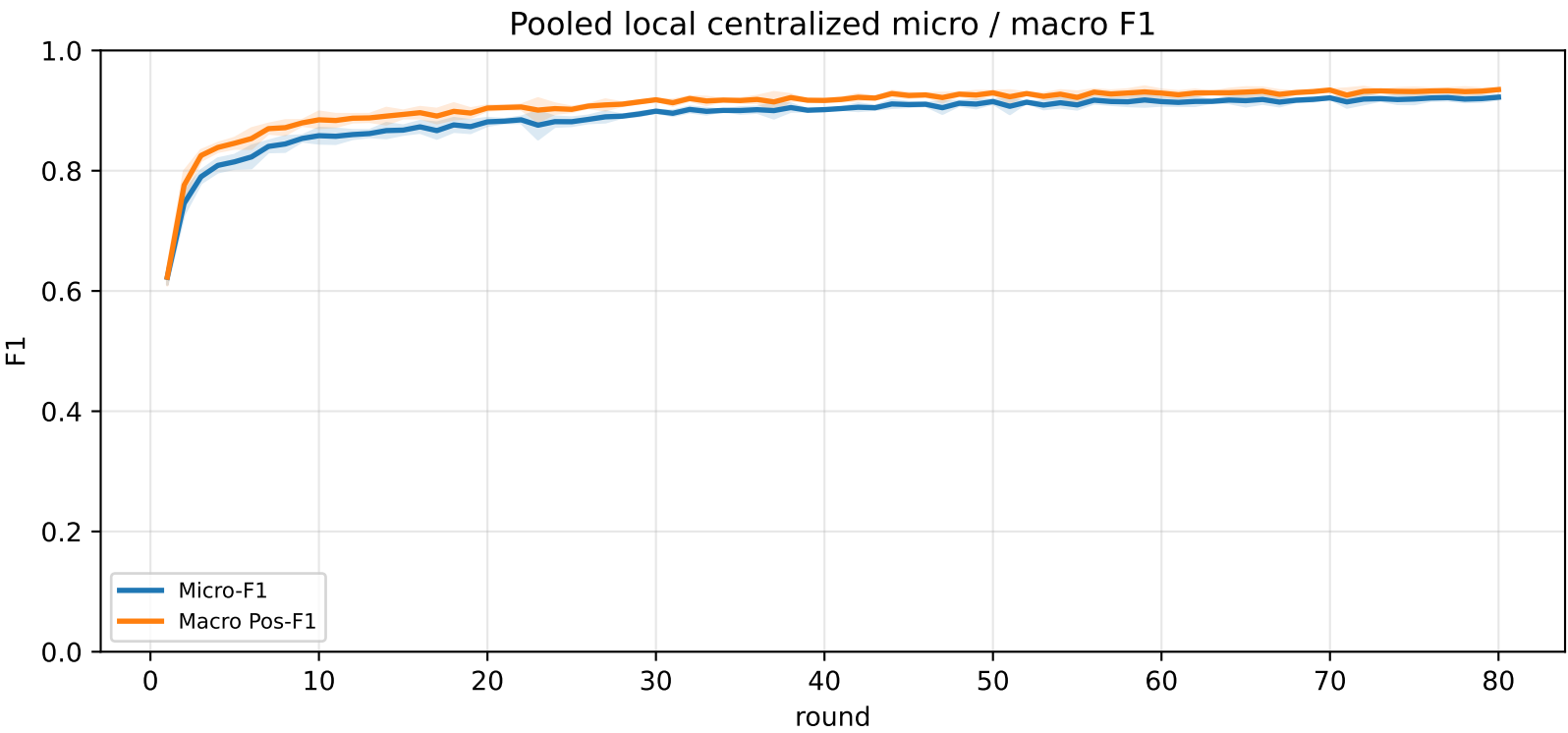
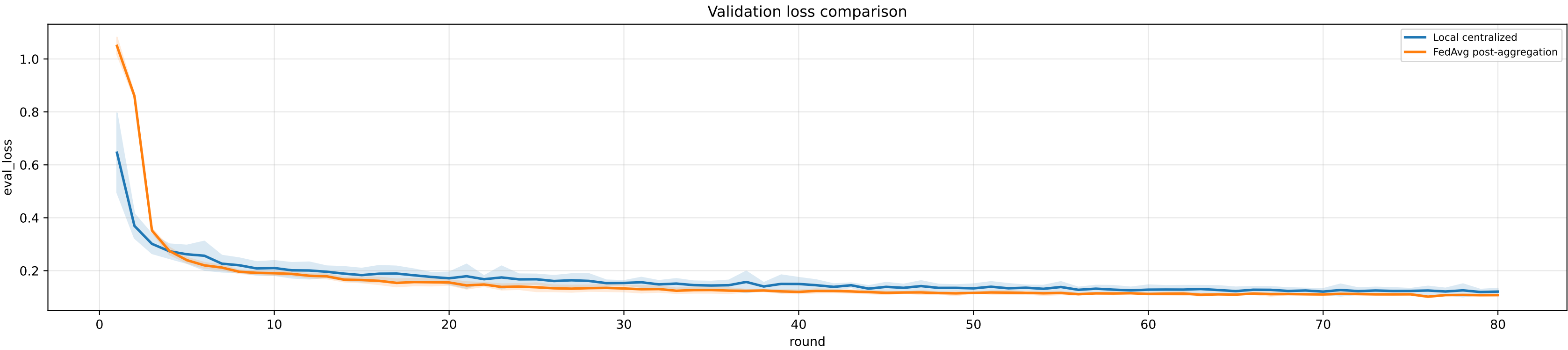
Delta vs FedAvg (local minus pre / post)



Heterogeneity level: Mid | FedAvg diagnostics | family=strong_increasing | n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=strong_increasing | cycle2=10.4%±0.6% | cycle3=15.5%±0.4% | cycle4=21.0%±0.3% | cycle5=24.5%±0.8% | cycle6=28.9%±0.9%
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

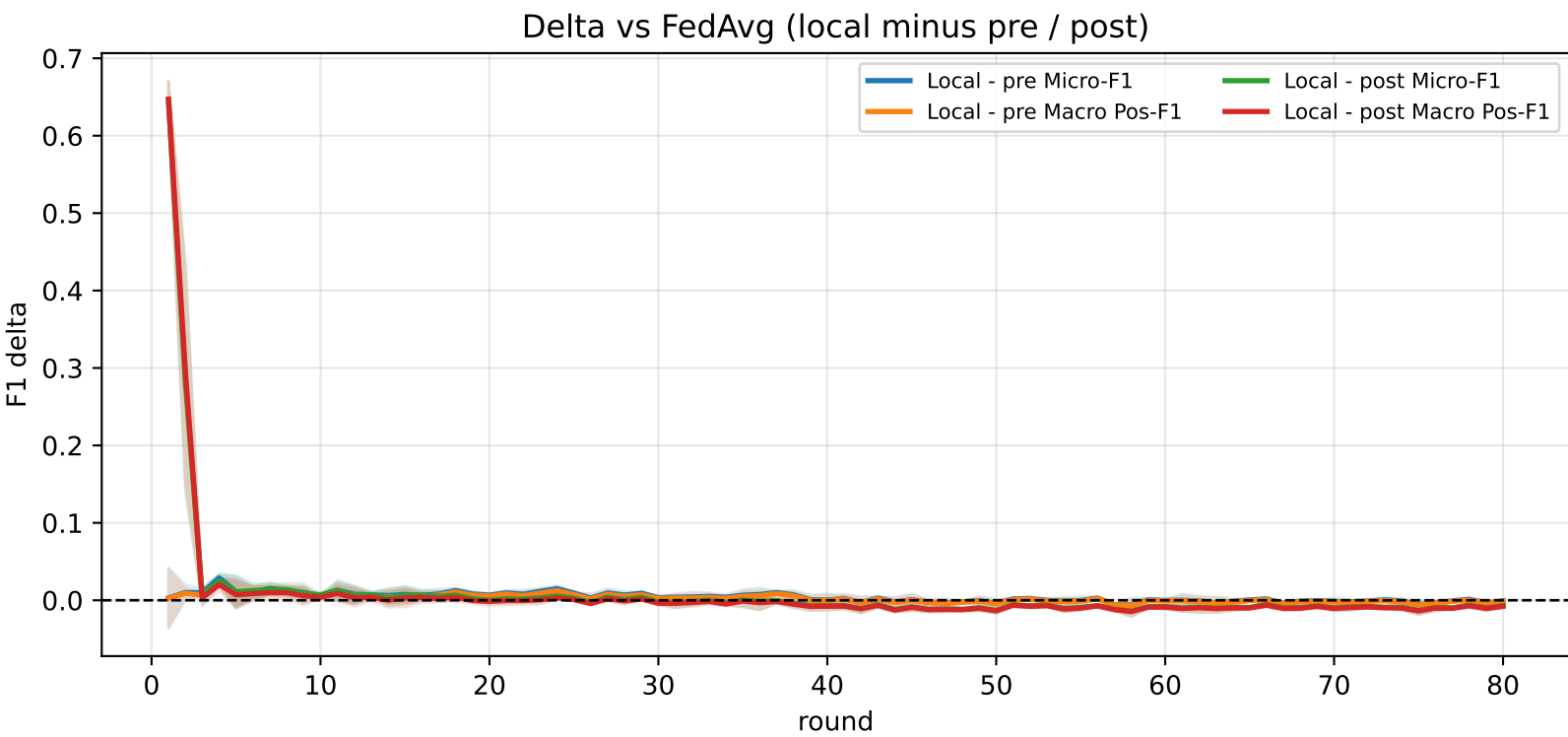
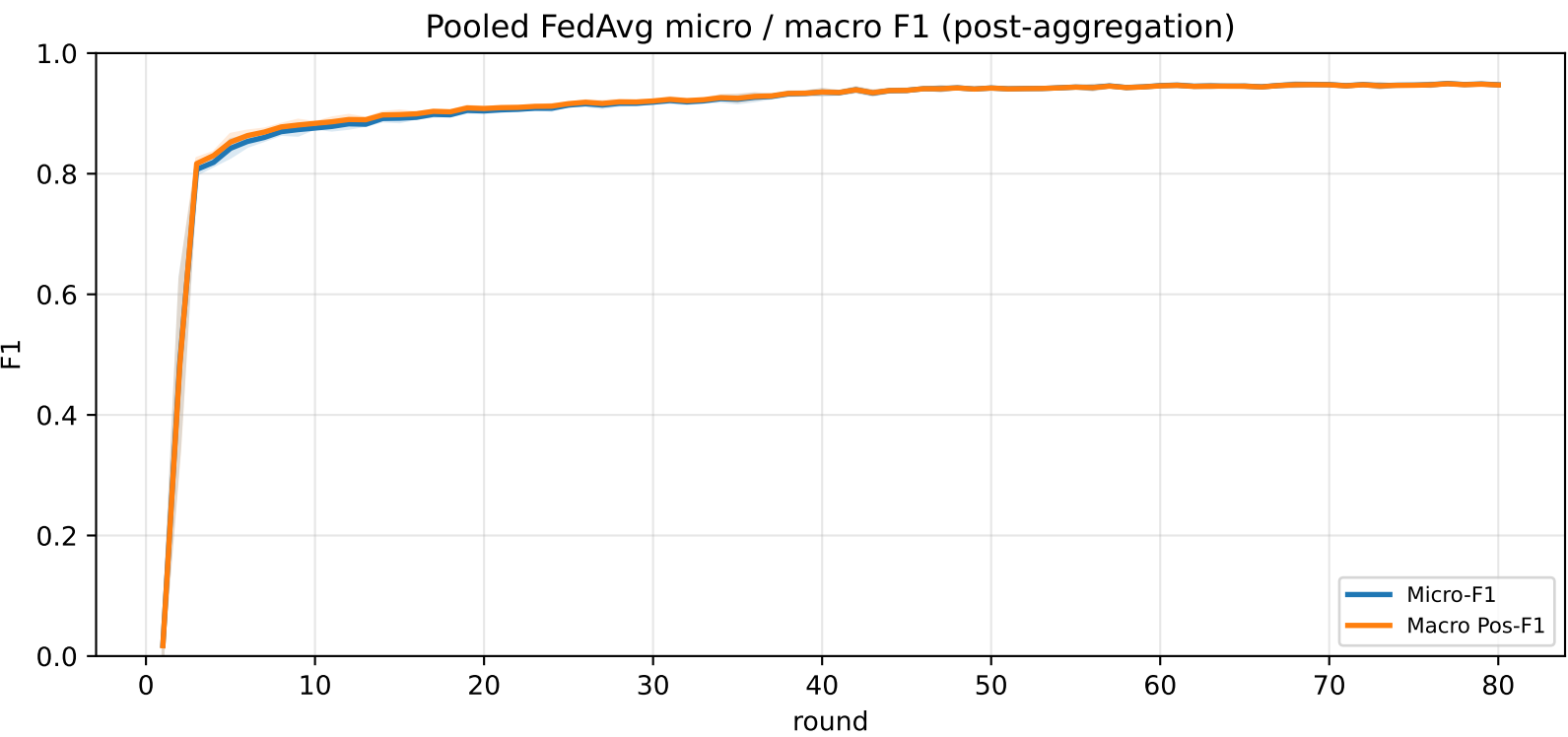
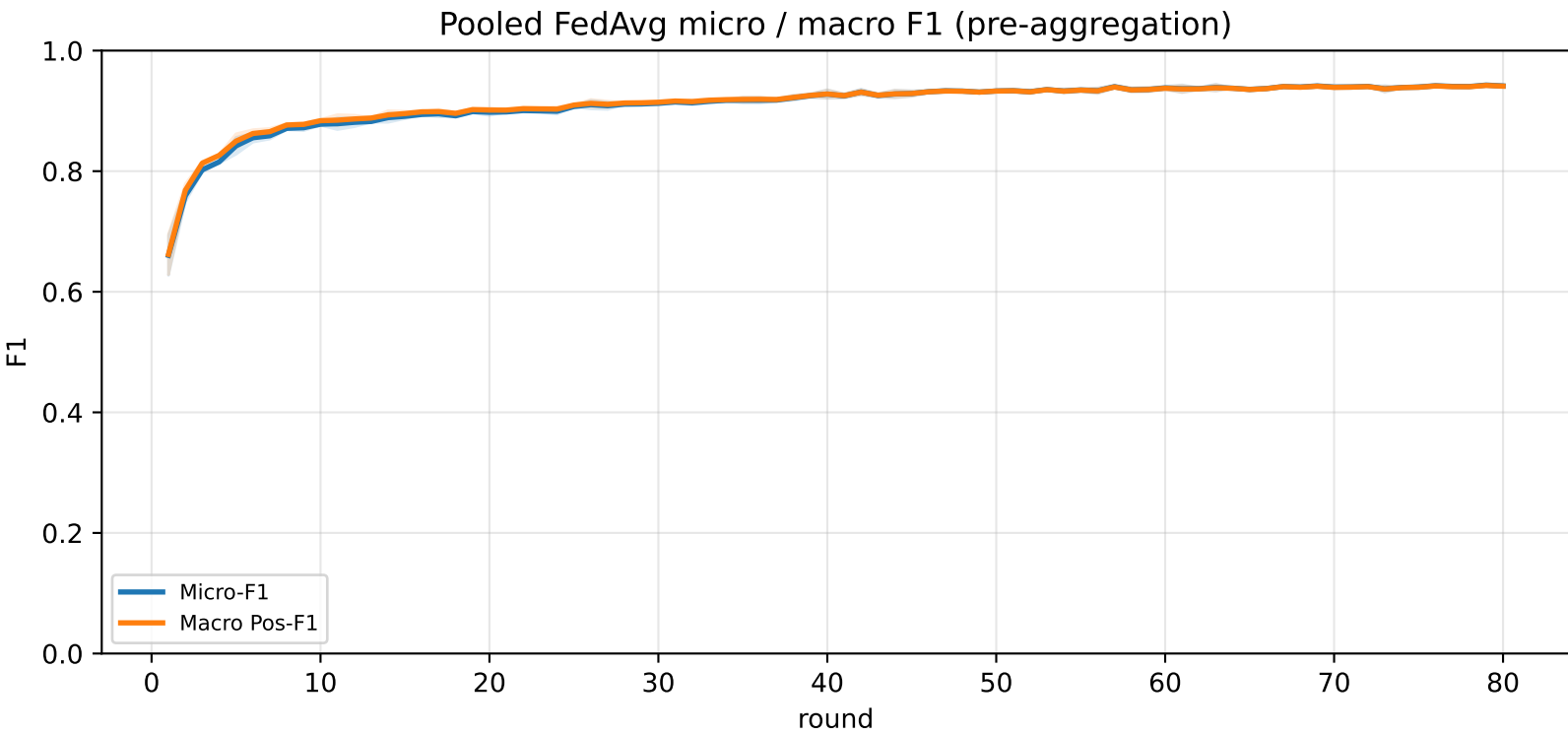
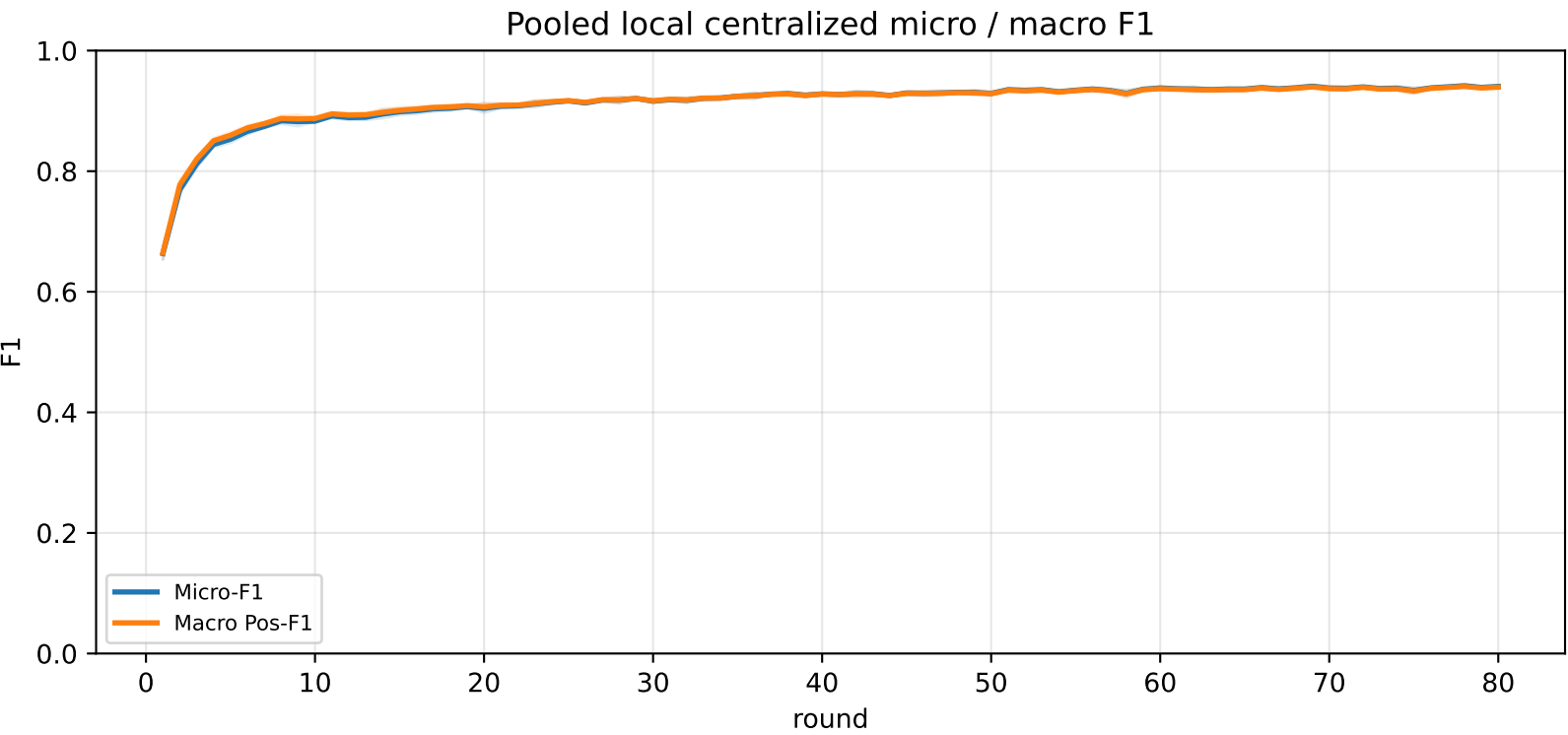
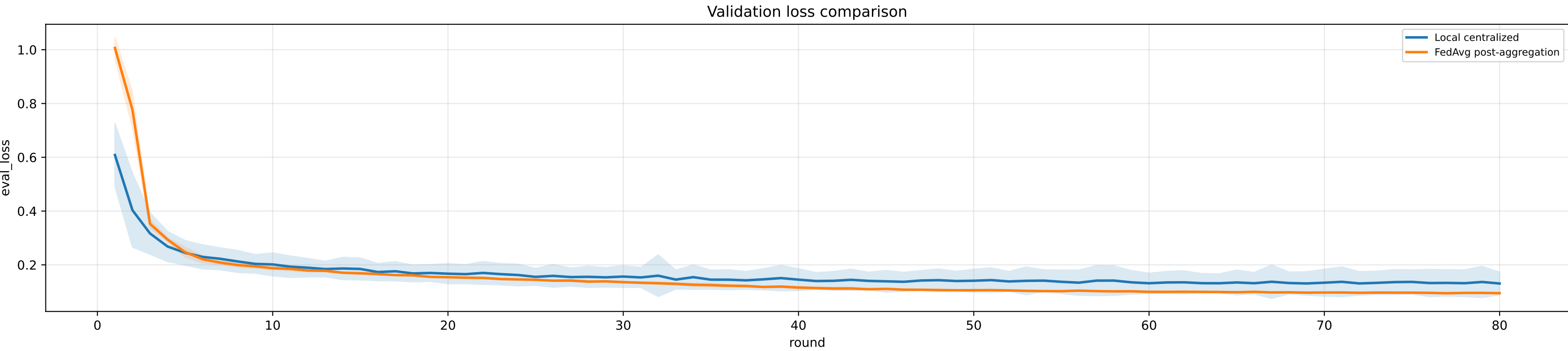
Best logged test performance (FedAvg vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	92.1% ± 0.3%	93.4% ± 0.2%	97.9% ± 0.3%	96.5% ± 0.2%	93.9% ± 0.2%	91.1% ± 0.7%	87.5% ± 0.7%
FedAvg	93.7% ± 0.1%	94.7% ± 0.1%	98.3% ± 0.1%	96.8% ± 0.3%	95.4% ± 0.4%	93.1% ± 0.3%	89.9% ± 0.1%



Heterogeneity level: High | FedAvg diagnostics | aggregated / pooled over 20 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

Best logged test performance (FedAvg vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	93.8% ± 0.1%	92.8% ± 0.1%	99.3% ± 0.1%	98.5% ± 0.1%	96.1% ± 0.1%	91.4% ± 0.2%	84.2% ± 0.3%
FedAvg	95.0% ± 0.1%	94.1% ± 0.1%	99.4% ± 0.0%	98.7% ± 0.0%	96.9% ± 0.1%	93.5% ± 0.1%	86.7% ± 0.3%



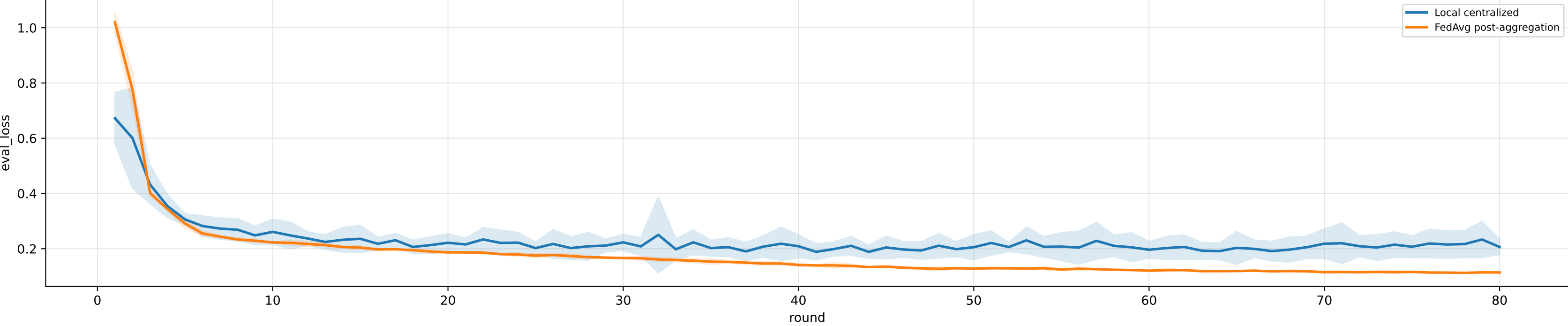
Heterogeneity level: High | FedAvg diagnostics | family=strong_decreasing | n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=strong_decreasing | cycle2=44.0%±0.3% | cycle3=34.3%±0.7% | cycle4=22.7%±0.7% | cycle5=13.9%±0.4% | cycle6=7.8%±0.6%
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

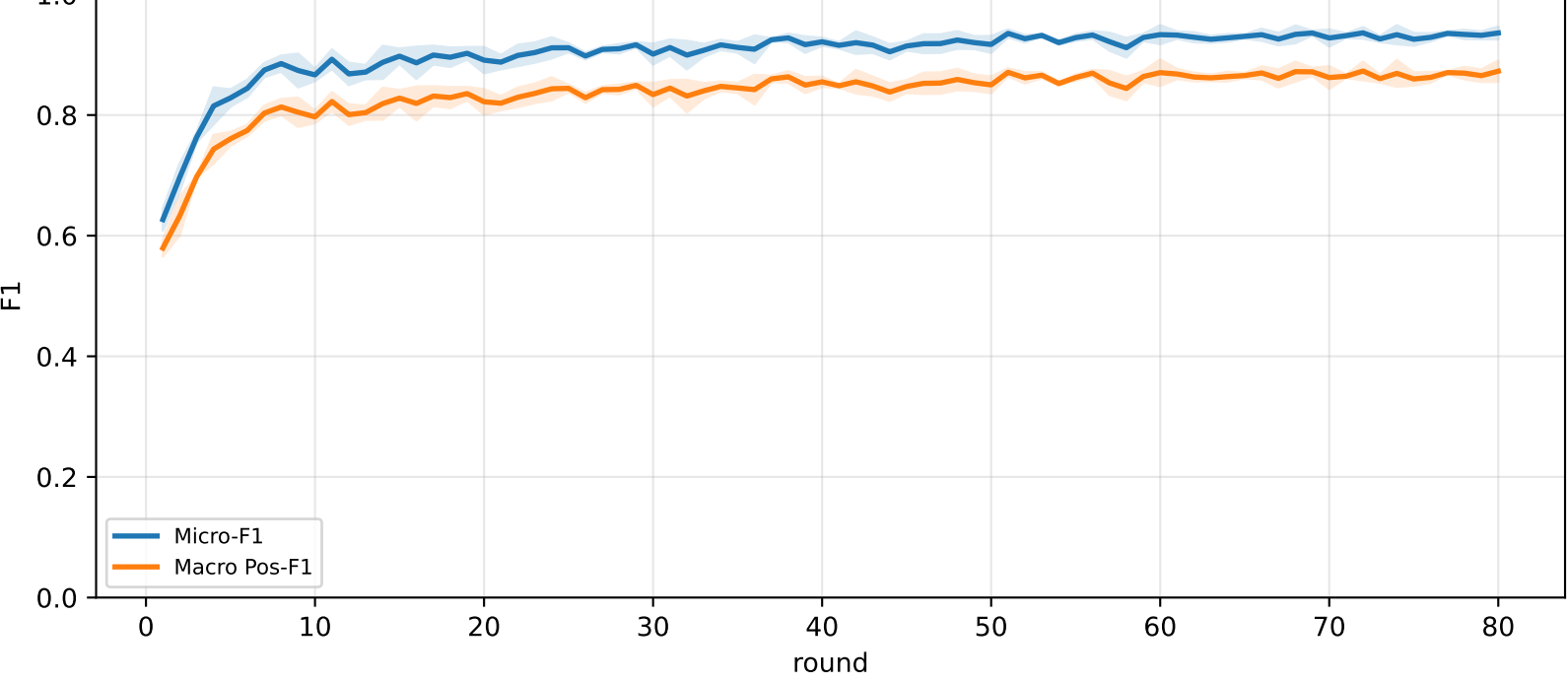
Best logged test performance (FedAvg vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	92.1% ± 0.6%	85.7% ± 0.7%	99.6% ± 0.1%	98.9% ± 0.1%	95.0% ± 0.4%	79.3% ± 1.4%	55.6% ± 2.3%
FedAvg	94.1% ± 0.2%	88.5% ± 0.3%	99.8% ± 0.0%	99.4% ± 0.1%	96.9% ± 0.4%	85.2% ± 0.8%	61.5% ± 1.0%

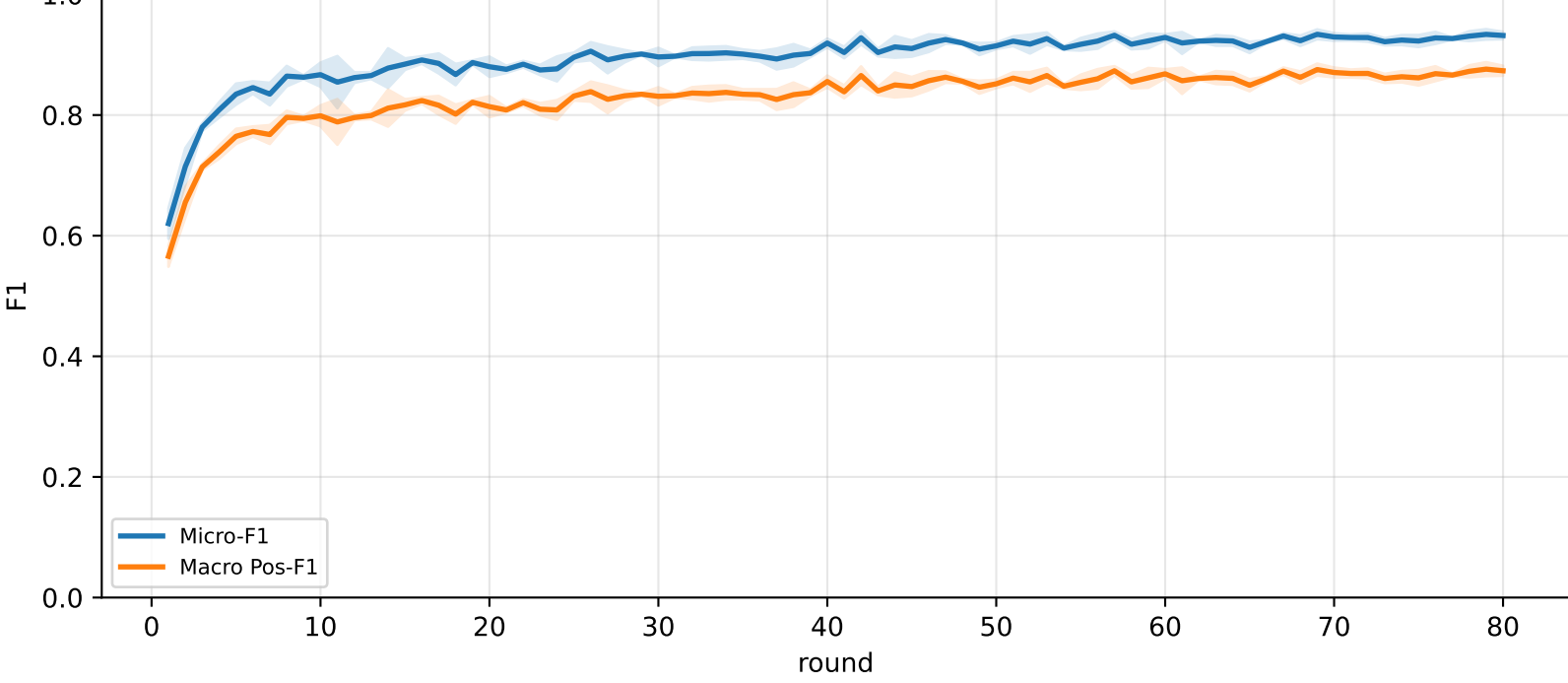
Validation loss comparison



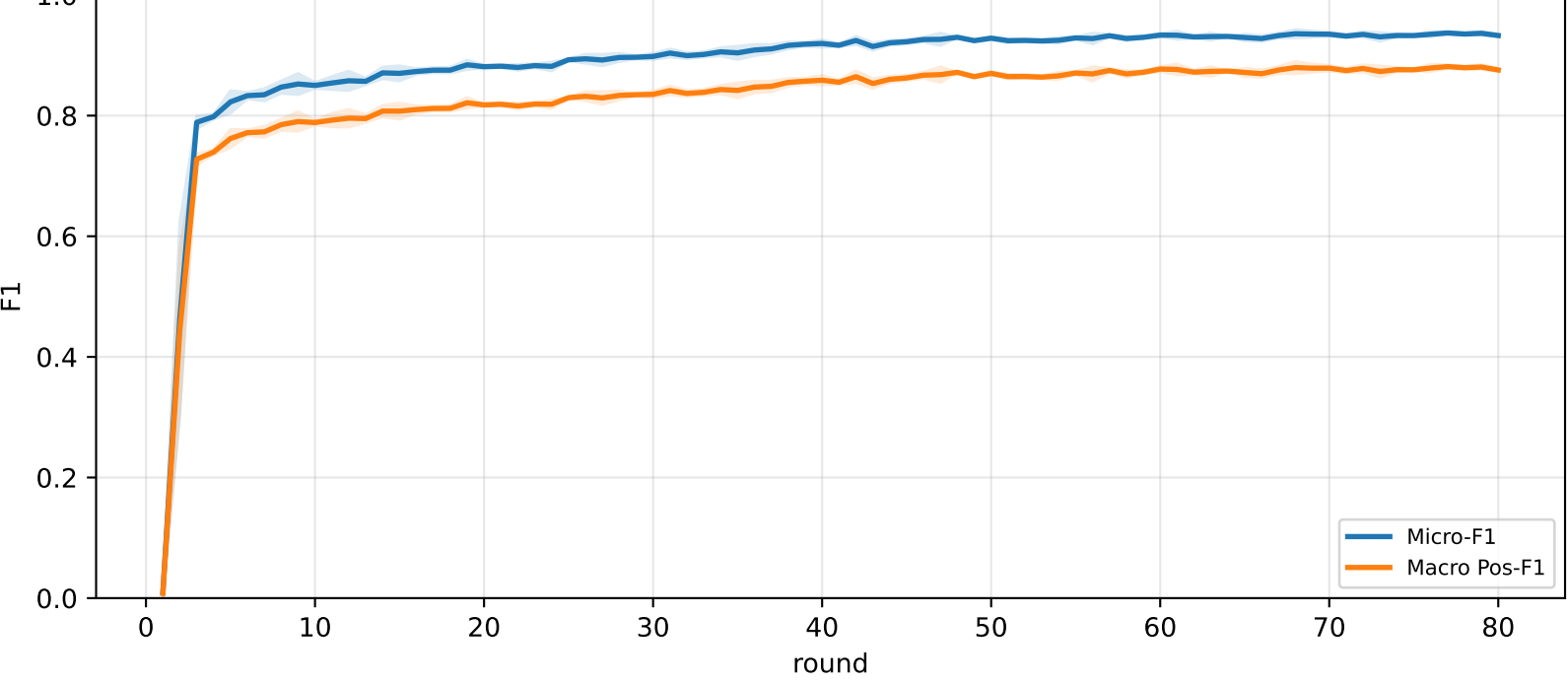
Pooled local centralized micro / macro F1



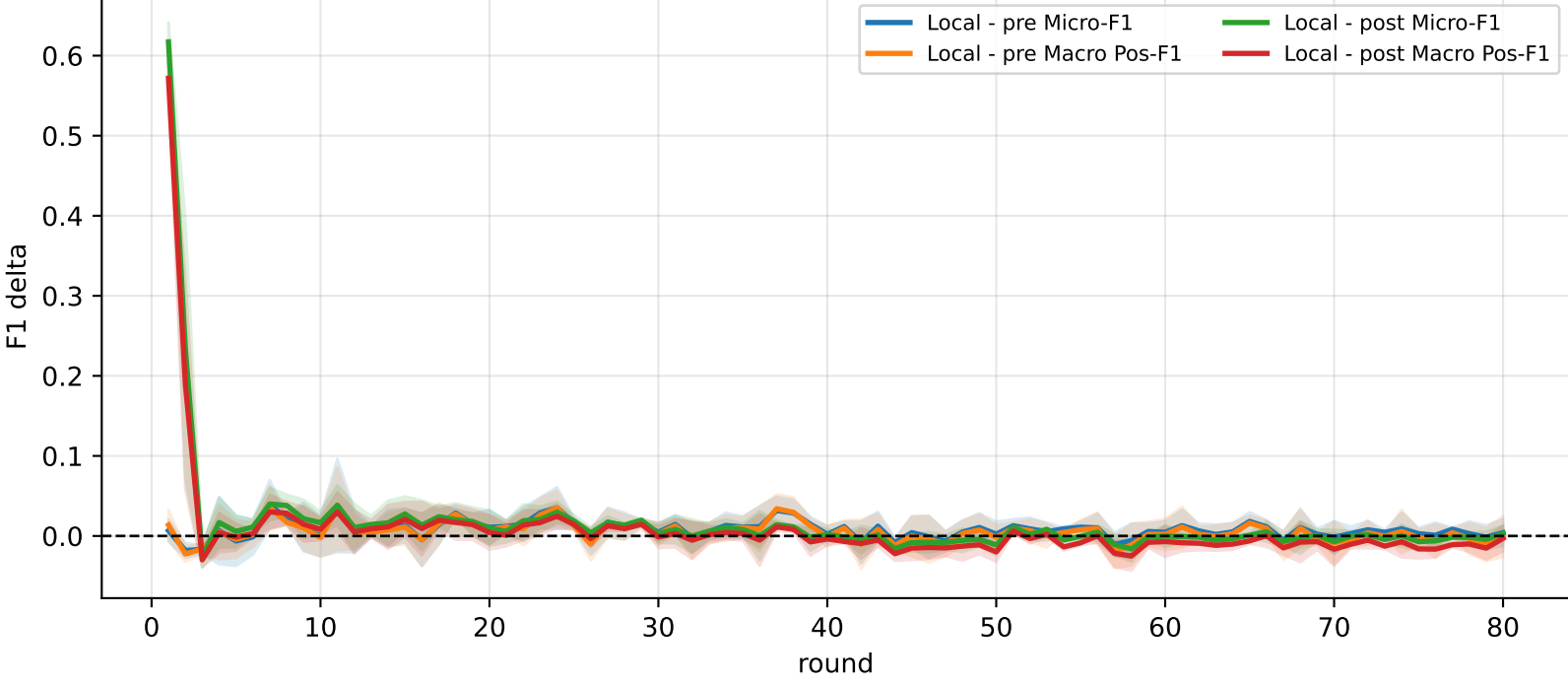
Pooled FedAvg micro / macro F1 (pre-aggregation)



Pooled FedAvg micro / macro F1 (post-aggregation)



Delta vs FedAvg (local minus pre / post)



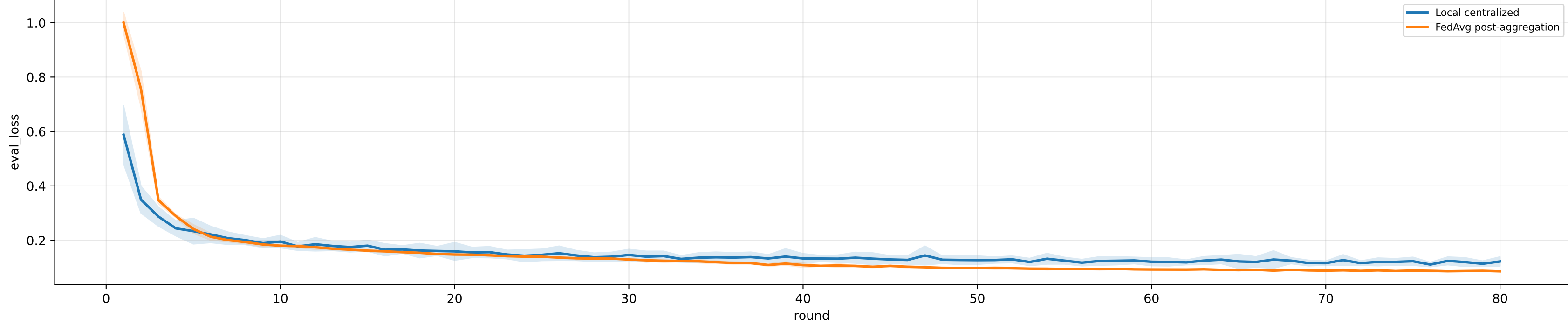
Heterogeneity level: High | FedAvg diagnostics | family=mild_decreasing | n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=mild_decreasing | cycle2=32.5%±0.7% | cycle3=30.7%±1.0% | cycle4=26.8%±1.1% | cycle5=24.5%±1.0% | cycle6=19.2%±1.0%
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

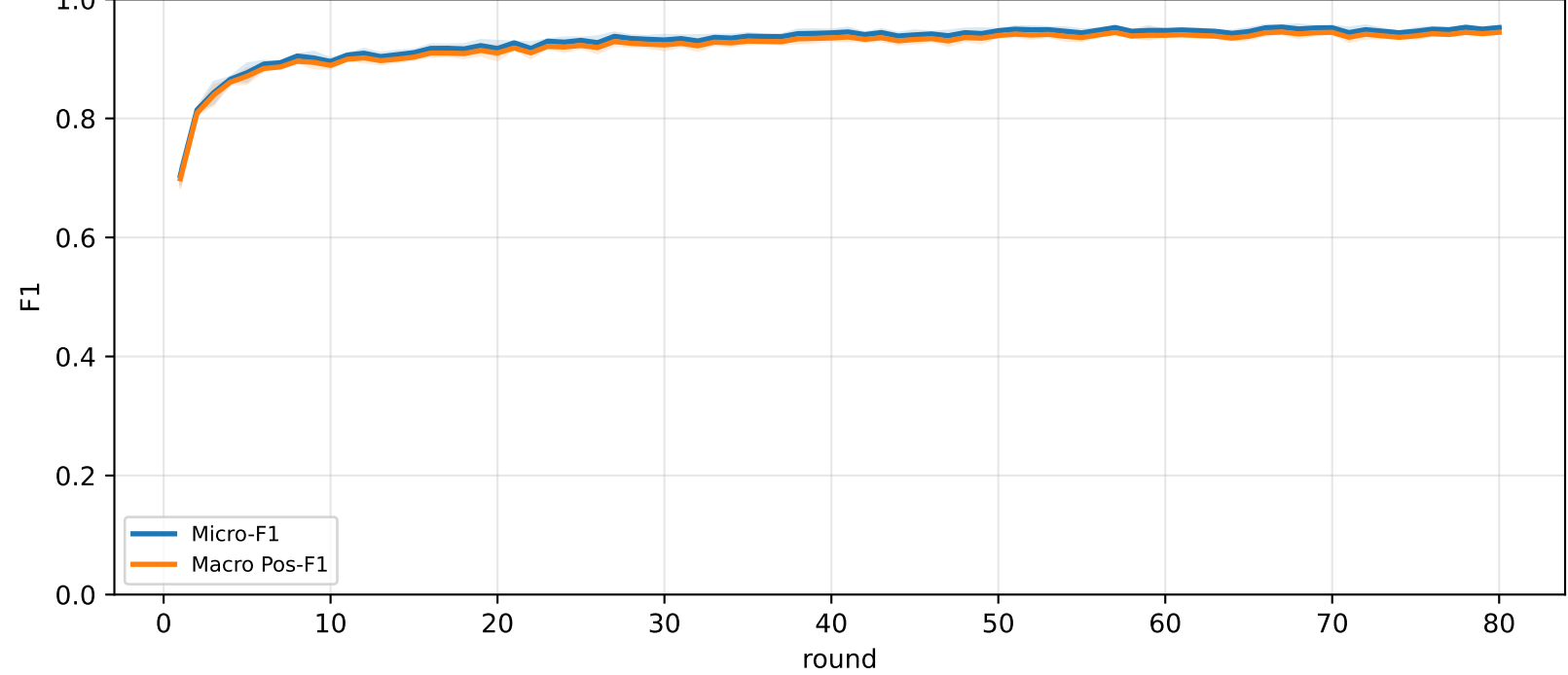
Best logged test performance (FedAvg vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	95.2% ± 0.2%	94.4% ± 0.2%	99.5% ± 0.1%	99.1% ± 0.1%	97.3% ± 0.1%	92.9% ± 0.1%	83.2% ± 0.7%
FedAvg	95.9% ± 0.2%	95.2% ± 0.2%	99.6% ± 0.0%	99.3% ± 0.1%	98.0% ± 0.2%	94.5% ± 0.3%	84.7% ± 0.6%

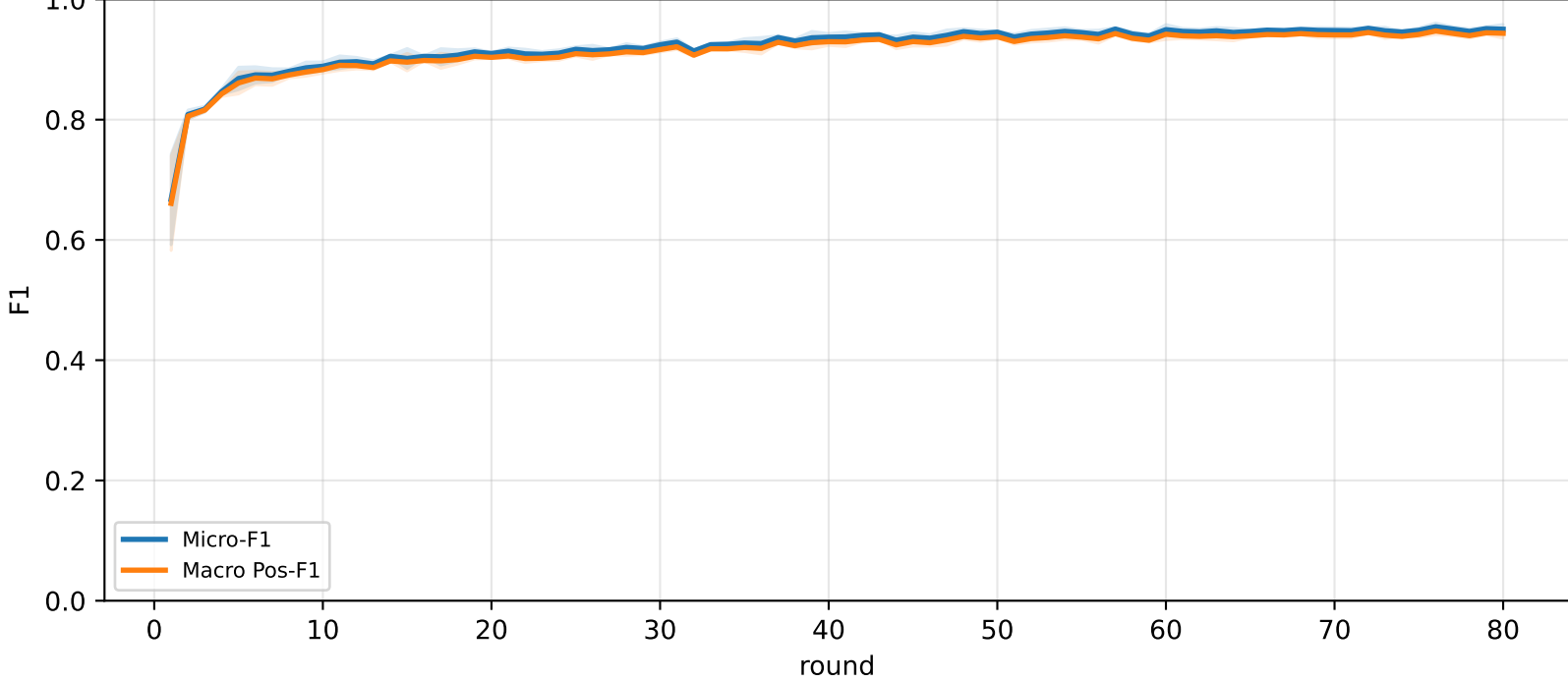
Validation loss comparison



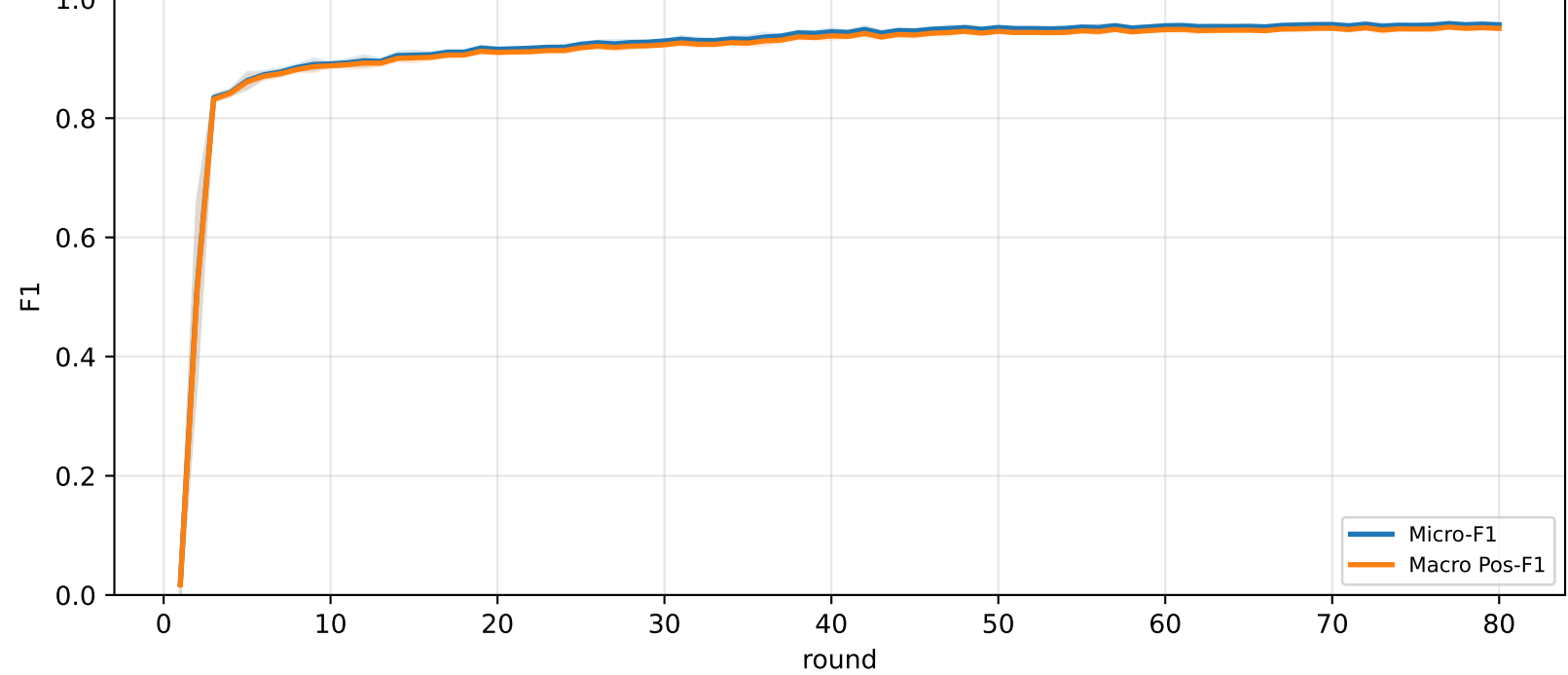
Pooled local centralized micro / macro F1



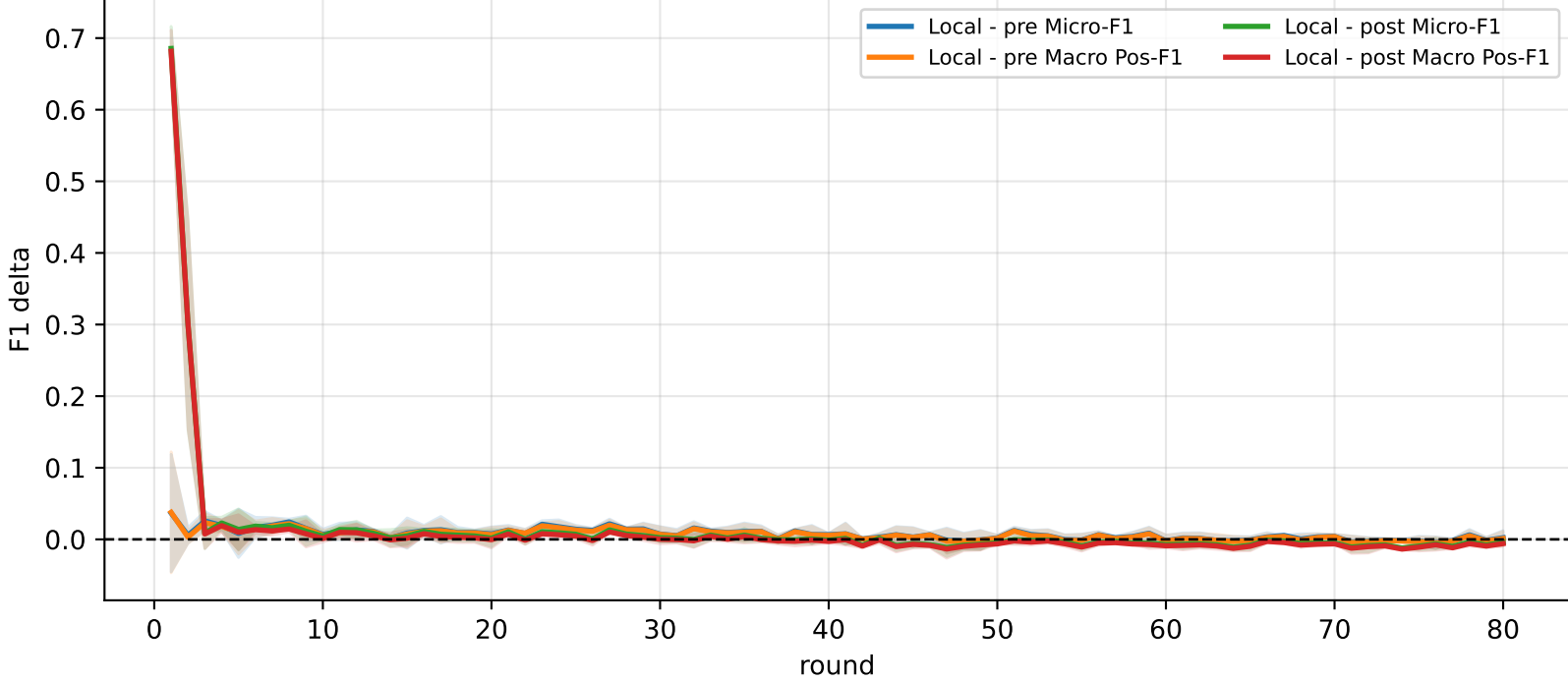
Pooled FedAvg micro / macro F1 (pre-aggregation)



Pooled FedAvg micro / macro F1 (post-aggregation)



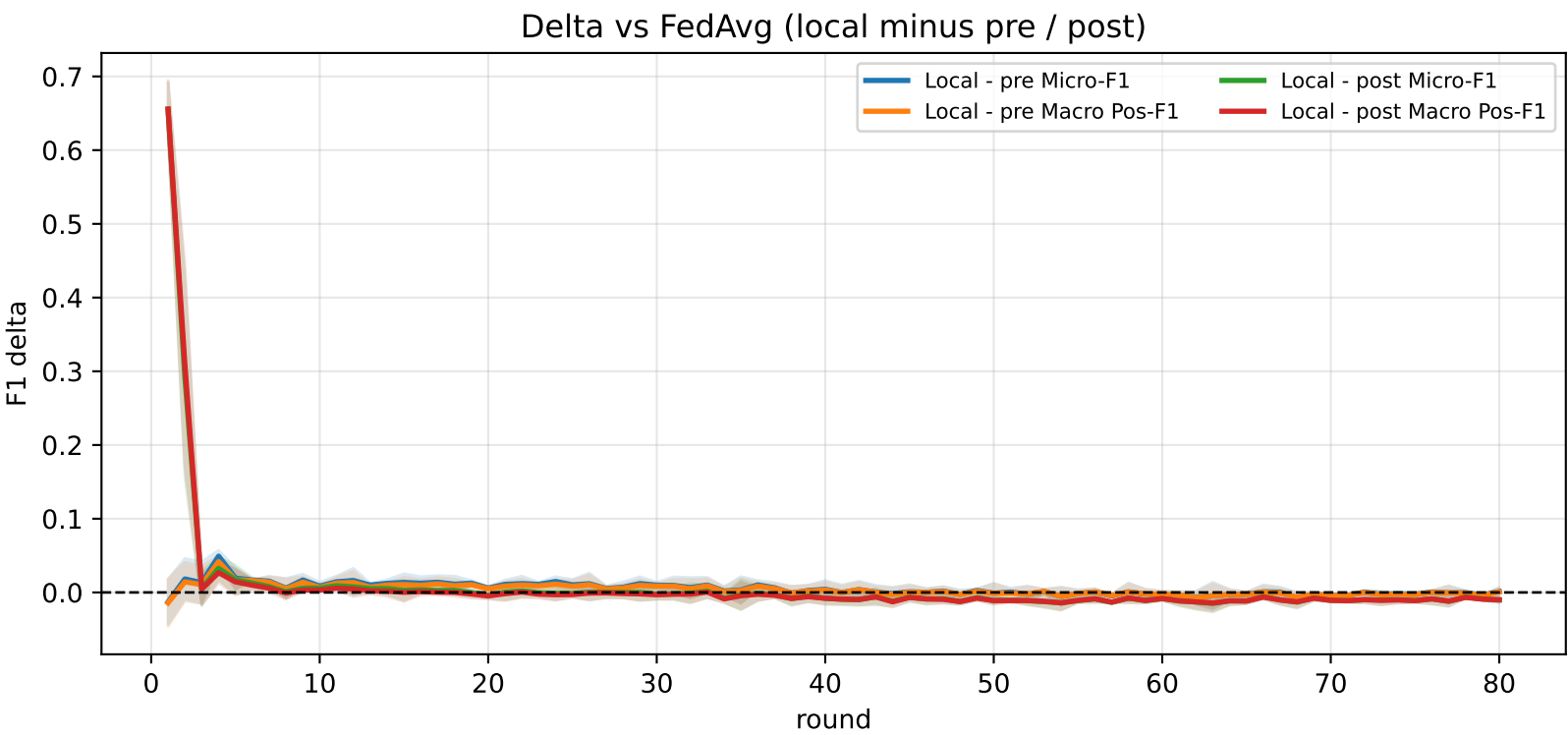
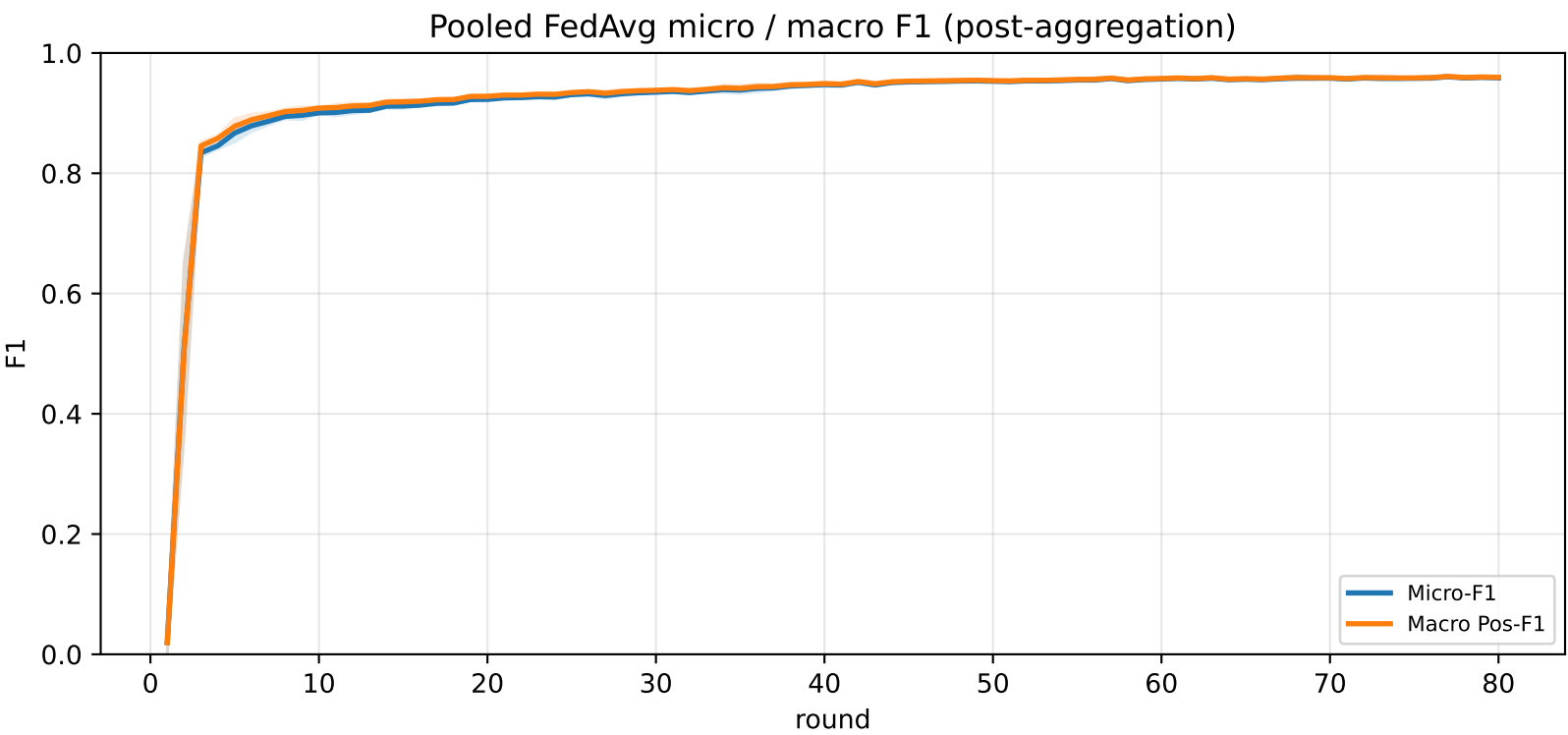
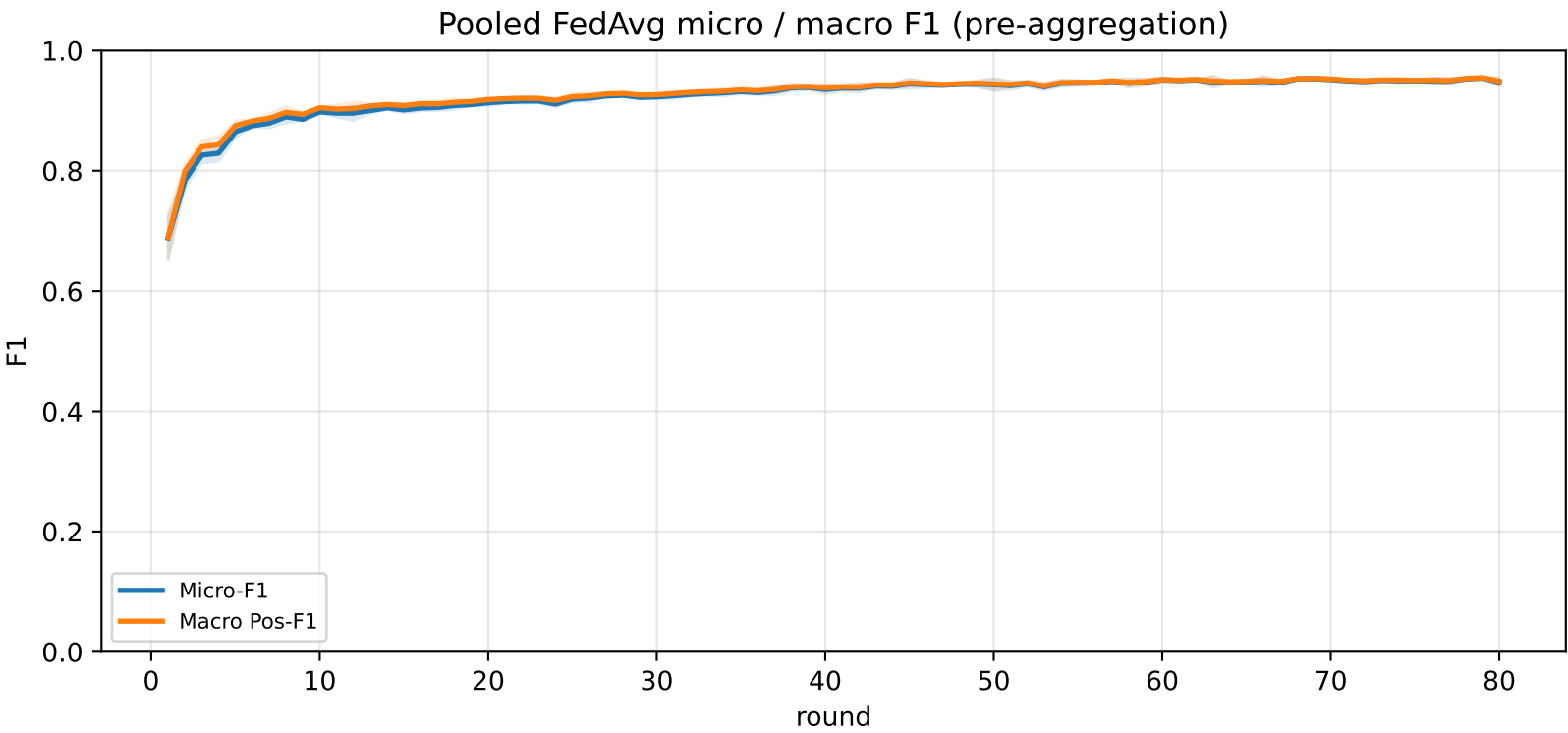
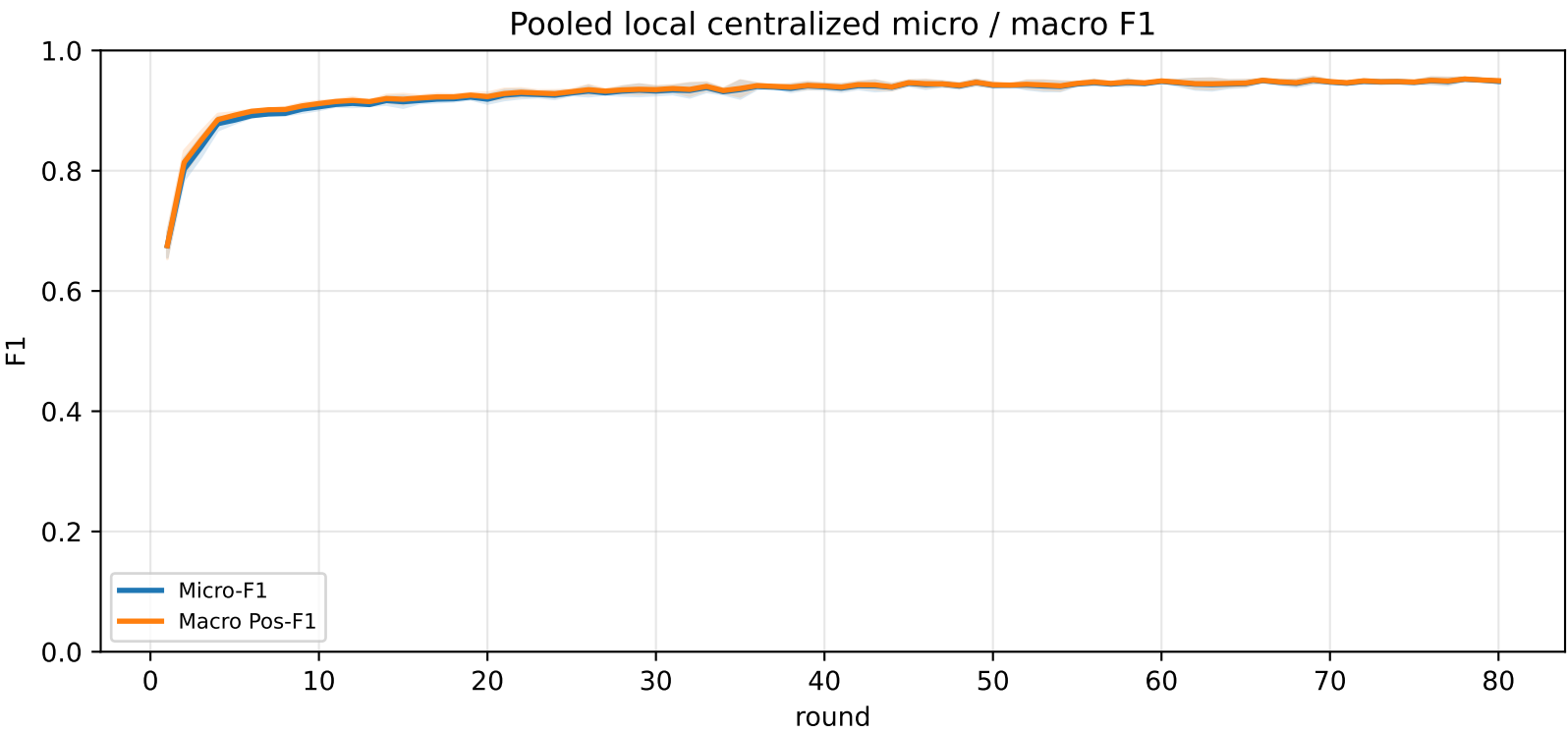
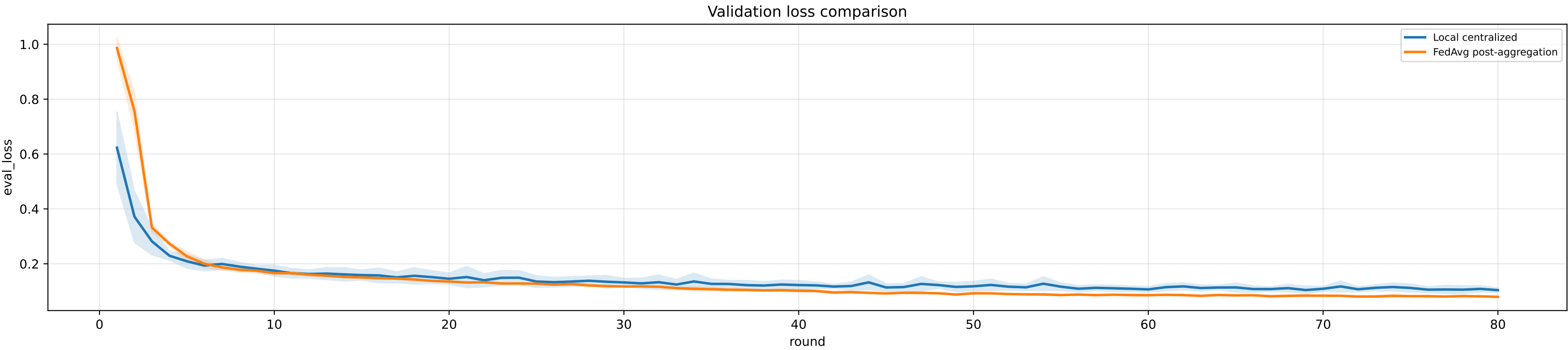
Delta vs FedAvg (local minus pre / post)



Heterogeneity level: High | FedAvg diagnostics | family=flat_balanced |
n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=flat_balanced | cycle2=25.1%±0.7% | cycle3=26.6%±0.8% | cycle4=27.5%±0.8% | cycle5=27.8%±1.2% | cycle6=25.6%±0.3%
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

Best logged test performance (FedAvg vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	95.0% ± 0.1%	95.2% ± 0.1%	99.1% ± 0.1%	98.8% ± 0.2%	97.1% ± 0.1%	94.2% ± 0.2%	86.6% ± 0.5%
FedAvg	96.1% ± 0.1%	96.2% ± 0.1%	99.3% ± 0.0%	98.7% ± 0.0%	97.4% ± 0.1%	95.7% ± 0.2%	89.7% ± 0.5%



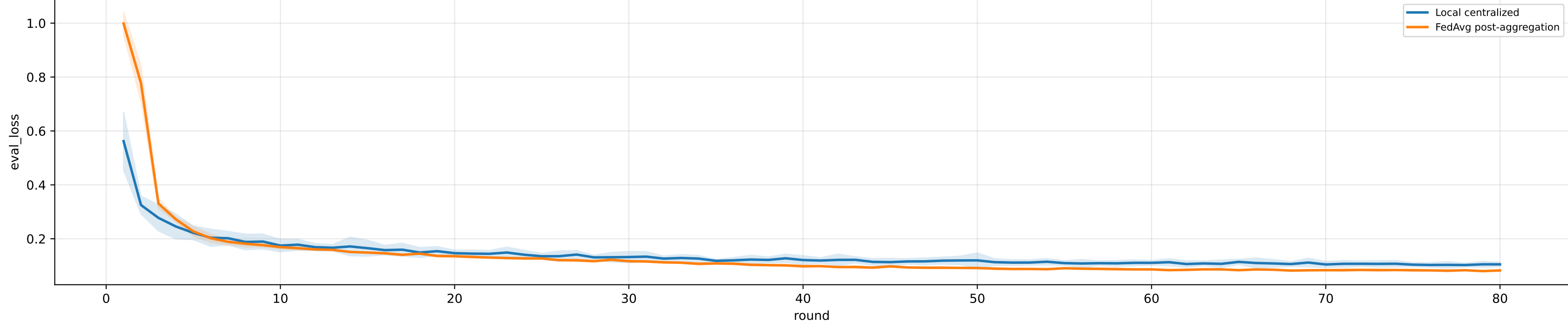
Heterogeneity level: High | FedAvg diagnostics | family=mild_increasing | n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=mild_increasing | cycle2=16.7%±0.8% | cycle3=21.4%±0.7% | cycle4=25.6%±1.2% | cycle5=28.8%±0.7% | cycle6=30.0%±0.4%
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

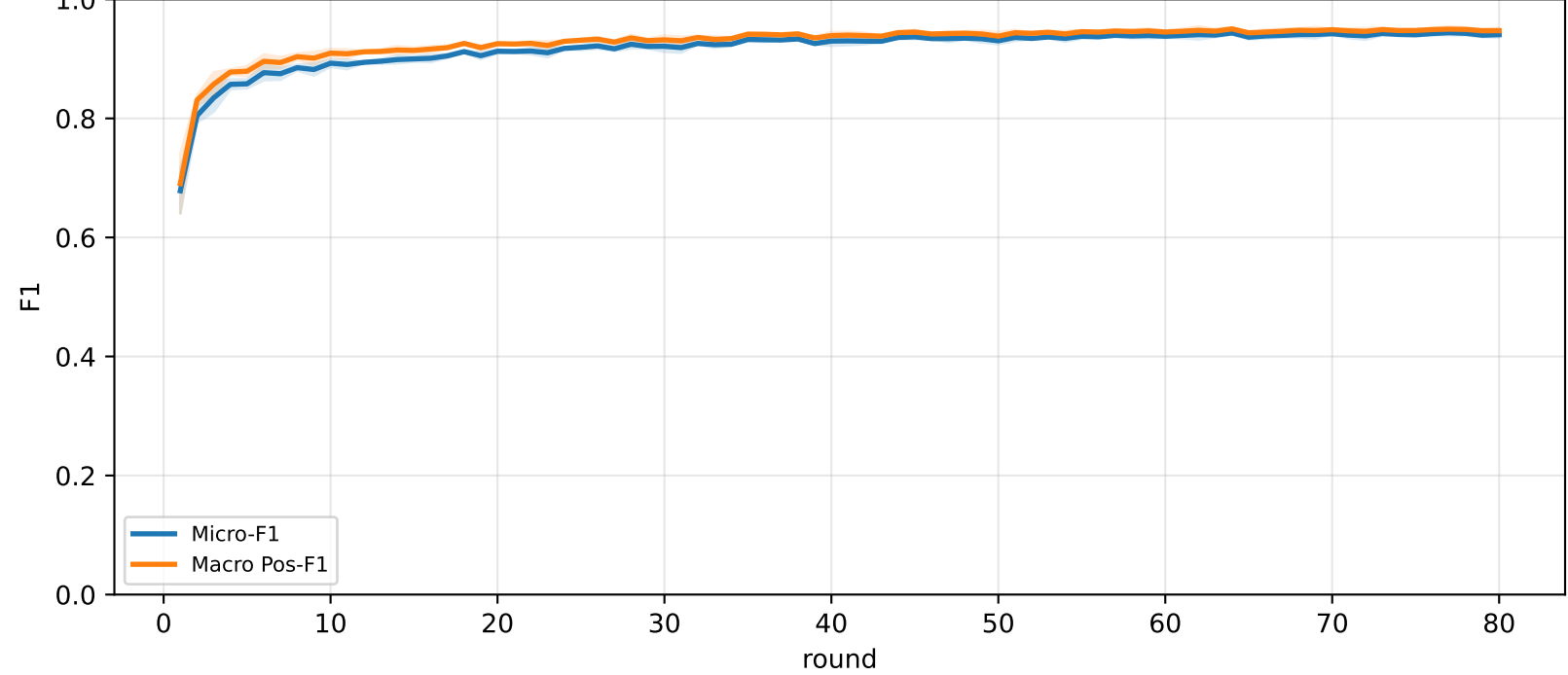
Best logged test performance (FedAvg vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.4% ± 0.1%	95.1% ± 0.1%	98.7% ± 0.2%	97.8% ± 0.2%	96.2% ± 0.2%	93.8% ± 0.4%	89.2% ± 0.5%
FedAvg	95.7% ± 0.1%	96.2% ± 0.1%	98.8% ± 0.2%	98.2% ± 0.2%	96.8% ± 0.1%	95.2% ± 0.2%	92.1% ± 0.0%

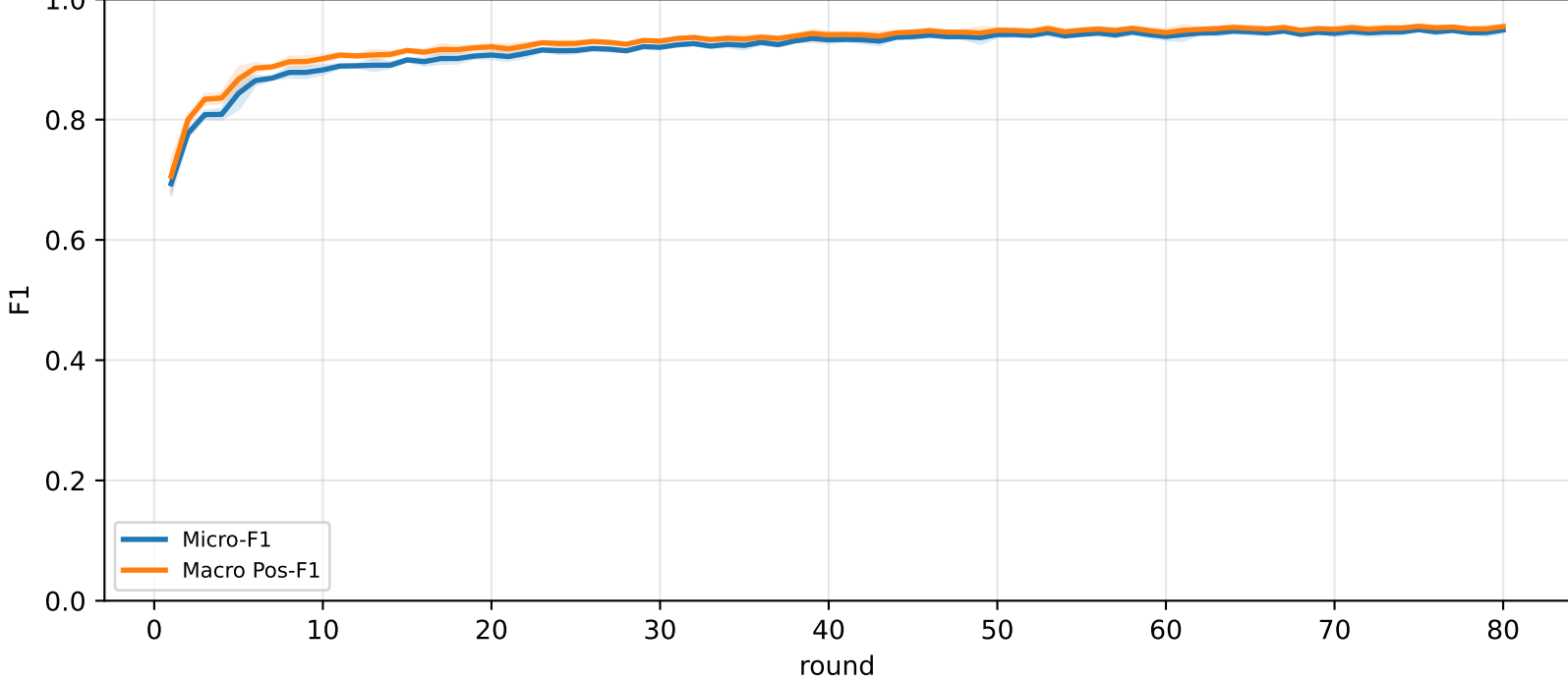
Validation loss comparison



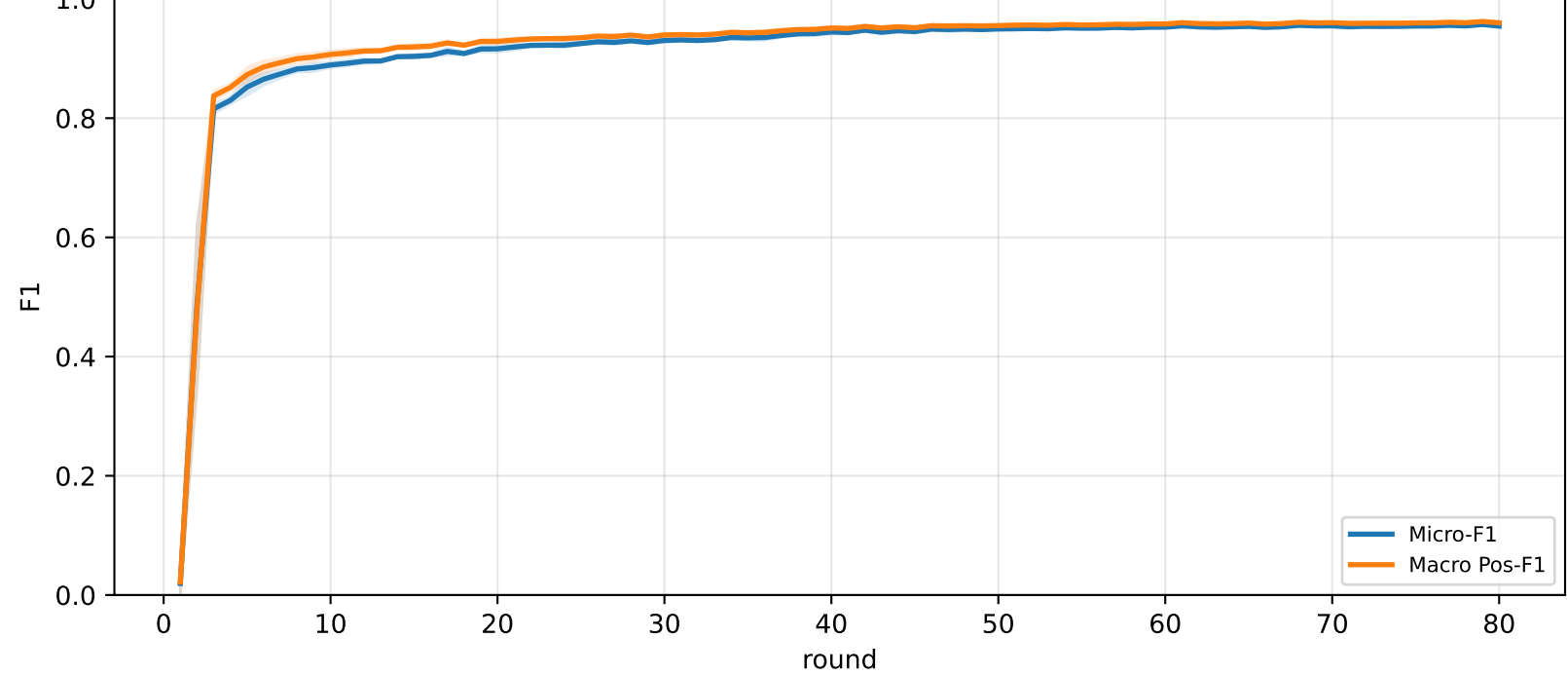
Pooled local centralized micro / macro F1



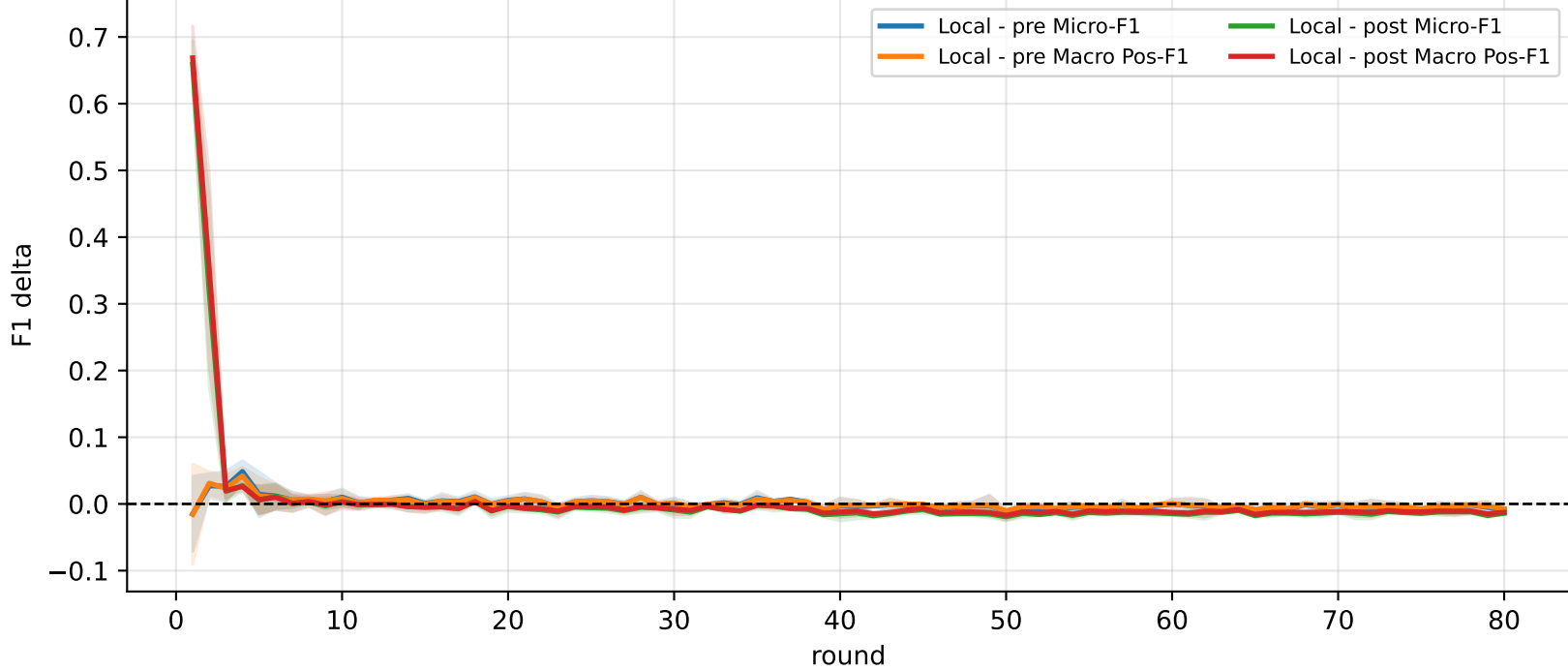
Pooled FedAvg micro / macro F1 (pre-aggregation)



Pooled FedAvg micro / macro F1 (post-aggregation)



Delta vs FedAvg (local minus pre / post)



Heterogeneity level: High | FedAvg diagnostics | family=strong_increasing | n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=strong_increasing | cycle2=9.9%±1.2% | cycle3=14.1%±0.8% | cycle4=19.4%±1.7% | cycle5=25.4%±1.6% | cycle6=31.3%±1.0%
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

Best logged test performance (FedAvg vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	92.3% ± 0.2%	93.6% ± 0.2%	98.2% ± 0.1%	96.3% ± 0.1%	94.2% ± 0.0%	91.2% ± 0.2%	88.2% ± 0.7%
FedAvg	93.1% ± 0.1%	94.2% ± 0.1%	98.2% ± 0.1%	96.5% ± 0.1%	94.9% ± 0.1%	92.7% ± 0.4%	89.0% ± 0.1%

